

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\108-CA-CBStack-2026y-4m-20d-51s-9h-32m.pvdb

Date of Analysis : Sep 8, 2026 6:07pm

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User
 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 2 of 285
 Vessel Design Summary: Step: 40 6:07pm Sep 8,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2480.000 mm ID
 Vessel Design Length, Tangent to Tangent 10346.50 mm
 Specified Datum Line Distance 0.00 mm
 Shell Side Design Temperature 130 °C
 Channel Side Design Temperature 82 °C
 Shell Side Design Pressure 520.000 KPa.
 Channel Side Design Pressure 1000.000 KPa.
 Shop Shell Side Test Pressure 1489.714 KPa.
 Field Shell Side Test Pressure 676.000 KPa.
 Shop Channel Side Test Pressure 1489.379 KPa.
 Field Channel Side Test Pressure 676.000 KPa.
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-266 2	...	...	K03506	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 2	...	...	K03506	
Flo. Head	SA-516 70	...	...	K02700	
Bk. Bolting	SA-193 B7M	...	<= 100	G41400	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-266 2	...	...	K03506	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Blind Flang	1027.936	103.42	1088.000	3.0000	N/A	No
LHS CH flange-1	1024.897	103.42	1819.000	3.0000	N/A	No
LHS Ch Shell	1024.142	103.42	1848.000	3.0000	N/A	No
LHS Ch Flg-2	1024.897	103.42	1806.000	3.0000	N/A	No
LHS Shell Flg-1	544.897	103.42	1409.000	3.0000	N/A	No
Main Shell	544.319	103.42	2718.000	3.0000	N/A	No
Main Shell	544.319	103.42	1629.000	3.0000	N/A	No
Main Shell	544.142	103.42	2718.000	3.0000	N/A	No
RHS Shell Flg-1	544.897	103.42	1409.000	3.0000	N/A	No



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RHS Shell Flg-2	544.897	103.42	1611.000	3.0000	N/A	No
RHS	544.142	103.42	1848.000	3.0000	N/A	No
RHS dish end	544.319	103.42	1422.000	3.0000	No	No

Liquid Level: 2850.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	139.2	136.0	144.0	136.0	130.7	64.9	1.00	1.00
Body Flg	426.2	287.0	220.0	212.0	207.9	207.9	1.00	1.00
Cylinder	1926.2	1500.0	20.0	20.0	12.3	9.5	1.00	1.00
Body Flg	2216.3	287.0	220.0	212.0	207.5	207.5	1.00	1.00
Body Flg	2629.5	295.0	228.0	220.0	216.8	216.8	1.00	1.00
Cylinder	4429.5	1800.0	28.0	28.0	7.9	14.3	1.00	1.00
Cylinder	6429.5	2000.0	18.0	18.0	7.9	14.2	1.00	1.00
Cylinder	8229.5	1800.0	28.0	28.0	7.9	14.3	1.00	1.00
Body Flg	8518.7	286.0	219.0	211.0	206.4	206.4	1.00	1.00
Body Flg	8795.7	277.0	210.0	202.0	196.9	196.9	1.00	1.00
Cylinder	10295.7	1500.0	20.0	20.0	7.9	9.9	1.00	1.00
Ellipse	10346.5	50.8	20.0	16.0	7.9	9.5	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3-Tube Inlet	WRC-297	0.993	Shell	Passed
N1-Shell Inlet	WRC-297	0.767	Nozzle	Passed
N2-Shell Outle	WRC-297	0.767	Nozzle	Passed
N4-Tube Outlet	WRC-297	0.988	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

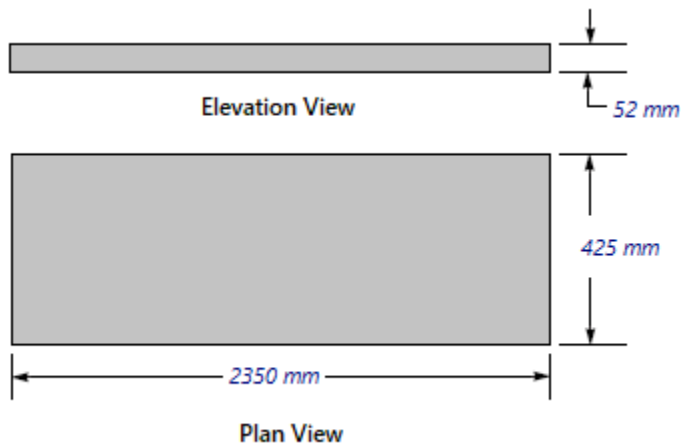
Saddle Width 400.000 mm
 Saddle Bearing Angle 120.000 deg.
 Centerline Dimension 1950.000 mm
 Wear Pad Width 700.000 mm
 Wear Pad Thickness 20.000 mm
 Wear Pad Bearing Angle 132.000 deg.
 Distance from Saddle to Tangent 1120.000 mm

Baseplate Length 2350.000 mm
 Baseplate Thickness 52.000 mm
 Baseplate Width 425.000 mm
 Number of Ribs (including outside ribs) 5
 Rib Thickness 20.000 mm
 Web Thickness 20.000 mm
 Height of Center Web 680.000 mm
 Number of Bolts in Baseplate 8

Baseplate Sketch



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Maximum Tensile Bolt Load 2162. Kg

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	284662.47	Kgf
Maximum Transverse Saddle Shear Load	43098.49	Kgf
Maximum Longitudinal Saddle Shear Load	15263.65	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	170297.81	Kgf
Maximum Transverse Saddle Shear Load	1128.77	Kgf
Maximum Longitudinal Saddle Shear Load	312.52	Kgf

Weights:

Fabricated - Bare w/o removable internals	161533.1	kg.
Shop Test - Fabricated + water (full)	260089.5	kg.
Shipping - Fab. + rem. intls.+ shipping app.	161533.1	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	161533.1	kg.
Empty - Fab. + intls. + details + wghts.	164119.0	kg.
Operating - Empty + operating liquid (no ca)	262657.7	kg.
Field Test - Empty weight + test medium (full)	260690.1	kg.

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 Nozzle Summary: Step: 38 6:07pm Sep 8,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3-Tube Inlet	1848	OK	...	OK	11.33	OK	Passed
N1-Shell Inlet	2160	OK	...	OK	11.33	OK	Passed
Drain	1629	...	...	OK	7.52	OK	No Calc[*]
Vent	1629	...	...	OK	7.52	OK	No Calc[*]
N2-Shell Outlet	2160	OK	...	OK	11.33	OK	Passed
N4-Tube Outlet	1848	OK	...	OK	11.33	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1629 Nozzle : Drain [Shellside]
 Minimum MAWP Nozzles : 1848 Nozzle : N3-Tube Inlet [Tubeside]

[*] - This was a small opening and the areas were not computed.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N3-Tube Inlet	1176.175	0.000	1162.198
60	N1-Shell Inlet	3453.754	0.000	662.850
70	Drain	5878.333	180.000	110.000
70	Vent	5753.083	0.000	110.000
80	N2-Shell Outlet	7747.554	180.000	662.850
110	N4-Tube Outlet	9545.702	180.000	1162.198

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
Drain	1.929 in	Actual	LWN	2.638	9.000	...	...	318.45	300
Vent	1.929 in	Actual	LWN	2.638	9.000	...	...	318.45	300
N1-Shell Inlet	14.000 in	60	WNF	14.000	15.088	595.60	16.00	340.81	150
N2-Shell Outlet	14.000 in	60	WNF	14.000	15.088	595.60	16.00	340.81	150
N3-Tube Inlet	24.016 in	Actual	WNF	24.016	17.450	1020.00	24.00	458.10	150
N4-Tube Outlet	24.016 in	Actual	WNF	24.016	17.450	1030.00	20.00	458.10	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Drain	SA-266 2	16.000	10.000	...	...	...
Vent	SA-266 2	16.000	10.000	...	...	...
N1-Shell In	SA-106 B	16.000	10.000	14.000	16.000	...
N2-Shell Ou	SA-106 B	16.000	10.000	14.000	16.000	...
N3-Tube Inl	SA-516 70	16.000	16.000	16.000	16.000	...
N4-Tube Out	SA-516 70	16.000	10.000	16.000	20.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Drain	5878.333	180.0	300.00	0.00	Main Shell
Vent	5753.083	0.0	300.00	0.00	Main Shell
N1-Shell Inlet	3453.754	0.0	300.00	0.00	Main Shell
N2-Shell Outlet	7747.554	180.0	300.00	0.00	Main Shell
N3-Tube Inlet	1176.175	0.0	400.00	0.00	LHS Ch Shell
N4-Tube Outlet	9545.701	180.0	400.00	0.00	RHS



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 Bill of Material: Step: 36 6:07pm Sep 8,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 144.0mm THK X 2748.0mm OD	SA-266 2
2	WELD NECK FLANGE: 220.0mm THK X 2748.0mm OD	SA-266 2
1	WELD NECK FLANGE: 228.0mm THK X 2748.0mm OD	SA-266 2
1	CYLINDER: 28.0mm THK X 2480.0mm ID X 1800.0mm	SA-516 70
1	WELD NECK FLANGE: 219.0mm THK X 2748.0mm OD	SA-266 2
1	WELD NECK FLANGE: 210.0mm THK X 2748.0mm OD	SA-266 2
1	CYLINDER: 20.0mm THK X 2480.0mm ID X 1500.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 20.0mm THK X 2480.0mm ID X 50.8mm	SA-516 70
9	BAFFLES: 2455.0mm DIA X 10.0mm THK	SA-516 70
12	TIE RODS: 16mm DIA X 4964mm	SA-193 B7M
1	TUBESHEET: 2562.0mm X 115.0mm THK	SA-266 2
6000	TUBES: 6000.0mm X 19.0mm DIA X 2.1mm THK	SA-179
1	FLOATING TUBESHEET: 2465.0mm X 115.0mm THK	SA-266 2
72	FLOATING HEAD FLANGE BOLTS: 38.1mm DIA	SA-193 B7M
144	NUTS FOR FLOATING HEAD FLANGE BOLTS: 38.1mm DIA	...
1	FLOATING HEAD FLANGE GASKET: 2323.0mm OD X 2297.0mm ID	...
1	INSULATION: 295mm X 100mm THK	Personal Protec
2	SADDLE: 400.0mm X 120 DEG	SA-266 2
1	INSULATION: 1800mm X 100mm THK	
1	INSULATION: 2000mm X 100mm THK	
1	INSULATION: 1800mm X 100mm THK	Personal Protec
2	INSULATION: 312mm X 100mm THK	Personal Protec
1	INSULATION: 1500mm X 100mm THK	Personal Protec
1	INSULATION: 50mm X 100mm THK	Personal Protec
3	GASKET: 2554.0mm OD X 2510.0mm ID	...
208	BODY FLANGE BOLTS: 38.1mm DIA	SA-193 B7
416	NUTS FOR BODY FLANGE BOLTS: 38.1mm DIA	...
2	CYLINDRICAL SEGMENT 20.0mm THK X 3149.5mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 1491.9mm X 1500.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 28.0mm THK X 3050.4mm X 1800.0mm	SA-516 70



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Bill of Material: Step: 36 6:07pm Sep 8,2026

1	CYLINDRICAL SEGMENT 28.0mm THK X 1690.2mm X 1800.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 18.0mm THK X 3050.4mm X 2000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm THK X 1690.2mm X 2000.0mm	SA-516 70
1	NAMEPLATE	...

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 MDMT Summary: Step: 39 6:07pm Sep 8,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
LHS Shell Flg[11]	B	-14	-37	-29	0.598	16.000	1.00	No
RHS Shell Flg[11]	B	-31	-48		0.385	16.000	1.00	No
Bolting [21]		-48						
RHS Shell Flg[11]	B	-24	-48	-29	0.564	20.000	1.00	No
Main Shell [8]	B	2	-48		0.382	28.000	1.00	No
Main Shell [8]	B	-11	-31	-29	0.637	18.000	1.00	No
Main Shell [8]	B	2	-48		0.382	28.000	1.00	No
Left Saddle [24]	B	-7	-48	-29	0.382	20.000	1.00	No
Sdl 2 Fr60 [24]	B	-7	-48	-29	0.382	20.000	1.00	No
N1-Shell Inlet[1]	B	-14	-48	-29	0.382	16.000	1.00	No
Nozzle Flg [5]	A	-18	-46					
Drain [1]	B	-29	-104	-29	0.036	9.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
Vent [1]	B	-29	-104	-29	0.035	9.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
N2-Shell Outle[1]	B	-14	-48	-29	0.391	16.000	1.00	No
Nozzle Flg [5]	A	-18	-44					
Tubesheet: SS[13]	B	3	-5		0.509	28.750	1.00	No
Warmest MDMT:		3	-5					
LHS Blind Fla[11]	B	7	2		0.906	34.000	1.00	No
LHS CH flange[11]	B	-7	-38	-29	0.514	20.000	1.00	No
LHS Ch Flg-2 [11]	B	-7	-38	-29	0.514	20.000	1.00	No
LHS Ch Shell [8]	B	-7	-33	-29	0.562	20.000	1.00	No
RHS [8]	B	-24	-48	-29	0.562	20.000	1.00	No
RHS dish end [10]	B	-14	-29	-29	0.730	16.000	1.00	No
RHS dish end [7]	B	-7	-33	-29	0.562	20.000	1.00	No
N3-Tube Inlet [1]	B	-7	-33	-29	0.562	20.000	1.00	No
Nozzle Flg [5]	A	-18	-46					
N4-Tube Outlet[1]	B	-24	-48	-29	0.575	20.000	1.00	No
Nozzle Flg [5]	A	-18	-44					
Tubesheet: CS[14]	B	3	-5		0.509	28.750	1.00	No
Bolting [21]		-48						
Bolting [21]		-48						
Warmest MDMT:		7	2					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-5.0	-12.0	Fail
Channel/Tube	2.0	-12.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable



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MDMT Summary:

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- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors: Step: 0 6:07pm Sep 8, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning: LHS Ch Flg-2 and LHS Shell Flg-1

There are mating flanges in this model that have different design pressures. This will generate different bolt loads. If a tubesheet is in between these flanges, PV Elite will compute the mating flange loads for you. If there is not a tubesheet physically separating these flanges, you must enter in the mating flange loads into the flange dialog.

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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 Input Echo: Step: 1 6:07pm Sep 8,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	520	KPa.
Channel Side Design Pressure	999.99	KPa.
Shell Side Design Temperature	130.0	°C
Channel Side Design Temperature	82.0	°C
Radiography, Shell Side	RT-2	
Radiography, Channel Side	RT-2	
Service Type, Shell Side	Sour	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	92.7 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Use ASME VIII-1 Appendix 1-9 N
 Perform Blast Load Analysis No
 Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No
 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/12	
Element From Node	10	
Element To Node	20	
Element Type	Flange	
Description	LHS Blind Flang	
Distance "FROM" to "TO"	136	mm
Flange Outside Diameter	2748	mm
Element Thickness	136	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	144	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	82	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	82	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2850	mm
Liquid Density	0.0009984	kg./cm ³

Element#	2/12
Element From Node	20



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Element To Node	30
Element Type	Flange
Description	LHS CH flange-1
Distance "FROM" to "TO"	287 mm
Flange Inside Diameter	2480 mm
Element Thickness	212 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	220 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2540 mm
Liquid Density	0.0009984 kg./cm ³

Element#	3/12
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	LHS Ch Shell
Distance "FROM" to "TO"	1500 mm
Inside Diameter	2480 mm
Element Thickness	20 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	20 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	242.6 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2460 mm
Liquid Density	0.0009984 kg./cm ³



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Element From Node	30
Detail Type	Nozzle
Detail ID	N3-Tube Inlet
Dist. from "FROM" Node / Offset dist	750 mm
Nozzle Diameter	24.02 in.
Nozzle Schedule	40
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	200.41 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element#	4/12
Element From Node	40
Element To Node	50
Element Type	Flange
Description	LHS Ch Flg-2
Distance "FROM" to "TO"	287 mm
Flange Inside Diameter	2478 mm
Element Thickness	212 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	220 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.08 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2540 mm
Liquid Density	0.0009984 kg./cm ³

Element#	5/12
Element From Node	50
Element To Node	60
Element Type	Flange
Description	LHS Shell Flg-1
Distance "FROM" to "TO"	295 mm
Flange Inside Diameter	2480 mm
Element Thickness	220 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	228 mm
External Corrosion Allowance	0 mm



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Design Internal Pressure	520	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2540	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	295	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	50	
Detail Type	For./Mom.	
Detail ID	TOP HE FORCES	
Dist. from "FROM" Node / Offset dist	295	mm
Force in X Direction	950	Kgf
Force in Y Direction	0	Kgf
Force in Z Direction	2100	Kgf
Moment about X Axis	0	Kg-m.
Moment about Y Axis	0	Kg-m.
Moment about Z Axis	0	Kg-m.
Force/Moment Combination Method	SRSS	

Element From Node	50	
Detail Type	For./Mom.	
Detail ID	TOP HE FORCES	
Dist. from "FROM" Node / Offset dist	295	mm
Force in X Direction	950	Kgf
Force in Y Direction	0	Kgf
Force in Z Direction	2100	Kgf
Moment about X Axis	0	Kg-m.
Moment about Y Axis	0	Kg-m.
Moment about Z Axis	0	Kg-m.
Force/Moment Combination Method	SRSS	

Element From Node	50	
Detail Type	For./Mom.	
Detail ID	TOP HE FORCES	
Dist. from "FROM" Node / Offset dist	295	mm
Force in X Direction	11200	Kgf
Force in Y Direction	0	Kgf
Force in Z Direction	11200	Kgf
Moment about X Axis	0	Kg-m.
Moment about Y Axis	0	Kg-m.
Moment about Z Axis	0	Kg-m.
Force/Moment Combination Method	SRSS	

Element From Node	50	
Detail Type	For./Mom.	
Detail ID	TOP HE FORCES	
Dist. from "FROM" Node / Offset dist	295	mm
Force in X Direction	11200	Kgf



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Force in Y Direction	0	Kgf
Force in Z Direction	11200	Kgf
Moment about X Axis	0	Kg-m.
Moment about Y Axis	0	Kg-m.
Moment about Z Axis	0	Kg-m.
Force/Moment Combination Method	SRSS	

Element#	6/12
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	1800 mm
Inside Diameter	2480 mm
Element Thickness	28 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	28 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	234.4 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	1120 mm
Width of Saddle	400 mm
Height of Saddle at Bottom	1950 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	700 mm
Thickness of Wear Plate	20 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.30000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2478 mm
Liquid Density	0.0009984 kg./cm ³



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Element From Node	60
Detail Type	Insulation
Detail ID	IN
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1800 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	60
Detail Type	Nozzle
Detail ID	N1-Shell Inlet
Dist. from "FROM" Node / Offset dist	824.23 mm
Nozzle Diameter	14 in.
Nozzle Schedule	60
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	114.22 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	60
Detail Type	Weight
Detail ID	Fireproofing we
Dist. from "FROM" Node / Offset dist	1120 mm
Miscellaneous Weight	500 Kg
Offset from Element Centerline	1900 mm

Element#	7/12
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	2000 mm
Inside Diameter	2480 mm
Element Thickness	18 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2478 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	70
Detail Type	Insulation
Detail ID	iN
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2000 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³



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Element From Node	70
Detail Type	Nozzle
Detail ID	Drain
Dist. from "FROM" Node / Offset dist	1448.8 mm
Nozzle Diameter	1.929 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	7.4236 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2

Element From Node	70
Detail Type	Nozzle
Detail ID	Vent
Dist. from "FROM" Node / Offset dist	1323.6 mm
Nozzle Diameter	1.929 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	7.4236 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2

Element From Node	70
Detail Type	Weight
Detail ID	TOP HE EMPTY
Dist. from "FROM" Node / Offset dist	390.9 mm
Miscellaneous Weight	79500 Kg
Offset from Element Centerline	3900 mm

Element From Node	70
Detail Type	Weight
Detail ID	TOP HE OPERATING
Dist. from "FROM" Node / Offset dist	390.9 mm
Miscellaneous Weight	50000 Kg
Offset from Element Centerline	3900 mm

Element From Node	70
Detail Type	Weight
Detail ID	TOP HE TEST
Dist. from "FROM" Node / Offset dist	390.9 mm
Miscellaneous Weight	50000 Kg
Offset from Element Centerline	3900 mm

Element#	8/12
Element From Node	80
Element To Node	90
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	1800 mm
Inside Diameter	2480 mm
Element Thickness	28 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	28 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	130 °C



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Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 80
 Detail Type Saddle
 Detail ID Sd1 2 Fr60
 Dist. from "FROM" Node / Offset dist 680 mm
 Width of Saddle 400 mm
 Height of Saddle at Bottom 1950 mm
 Saddle Contact Angle 120.0
 Height of Composite Ring Stiffener 0 mm
 Width of Wear Plate 700 mm
 Thickness of Wear Plate 20 mm
 Contact Angle, Wear Plate (degrees) 132.0
 Friction coefficient 0.0
 Moment Factor 3.0
 Dimension E at base (optional) 0 mm
 Circumferential Eff. over Saddle 1.0
 Circumferential Eff. at Midspan 1.0
 Tangent to Tangent dist. (optional) 0 mm

Element From Node 80
 Detail Type Liquid
 Detail ID Liquid: 60
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Liquid 2460 mm
 Liquid Density 0.0009984 kg./cm³

Element From Node 80
 Detail Type Insulation
 Detail ID Insulation
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 1800 mm
 Thickness of Insulation 100 mm
 Density 0.000352 kg./cm³

Element From Node 80
 Detail Type Nozzle
 Detail ID N2-Shell Outlet
 Dist. from "FROM" Node / Offset dist 1318 mm
 Nozzle Diameter 14 in.
 Nozzle Schedule 60
 Nozzle Class 150
 Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 114.22 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B

Element From Node 80
 Detail Type Weight
 Detail ID Wght:[2 of 2]
 Dist. from "FROM" Node / Offset dist 680 mm
 Miscellaneous Weight 500 Kg
 Offset from Element Centerline 1900 mm

Element# 9/12
 Element From Node 90
 Element To Node 100
 Element Type Flange



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Description	RHS Shell Flg-1	
Distance "FROM" to "TO"	286	mm
Flange Inside Diameter	2480	mm
Element Thickness	211	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	219	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	520	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	222.2	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	Yes	
Element From Node	90	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2540	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	90	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	312	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element#	10/12	
Element From Node	100	
Element To Node	110	
Element Type	Flange	
Description	RHS Shell Flg-2	
Distance "FROM" to "TO"	277	mm
Flange Inside Diameter	2480	mm
Element Thickness	202	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	210	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	520	KPa.
Design Temperature Internal Pressure	130	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	Yes	
Element From Node	100	
Detail Type	Liquid	



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Detail ID	Liquid: 80
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2540 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	100
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	312 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	11/12
Element From Node	110
Element To Node	120
Element Type	Cylinder
Description	RHS
Distance "FROM" to "TO"	1500 mm
Inside Diameter	2480 mm
Element Thickness	20 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	20 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	234.4 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	Yes

Element From Node	110
Detail Type	Liquid
Detail ID	Liquid: 90
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2460 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	110
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1500 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	110
Detail Type	Nozzle
Detail ID	N4-Tube Outlet
Dist. from "FROM" Node / Offset dist	750 mm



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Nozzle Diameter	24.02	in.
Nozzle Schedule	40	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	184.63	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element#	12/12
Element From Node	120
Element To Node	130
Element Type	Elliptical
Description	RHS dish end
Distance "FROM" to "TO"	50.8 mm
Inside Diameter	2480 mm
Element Thickness	16 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	20 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	130 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	120
Detail Type	Liquid
Detail ID	Liquid: 100
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2478 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	120
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	50.5 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³



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XY Coordinate Calculations:

Step: 2 6:07pm

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Blind Flang		139.175	...	136	...
LHS CH flange-1		426.175	...	287	...
LHS Ch Shell		1926.17	...	1500	...
LHS Ch Flg-2		2216.35	...	287	...
LHS Shell Flg-1		2629.53	...	295	...
Main Shell		4429.52	...	1800	...
Main Shell		6429.52	...	2000	...
Main Shell		8229.52	...	1800	...
RHS Shell Flg-1		8518.7	...	286	...
RHS Shell Flg-2		8795.7	...	277	...
RHS		10295.7	...	1500	...
RHS dish end		10346.5	...	50.8	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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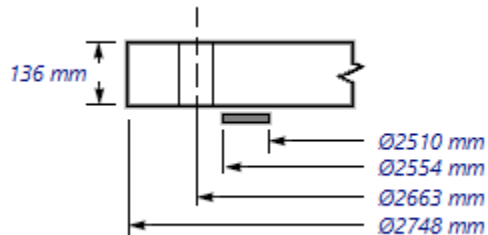
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 Flg Calc [Int P]: LHS Blind Fl Flng: 31 6:07pm Sep 8,2026

Flange Input Data Values

Description: LHS Blind Fl :

LHS Blind Flang

Flange Type		Blind	
Design Pressure	P	1027.94	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2748.000	mm
Flange Thickness	t	136.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	8366.062	mm
Diameter of Bolt Circle	C	2663.000	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		72	
Flange Face Outside Diameter	Fod	2565.000	mm
Flange Face Inside Diameter	Fid	2478.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2554.000	mm
Gasket Inside Diameter	Gi	2510.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	5118.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	0.0000	
Partition Gasket Design Seating Stress	yPart	0.00	N./mm ²



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Gasket Contact Width,	$N = (Go - Gi)/2$	22.000 mm
Basic Gasket Width,	$bo = N/2$	11.000 mm



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 Flg Calc [Int P]: LHS Blind Fl Flng: 31 6:07pm Sep 8,2026

Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$ 8.358 mm
 Gasket Reaction Diameter, $G = Go - 2 \cdot b$ 2537.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 2537^2 \cdot 1028$
 $= 530018.375 \text{ Kgf}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:
 $= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot lp \cdot bPart \cdot mPart \cdot P$
 $= 2 \cdot 8.36 \cdot \pi \cdot 2537 \cdot 3 \cdot 1028$
 $= 41900.309 \text{ Kgf}$

Design Bolt Load for the Operating Condition [Wo]:
 $= \max(Wo,1 + Wo,2 + H'p, 0)$
 $= \max(530018 + 41900 + 0, 0)$
 $= 571918.688 \text{ Kgf}$

Design Bolt Load for the Gasket Seating Condition [Wg]:
 $= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$
 $= 68.9 \cdot 8.36 \cdot \pi \cdot 2537 + 0 \cdot 0 \cdot 5118$
 $= 468391.500 \text{ Kgf}$

Required Bolt Area [Am]:
 $= \max((Wo + Fa + 4Me/G) / Sb, Wg / Sa)$
 $= \max((571919 + 0 + 4 \cdot 0 / 2537) / 172, 468392 / 172)$
 $= 326.094 \text{ cm}^2$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	326.094	652.644	
Radial Distance between Bolts and Edge:	38.100	42.500	
Circ. Spacing between the Bolts:	82.550	116.158	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
 $= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$
 $= 653 \cdot 172 / (68.9 \cdot \pi (2554 + 2510))$
 $= 10.233 \text{ mm}$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:
 $= Sa \cdot Ab$
 $= 172 \cdot 653$
 $= 1144638.00 \text{ Kgf}$

Gasket Load for the Operating Condition [HG]:
 $= Wo$
 $= 571918.69 \text{ Kgf}$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (2663 - 2537) / 2$
 $= 62.8575 \text{ mm}$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:
 $= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$
 $= 1.9 \cdot 571919 \cdot 62.9 \cdot 1 / (133^2 \cdot 2537) + 0.3 \cdot 1 \cdot 1028 \cdot 2537^2 / 133^2$



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 Flg Calc [Int P]: LHS Blind Fl Flng: 31 6:07pm Sep 8,2026

= 127.16 N./mm²

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

= $1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$
 = $1.9 \cdot 1144638 \cdot 62.9 \cdot 1 / (133^2 \cdot 2537)$
 = 29.87 N./mm²

Bolt Stress, Operating [BSO]:

= $Wm1 / Ab$
 = 571919/653
 = 85.94 N./mm²

Bolt Stress, Seating [BSa]:

= $(Wm2 / Ab)$
 = (468392/653)
 = 70.38 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	127.16	138.00	29.87	138.00
Bolting	85.94	172.00	70.38	172.00

Reqd. Blind Flange Thickness at Center 130.673 mm
 Reqd. Blind Flange Thickness at Gasket 64.878 mm

Estimated M.A.W.P. (Operating) 1115 KPa.
 Estimated Finished Weight of Flange at given Thk. 6251.6 kg.
 Estimated Unfinished Weight of Forging at given Thk 6251.6 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.91, Temperature Reduction per Fig. UCS 66.1 = 5 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 2 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	121.62	138.00	28.57	138.00
Bolting	85.94	172.00	70.38	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 127.686 mm

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Flange Input Data Values

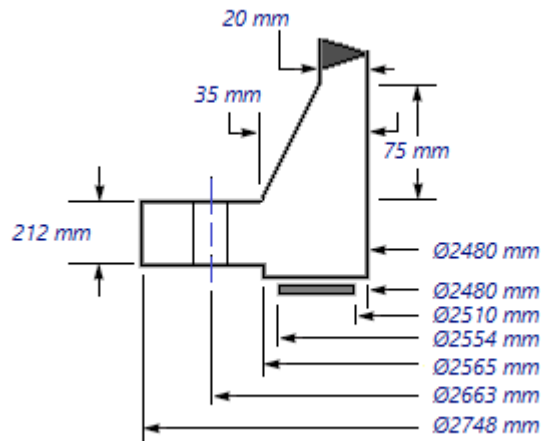
Description: LHS CH flang :

LHS CH flange-1

Flange Type	Integral Weld Neck		
Design Pressure	P	1024.90	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2480.000	mm
Flange Outside Diameter	A	2748.000	mm
Flange Thickness	t	212.0000	mm
Thickness of Hub at Small End	go	20.0000	mm
Thickness of Hub at Large End	gl	35.0000	mm
Length of Hub	h	75.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2663.000	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		72	
Flange Face Outside Diameter	Fod	2565.000	mm
Flange Face Inside Diameter	Fid	2480.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2554.000	mm
Gasket Inside Diameter	Gi	2510.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	5118.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	0.0000	
Partition Gasket Design Seating Stress	yPart	0.00	N./mm ²



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1025 \cdot (2480/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 1025) + C_a$$

$$= 12.2735 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 17) / (1243 + 0.6 \cdot 17)$$

$$= 1871.865 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 20) / (1240 + 0.6 \cdot 20)$$

$$= 2204.305 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	209.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2486.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	32.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	17.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	56.500 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2537.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2537^2 \cdot 1025$$

$$= 528451.625 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2537 \cdot 3 \cdot 1025$$

$$= 41776.449 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2486^2 \cdot 1025$$

$$= 507304.750 \text{ Kg}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 528452 - 507305$$

$$= 21146.889 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(528452 + 41776 + 0, 0)$$

$$= 570228.062 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2537 + 0 \cdot 0 \cdot 5118$$

$$= 468391.500 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((570228 + 0 + 4 \cdot 0 / 2537) / 172, 468392 / 172)$$

$$= 325.130 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 209 / (3 + 0.5)$$

$$= 434.486 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2663 \cdot \sin(\pi / 72)$$

$$= 116.158 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 116 / (2 \cdot 38.1 + 209) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	325.130	652.644	
Radial Distance between Hub and Bolts:	50.800	56.500	
Radial Distance between Bolts and Edge:	38.100	42.500	
Circ. Spacing between the Bolts:	82.550	116.158	434.486

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 653 \cdot 172 / (68.9 \cdot \pi (2554 + 2510))$$

$$= 10.233 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 653$$

$$= 1144638.00 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 570228 - 528452$$

$$= 41776.45 \text{ Kgf}$$

Moment Arm Calculations:



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Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (2663 - 2537) / 2$
 $= 62.8575 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5 ((C - B) / 2 + hg)$
 $= 0.5 ((2663 - 2486) / 2 + 62.9)$
 $= 75.6788 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (2663 - 2486 - 32) / 2$
 $= 72.5000 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	507305.	72.5000	1.0000	36780.
Face Pressure, Mt	21147.	75.6788	1.0000	1600.
Gasket Load, Mg	41776.	62.8575	1.0000	2626.
Gasket Seating, Matm	1144638.	62.8575	1.0000	71951.

Total Moment for Operation, Mo 41007. Kg-m.
 Total Moment for Gasket seating, Mg 71951. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 205.577 mm
 Hub Ratio, h/h0 = HL / H0 0.365
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.882

Flange Factors for Integral Flange:
 Factor F per 4.16.5 0.857
 Factor V per 4.16.5 0.288
 Factor f per 4.16.5 1.564

Stress factor from Table 4.16.4 [K]:
 $= A / B$
 $= 2748 / 2486$
 $= 1.105$

Stress factor from Table 4.16.4 [Y]:
 $= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.11^2 \cdot \log_{10}(1.11))) / (1.11^2 - 1)) / (1.11 - 1)$
 $= 19.341$

Stress factor from Table 4.16.4 [T]:
 $= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.11^2 (1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / ((1.0472 + 1.9448 \cdot 1.11^2) \cdot (1.11 - 1))$
 $= 1.875$

Stress factor from Table 4.16.4 [U]:
 $= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.11^2 (1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / (1.36136 (1.11^2 - 1) \cdot (1.11 - 1))$
 $= 21.254$

Stress factor from Table 4.16.4 [Z]:
 $= (K^2 + 1) / (K^2 - 1)$
 $= (1.11^2 + 1) / (1.11^2 - 1)$
 $= 10.014$

Stress factor from Table 4.16.4 for integral flanges [d]:
 $= U \cdot g0^2 \cdot h0 / V$
 $= 21.3 \cdot 17^2 \cdot 206 / 0.29$



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$$= 4390807.500 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.86/206 \\ &= 0.004 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3/d \\ &= (209 \cdot 0.0042 + 1) / 1.88 + 209^3/4390807 \\ &= 3.077 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g^{1^2}) \\ &= (1.56 \cdot 41007/2486) / (3.08 \cdot 32^2) \\ &= 80.31 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g^{1^2}) \\ &= (1.56 \cdot 71951/2486) / (3.08 \cdot 32^2) \\ &= 140.91 \text{ N./mm}^2 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.16 \cdot 41007/2486) / (3.08 \cdot 209^2) \\ &= 2.60 \text{ N./mm}^2 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.16 \cdot 71951/2486) / (3.08 \cdot 209^2) \\ &= 4.56 \text{ N./mm}^2 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (19.3 \cdot 41007 / (209^2 \cdot 2486)) - 10 \cdot 2.6 \\ &= 45.58 \text{ N./mm}^2 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (19.3 \cdot 71951 / (209^2 \cdot 2486)) - 10 \cdot 4.56 \\ &= 79.97 \text{ N./mm}^2 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (SHo + \max(SRo, STo)) / 2 \\ &= (80.3 + \max(2.6, 45.6)) / 2 \\ &= 62.94 \text{ N./mm}^2 \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned} &= (SHa + \max(SRa, STa)) / 2 \\ &= (141 + \max(4.56, 80)) / 2 \\ &= 110.44 \text{ N./mm}^2 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= Wm1 / Ab \\ &= 570228/653 \\ &= 85.68 \text{ N./mm}^2 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm2 / Ab) \\ &= (468392/653) \\ &= 70.38 \text{ N./mm}^2 \end{aligned}$$

Flange Stress Analysis Results: N./mm²



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	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	80.31	207.00	140.91	207.00
Radial Flange	2.60	138.00	4.56	138.00
Tangential Flange	45.58	138.00	79.97	138.00
Maximum Average	62.94	138.00	110.44	138.00
Bolting	85.68	172.00	70.38	172.00

Minimum Required Flange Thickness [Rigidity] 207.950 mm

Estimated M.A.W.P. (Operating) 1844 KPa.

Estimated Finished Weight of Flange at given Thk. 1934.0 kg.

Estimated Unfinished Weight of Forging at given Thk 2447.8 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	283.10	374.99	0.755	283.10	374.99	0.755
Tangential Flange	Sto	160.58	250.00	0.642	160.58	250.00	0.642
Combined	Sh/f + St	602.00	749.99	0.803	602.00	749.99	0.803
Combined	Sr + St	299.26	749.99	0.399	299.26	749.99	0.399
Combined	Sh + Sto	660.57	749.99	0.881	660.57	749.99	0.881
Radial Flange	Sr	16.16	499.99	0.032	16.16	499.99	0.032
Maximum Stress Ratio				1.000			1.000



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)
 = max(1; 2.09/1.33)
 = 1.5678

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)
 = max(0.25; 0.63/1.26) if $h/h0 < 0.35$, else max (0.25; 0.38/0.75)
 = 0.5035

In this case, the hub ratio $h/h0$, 0.3648 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	608.9 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	9.0645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.679 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (2554^2 - 2510^2) + (0.5 \cdot 5118 \cdot 0)$
 $= 1750.0 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 207 \cdot 1750 / (72 \cdot 9.06)$
 $= 555.1 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(555; 508)$
 $= 507.5 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(508; 290)$
 $= 507.5 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(508; 609)$
 $= 507.5 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin \cdot S(Ag / (nb \cdot Abs))$
 $:: 508 \geq 140 (1750 / (72 \cdot 9.06))$
 $:: 507.5 \geq 375.4 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin \cdot O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 508 \geq (97 \cdot 1750 + \pi/4 \cdot 1025 \cdot 2510^2) / (0.7 \cdot 9.06 \cdot 72)$
 $:: 507.5 \geq 482.6 \text{ N./mm}^2$ Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 508 \leq 276 (1750 / (72 \cdot 9.06))$
 $:: 507.5 \leq 740.1 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(thetagmax / thetafmax)$
 $:: 508 \leq 609 (1/0.68)$
 $:: 507.5 \leq 897.2 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 508 \cdot 0.2 \cdot 9.06 \cdot 38.1$
 $= 357.4 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 71951 / 1 \cdot 9807 \cdot 0.29 / (2.95 \cdot 201334 \cdot 17^2 \cdot 206 \cdot 0.3)$
 $= 0.999$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 41007 / 1 \cdot 9807 \cdot 0.29 / (2.95 \cdot 197847 \cdot 17^2 \cdot 206 \cdot 0.3)$
 $= 0.579$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 71951 / 1 \cdot 9807 \cdot 0.29 / (3.08 \cdot 201334 \cdot 17^2 \cdot 206 \cdot 0.3)$
 $= 0.958$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 41007 / 1 \cdot 9807 \cdot 0.29 / (3.08 \cdot 197847 \cdot 17^2 \cdot 206 \cdot 0.3)$
 $= 0.555$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.51, Temperature Reduction per Fig. UCS 66.1 = 31 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-38 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	75.37	207.00	129.71	207.00
Radial Flange	2.97	138.00	5.11	138.00
Tangential Flange	40.43	138.00	69.59	138.00
Maximum Average	57.90	138.00	99.65	138.00
Bolting	85.68	172.00	70.38	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 192.862 mm

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Flange Input Data Values

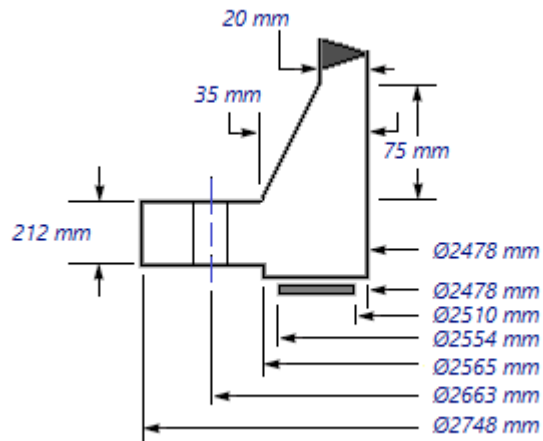
Description: LHS Ch FI-2 :

LHS Ch Flg-2

Flange Type	Integral Weld Neck		
Design Pressure	P	1024.90	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2478.000	mm
Flange Outside Diameter	A	2748.000	mm
Flange Thickness	t	212.0000	mm
Thickness of Hub at Small End	go	20.0000	mm
Thickness of Hub at Large End	gl	35.0000	mm
Length of Hub	h	75.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2663.000	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		72	
Flange Face Outside Diameter	Fod	2565.000	mm
Flange Face Inside Diameter	Fid	2478.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2554.000	mm
Gasket Inside Diameter	Gi	2510.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	5118.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	0.0000	
Partition Gasket Design Seating Stress	yPart	0.00	N./mm ²



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1025 \cdot (2478/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 1025) + C_a$$

$$= 12.2661 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 17) / (1242 + 0.6 \cdot 17)$$

$$= 1873.360 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 20) / (1239 + 0.6 \cdot 20)$$

$$= 2206.067 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	209.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2484.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	32.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	17.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	57.500 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2537.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2537^2 \cdot 1025$$

$$= 528451.625 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2537 \cdot 3 \cdot 1025$$

$$= 41776.449 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2484^2 \cdot 1025$$

$$= 506488.906 \text{ Kg}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 528452 - 506489$$

$$= 21962.744 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(528452 + 41776 + 0, 0)$$

$$= 570228.062 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2537 + 0 \cdot 0 \cdot 5118$$

$$= 468391.500 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((570228 + 0 + 4 \cdot 0 / 2537) / 172, 468392 / 172)$$

$$= 325.130 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 209 / (3 + 0.5)$$

$$= 434.486 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2663 \cdot \sin(\pi / 72)$$

$$= 116.158 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ Bs / (2a + t) }, 1)$$

$$= \max(\sqrt{ 116 / (2 \cdot 38.1 + 209) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	325.130	652.644	
Radial Distance between Hub and Bolts:	50.800	57.500	
Radial Distance between Bolts and Edge:	38.100	42.500	
Circ. Spacing between the Bolts:	82.550	116.158	434.486

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 653 \cdot 172 / (68.9 \cdot \pi (2554 + 2510))$$

$$= 10.233 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 653$$

$$= 1144638.00 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 570228 - 528452$$

$$= 41776.45 \text{ Kg}$$

Moment Arm Calculations:



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Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (2663 - 2537) / 2$
 $= 62.8575 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5 ((C - B) / 2 + hg)$
 $= 0.5 ((2663 - 2484) / 2 + 62.9)$
 $= 76.1787 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (2663 - 2484 - 32) / 2$
 $= 73.5000 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	506489.	73.5000	1.0000	37228.
Face Pressure, Mt	21963.	76.1787	1.0000	1673.
Gasket Load, Mg	41776.	62.8575	1.0000	2626.
Gasket Seating, Matm	1144638.	62.8575	1.0000	71951.

Total Moment for Operation, Mo 41527. Kg-m.
 Total Moment for Gasket seating, Mg 71951. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 205.495 mm
 Hub Ratio, h/h0 = HL / H0 0.365
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.882

Flange Factors for Integral Flange:
 Factor F per 4.16.5 0.857
 Factor V per 4.16.5 0.288
 Factor f per 4.16.5 1.564

Stress factor from Table 4.16.4 [K]:
 $= A / B$
 $= 2748 / 2484$
 $= 1.106$

Stress factor from Table 4.16.4 [Y]:
 $= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.11^2 \cdot \log_{10}(1.11))) / (1.11^2 - 1)) / (1.11 - 1)$
 $= 19.190$

Stress factor from Table 4.16.4 [T]:
 $= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.11^2 (1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / ((1.0472 + 1.9448 \cdot 1.11^2) \cdot (1.11 - 1))$
 $= 1.875$

Stress factor from Table 4.16.4 [U]:
 $= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.11^2 (1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / (1.36136 (1.11^2 - 1) \cdot (1.11 - 1))$
 $= 21.087$

Stress factor from Table 4.16.4 [Z]:
 $= (K^2 + 1) / (K^2 - 1)$
 $= (1.11^2 + 1) / (1.11^2 - 1)$
 $= 9.934$

Stress factor from Table 4.16.4 for integral flanges [d]:
 $= U \cdot g0^2 \cdot h0 / V$
 $= 21.1 \cdot 17^2 \cdot 205 / 0.29$



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$$= 4355430.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.86/205 \\ &= 0.004 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3/d \\ &= (209 \cdot 0.0042 + 1) / 1.88 + 209^3/4355430 \\ &= 3.094 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot M_{op} / B_{cor}) / (L \cdot g^{1^2}) \\ &= (1.56 \cdot 41527/2484) / (3.09 \cdot 32^2) \\ &= 80.91 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot M_{atm} / B_{cor}) / (L \cdot g^{1^2}) \\ &= (1.56 \cdot 71951/2484) / (3.09 \cdot 32^2) \\ &= 140.18 \text{ N./mm}^2 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot M_{op} / B_{cor}) / (L \cdot t^2) \\ &= (2.16 \cdot 41527/2484) / (3.09 \cdot 209^2) \\ &= 2.62 \text{ N./mm}^2 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot M_{atm} / B_{cor}) / (L \cdot t^2) \\ &= (2.16 \cdot 71951/2484) / (3.09 \cdot 209^2) \\ &= 4.54 \text{ N./mm}^2 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot M_{op} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ro} \\ &= (19.2 \cdot 41527 / (209^2 \cdot 2484)) - 9.93 \cdot 2.62 \\ &= 45.97 \text{ N./mm}^2 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot M_{atm} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ra} \\ &= (19.2 \cdot 71951 / (209^2 \cdot 2484)) - 9.93 \cdot 4.54 \\ &= 79.65 \text{ N./mm}^2 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2 \\ &= (80.9 + \max(2.62, 46)) / 2 \\ &= 63.44 \text{ N./mm}^2 \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned} &= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2 \\ &= (140 + \max(4.54, 79.6)) / 2 \\ &= 109.91 \text{ N./mm}^2 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= W_{m1} / A_b \\ &= 570228/653 \\ &= 85.68 \text{ N./mm}^2 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (W_{m2} / A_b) \\ &= (468392/653) \\ &= 70.38 \text{ N./mm}^2 \end{aligned}$$

Flange Stress Analysis Results: N./mm²



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	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	80.91	207.00	140.18	207.00
Radial Flange	2.62	138.00	4.54	138.00
Tangential Flange	45.97	138.00	79.65	138.00
Maximum Average	63.44	138.00	109.91	138.00
Bolting	85.68	172.00	70.38	172.00

Minimum Required Flange Thickness [Rigidity] 207.467 mm

Estimated M.A.W.P. (Operating) 1831 KPa.
 Estimated Finished Weight of Flange at given Thk. 1946.7 kg.
 Estimated Unfinished Weight of Forging at given Thk 2465.1 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	283.43	374.99	0.756	283.43	374.99	0.756
Tangential Flange	Sto	160.63	250.00	0.643	160.63	250.00	0.643
Combined	Sh/f + St	602.46	749.99	0.803	602.46	749.99	0.803
Combined	Sr + St	299.60	749.99	0.399	299.60	749.99	0.399
Combined	Sh + Sto	660.62	749.99	0.881	660.62	749.99	0.881
Radial Flange	Sr	16.17	499.99	0.032	16.17	499.99	0.032
Maximum Stress Ratio				1.000			1.000



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1_{\text{num}} / F1_{\text{den}})$$

$$= \max(1; 2.09/1.33)$$

$$= 1.5672$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; f_{n2}) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; f_{n3})$$

$$= \max(0.25; 0.64/1.26) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 0.38/0.75)$$

$$= 0.5035$$

In this case, the hub ratio $h/h0$, 0.3650 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sb_{max}	507.5 N./mm ²
Minimum permissible bolt stress	Sb_{min}	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sf_{max}	612.1 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sg_{max}	276.0 N./mm ²
Minimum gasket operating stress	$Sg_{\text{min-O}}$	97.0 N./mm ²
Minimum gasket seating stress	$Sg_{\text{min-S}}$	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	$Phig$	0.700



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Root Area of a Single Bolt	Abs	9.0645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.679 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (2554^2 - 2510^2) + (0.5 \cdot 5118 \cdot 0)$
 $= 1750.0 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 208 \cdot 1750 / (72 \cdot 9.06)$
 $= 557.7 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(558; 508)$
 $= 507.5 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(508; 290)$
 $= 507.5 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(508; 612)$
 $= 507.5 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$
 $:: 508 \geq 140 (1750 / (72 \cdot 9.06))$
 $:: 507.5 \geq 375.4 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 508 \geq (97 \cdot 1750 + \pi/4 \cdot 1025 \cdot 2510^2) / (0.7 \cdot 9.06 \cdot 72)$
 $:: 507.5 \geq 482.6 \text{ N./mm}^2$ Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 508 \leq 276 (1750 / (72 \cdot 9.06))$
 $:: 507.5 \leq 740.1 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(thetagmax / thetafmax)$
 $:: 508 \leq 612 (1/0.68)$
 $:: 507.5 \leq 902.0 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 508 \cdot 0.2 \cdot 9.06 \cdot 38.1$
 $= 357.4 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 71951 / 1 \cdot 9807 \cdot 0.29 / (2.95 \cdot 201334 \cdot 17^2 \cdot 205 \cdot 0.3)$
 $= 0.999$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 41527 / 1 \cdot 9807 \cdot 0.29 / (2.95 \cdot 197847 \cdot 17^2 \cdot 205 \cdot 0.3)$
 $= 0.587$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 71951 / 1 \cdot 9807 \cdot 0.29 / (3.09 \cdot 201334 \cdot 17^2 \cdot 205 \cdot 0.3)$
 $= 0.953$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 41527 / 1 \cdot 9807 \cdot 0.29 / (3.09 \cdot 197847 \cdot 17^2 \cdot 205 \cdot 0.3)$
 $= 0.559$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.51, Temperature Reduction per Fig. UCS 66.1 = 31 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-38 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	75.95	207.00	129.11	207.00
Radial Flange	3.00	138.00	5.09	138.00
Tangential Flange	40.80	138.00	69.35	138.00
Maximum Average	58.38	138.00	99.23	138.00
Bolting	85.68	172.00	70.38	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 192.456 mm

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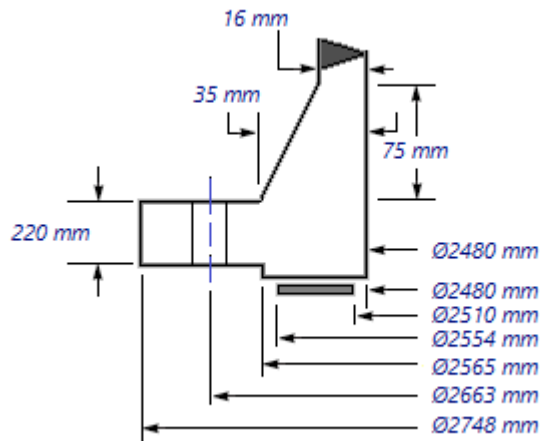
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Flange Input Data Values

Description: LHS Sh Fl 1 :

LHS Shell Flg-1

Flange Type	Integral Weld Neck		
Design Pressure	P	544.90	KPa.
Design Temperature		130	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2480.000	mm
Flange Outside Diameter	A	2748.000	mm
Flange Thickness	t	220.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	gl	35.0000	mm
Length of Hub	h	75.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2663.000	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		72	
Flange Face Outside Diameter	Fod	2565.000	mm
Flange Face Inside Diameter	Fid	2480.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2554.000	mm
Gasket Inside Diameter	Gi	2510.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	5118.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	0.0000	
Partition Gasket Design Seating Stress	yPart	0.00	N./mm ²



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (545 \cdot (2480/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 545) + C_a$$

$$= 7.9201 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 13) / (1243 + 0.6 \cdot 13)$$

$$= 1434.173 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 16) / (1240 + 0.6 \cdot 16)$$

$$= 1766.831 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	217.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2486.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	32.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	13.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	56.500 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2537.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2537^2 \cdot 545$$

$$= 280956.812 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2537 \cdot 3 \cdot 545$$

$$= 22210.883 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2486^2 \cdot 545$$

$$= 269713.844 \text{ Kg}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 280957 - 269714$$

$$= 11242.958 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(280957 + 22211 + 0, 0)$$

$$= 303167.688 \text{ Kgf}$$

$$= 570228.062 \text{ Kgf, Mating Flange Load Governs}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2537 + 0 \cdot 0 \cdot 5118$$

$$= 468391.500 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((570228 + 0 + 4 \cdot 0 / 2537) / 172, 468392 / 172)$$

$$= 325.130 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 217 / (3 + 0.5)$$

$$= 448.200 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2663 \cdot \sin(\pi / 72)$$

$$= 116.158 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ (Bs / (2a + t)) }, 1)$$

$$= \max(\sqrt{ (116 / (2 \cdot 38.1 + 217)) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	325.130	652.644	
Radial Distance between Hub and Bolts:	50.800	56.500	
Radial Distance between Bolts and Edge:	38.100	42.500	
Circ. Spacing between the Bolts:	82.550	116.158	448.200

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 653 \cdot 172 / (68.9 \cdot \pi (2554 + 2510))$$

$$= 10.233 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 653$$

$$= 1144638.00 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 570228 - 280957$$

$$= 289271.28 \text{ Kgf}$$



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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (2663 - 2537) / 2$
 $= 62.8575 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5 ((C - B) / 2 + hg)$
 $= 0.5 ((2663 - 2486) / 2 + 62.9)$
 $= 75.6788 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (2663 - 2486 - 32) / 2$
 $= 72.5000 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	269714.	72.5000	1.0000	19555.
Face Pressure, Mt	11243.	75.6788	1.0000	851.
Gasket Load, Mg	289271.	62.8575	1.0000	18183.
Gasket Seating, Matm	1144638.	62.8575	1.0000	71951.

Total Moment for Operation, Mo 38589. Kg-m.
 Total Moment for Gasket seating, Mg 71951. Kg-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 179.772 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 0.417
 Thickness Ratio, $g1/g_o = (g1_{cor}/g_{oCor})$ 2.462

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.835
 Factor V per 4.16.5 0.216
 Factor f per 4.16.5 2.372

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 2748 / 2486$
 $= 1.105$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.11^2 \cdot \log_{10}(1.11))) / (1.11^2 - 1)) / (1.11 - 1)$
 $= 19.341$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.11^2 (1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / ((1.0472 + 1.9448 \cdot 1.11^2) \cdot (1.11 - 1))$
 $= 1.875$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.11^2 (1 + 8.55246 \cdot \log_{10}(1.11)) - 1) / (1.36136 (1.11^2 - 1) \cdot (1.11 - 1))$
 $= 21.254$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.11^2 + 1) / (1.11^2 - 1)$
 $= 10.014$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_o^2 \cdot h_o / V$



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$$= 21.3 \cdot 13^2 \cdot 180 / 0.22$$

$$= 2982691.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.84 / 180$$

$$= 0.005 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$

$$= (217 \cdot 0.0046 + 1) / 1.88 + 217^3 / 2982691$$

$$= 4.497$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot M_{op} / B_{cor}) / (L \cdot g_1^2)$$

$$= (2.37 \cdot 38589 / 2486) / (4.5 \cdot 32^2)$$

$$= 78.43 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot M_{atm} / B_{cor}) / (L \cdot g_1^2)$$

$$= (2.37 \cdot 71951 / 2486) / (4.5 \cdot 32^2)$$

$$= 146.23 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot M_{op} / B_{cor}) / (L \cdot t^2)$$

$$= (2.34 \cdot 38589 / 2486) / (4.5 \cdot 217^2)$$

$$= 1.69 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot M_{atm} / B_{cor}) / (L \cdot t^2)$$

$$= (2.34 \cdot 71951 / 2486) / (4.5 \cdot 217^2)$$

$$= 3.14 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot M_o / (t^2 \cdot B_{cor})) - Z \cdot S_{Ro}$$

$$= (19.3 \cdot 38589 / (217^2 \cdot 2486)) - 10 \cdot 1.69$$

$$= 45.65 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot M_{atm} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ra}$$

$$= (19.3 \cdot 71951 / (217^2 \cdot 2486)) - 10 \cdot 3.14$$

$$= 85.11 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (78.4 + \max(1.69, 45.6)) / 2$$

$$= 62.04 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (146 + \max(3.14, 85.1)) / 2$$

$$= 115.67 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 570228 / 653$$

$$= 85.68 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (468392 / 653)$$

$$= 70.38 \text{ N./mm}^2$$



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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	78.43	207.00	146.23	207.00
Radial Flange	1.69	138.00	3.14	138.00
Tangential Flange	45.65	138.00	85.11	138.00
Maximum Average	62.04	138.00	115.67	138.00
Bolting	85.68	172.00	70.38	172.00

Minimum Required Flange Thickness [Rigidity] 216.840 mm

Estimated M.A.W.P. (Operating) 1434 KPa.
 Estimated Finished Weight of Flange at given Thk. 1993.0 kg.
 Estimated Unfinished Weight of Forging at given Thk 2516.0 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	291.58	374.99	0.778	291.58	374.99	0.778
Tangential Flange	Sto	164.45	250.00	0.658	164.45	250.00	0.658
Combined	Sh/f + St	502.75	749.99	0.670	502.75	749.99	0.670
Combined	Sr + St	302.35	749.99	0.403	302.35	749.99	0.403
Combined	Sh + Sto	664.44	749.99	0.886	664.44	749.99	0.886
Radial Flange	Sr	10.77	499.99	0.022	10.77	499.99	0.022
Maximum Stress Ratio				1.000			1.000



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1_{num} / F1_{den})$$

$$= \max(1; 3.28/1.38)$$

$$= 2.3678$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; f_{n2}) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; f_{n3})$$

$$= \max(0.25; 1.01/1.28) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 0.58/0.74)$$

$$= 0.7788$$

In this case, the hub ratio $h/h0$, 0.4172 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sb_{max}	507.5 N./mm ²
Minimum permissible bolt stress	Sb_{min}	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sf_{max}	589.2 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sg_{max}	276.0 N./mm ²
Minimum gasket operating stress	Sg_{min-O}	97.0 N./mm ²
Minimum gasket seating stress	Sg_{min-S}	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	$Phig$	0.700



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Root Area of a Single Bolt	Abs	9.0645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	222.2 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.591 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (2554^2 - 2510^2) + (0.5 \cdot 5118 \cdot 0)$
 $= 1750.0 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 208 \cdot 1750 / (72 \cdot 9.06)$
 $= 557.7 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(558; 508)$
 $= 507.5 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(508; 290)$
 $= 507.5 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(508; 589)$
 $= 507.5 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin \cdot S(Ag / (nb \cdot Abs))$
 $:: 508 \geq 140 (1750 / (72 \cdot 9.06))$
 $:: 507.5 \geq 375.4 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin \cdot O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 508 \geq (97 \cdot 1750 + \pi/4 \cdot 545 \cdot 2510^2) / (0.7 \cdot 9.06 \cdot 72)$
 $:: 507.5 \geq 430.6 \text{ N./mm}^2$ Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 508 \leq 276 (1750 / (72 \cdot 9.06))$
 $:: 507.5 \leq 740.1 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(thetagmax / thetafmax)$
 $:: 508 \leq 589 (1/0.59)$
 $:: 507.5 \leq 997.0 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 508 \cdot 0.2 \cdot 9.06 \cdot 38.1$
 $= 357.4 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 107 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 71951 / 1 \cdot 9807 \cdot 0.22 / (4.34 \cdot 201334 \cdot 13^2 \cdot 180 \cdot 0.3)$
 $= 0.999$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 38589 / 1 \cdot 9807 \cdot 0.22 / (4.34 \cdot 194922 \cdot 13^2 \cdot 180 \cdot 0.3)$
 $= 0.553$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 71951 / 1 \cdot 9807 \cdot 0.22 / (4.5 \cdot 201334 \cdot 13^2 \cdot 180 \cdot 0.3)$
 $= 0.965$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 38589 / 1 \cdot 9807 \cdot 0.22 / (4.5 \cdot 194922 \cdot 13^2 \cdot 180 \cdot 0.3)$
 $= 0.534$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.6, Temperature Reduction per Fig. UCS 66.1 = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -37 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	72.74	207.00	134.14	207.00
Radial Flange	2.01	138.00	3.70	138.00
Tangential Flange	40.60	138.00	74.87	138.00
Maximum Average	56.67	138.00	104.51	138.00
Bolting	85.68	172.00	70.38	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 203.784 mm



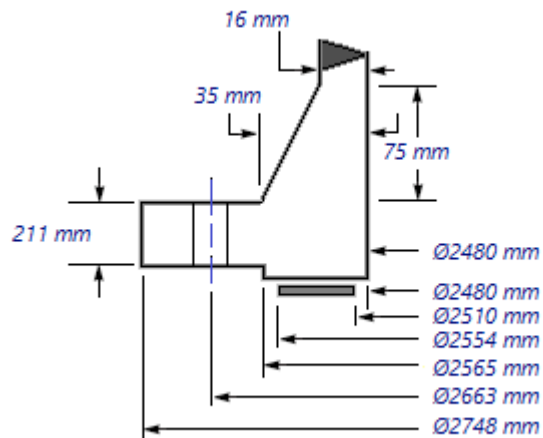
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Flange Input Data Values

Description: RHS Sh Fl 1 :

RHS Shell Flg-1

Flange Type	Integral Weld Neck		
Design Pressure	P	544.90	KPa.
Design Temperature		130	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2480.000	mm
Flange Outside Diameter	A	2748.000	mm
Flange Thickness	t	211.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	gl	35.0000	mm
Length of Hub	h	75.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2663.000	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		64	
Flange Face Outside Diameter	Fod	2565.000	mm
Flange Face Inside Diameter	Fid	2480.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2554.000	mm
Gasket Inside Diameter	Gi	2510.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (545 \cdot (2480/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 545) + Ca$$

$$= 7.9201 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 13) / (1243 + 0.6 \cdot 13)$$

$$= 1434.173 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 16) / (1240 + 0.6 \cdot 16)$$

$$= 1766.831 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	208.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2486.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	32.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	13.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	56.500 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2537.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2537^2 \cdot 545$$

$$= 280956.812 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2537 \cdot 3 \cdot 545$$

$$= 22210.883 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2486^2 \cdot 545$$

$$= 269713.844 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 280957 - 269714$$

$$= 11242.958 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o, 1 + W_o, 2 + H'p, 0)$$

$$= \max(280957 + 22211 + 0, 0)$$

$$= 303167.688 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2537 + 0 \cdot 0$$

$$= 468391.500 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((303168 + 0 + 4 \cdot 0 / 2537) / 172, 468392 / 172)$$

$$= 267.065 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 208/(3 + 0.5)$$

$$= 432.771 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2663 \cdot \sin(\pi/64)$$

$$= 130.667 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{131/(2 \cdot 38.1 + 208)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	267.065	580.128	
Radial Distance between Hub and Bolts:	50.800	56.500	
Radial Distance between Bolts and Edge:	38.100	42.500	
Circ. Spacing between the Bolts:	82.550	130.667	432.771

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 580 \cdot 172 / (68.9 \cdot \pi(2554 + 2510))$$

$$= 9.096 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 580$$

$$= 1017456.06 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 303168 - 280957$$

$$= 22210.89 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2663 - 2537) / 2$$

$$= 62.8575 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 (C - B) / 2 + hg$$

$$= 0.5 (2663 - 2486) / 2 + 62.9$$

$$= 75.6788 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2663 - 2486 - 32) / 2$$

$$= 72.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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End Pressure, Md	269714.	72.5000	1.0000	19555.
Face Pressure, Mt	11243.	75.6788	1.0000	851.
Gasket Load, Mg	22211.	62.8575	1.0000	1396.
Gasket Seating, Matm	1017456.	62.8575	1.0000	63956.

Total Moment for Operation, Mo 21802. Kg-m.
 Total Moment for Gasket seating, Mg 63956. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 179.772 mm
 Hub Ratio, h/h0 = HL / H0 0.417
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 2.462

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.835
 Factor V per 4.16.5 0.216
 Factor f per 4.16.5 2.372

Stress factor from Table 4.16.4 [K]:

= A / B
 = 2748/2486
 = 1.105

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
 = (0.66845 + (5.7169(1.11² • log10(1.11))) / (1.11² - 1))/(1.11 - 1)
 = 19.341

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K²) • (K - 1))
 = (1.11²(1 + 8.55246 • log10(1.11)) - 1)/((1.0472 + 1.9448 • 1.11²) • (1.11 - 1))
 = 1.875

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1)/(1.36136(K² - 1) • (K - 1))
 = (1.11²(1 + 8.55246 • log10(1.11)) - 1)/(1.36136(1.11² - 1) • (1.11 - 1))
 = 21.254

Stress factor from Table 4.16.4 [Z]:

= (K² + 1)/(K² - 1)
 = (1.11² + 1)/(1.11² - 1)
 = 10.014

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
 = 21.3 • 13² • 180/0.22
 = 2982691.000 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
 = 0.84/180
 = 0.005 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
 = (208 • 0.0046 + 1) / 1.88 + 208³/2982691
 = 4.066

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
 = (2.37•21802/2486)/(4.07•32²)
 = 49.01 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{12})$$

$$= (2.37 \cdot 63956 / 2486) / (4.07 \cdot 32^2)$$

$$= 143.77 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.29 \cdot 21802 / 2486) / (4.07 \cdot 208^2)$$

$$= 1.12 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.29 \cdot 63956 / 2486) / (4.07 \cdot 208^2)$$

$$= 3.28 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (19.3 \cdot 21802 / (208^2 \cdot 2486)) - 10 \cdot 1.12$$

$$= 27.24 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (19.3 \cdot 63956 / (208^2 \cdot 2486)) - 10 \cdot 3.28$$

$$= 79.92 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (49 + \max(1.12, 27.2)) / 2$$

$$= 38.13 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (144 + \max(3.28, 79.9)) / 2$$

$$= 111.84 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 303168 / 580$$

$$= 51.25 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (468392 / 580)$$

$$= 79.18 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	49.01	207.00	143.77	207.00
Radial Flange	1.12	138.00	3.28	138.00
Tangential Flange	27.24	138.00	79.92	138.00
Maximum Average	38.13	138.00	111.84	138.00
Bolting	51.25	172.00	79.18	172.00

Minimum Required Flange Thickness [Rigidity] 206.375 mm

Estimated M.A.W.P. (Operating) 1434 KPa.
 Estimated Finished Weight of Flange at given Thk. 1916.3 kg.
 Estimated Unfinished Weight of Forging at given Thk 2439.3 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):



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		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	278.48	374.99	0.743	278.48	374.99	0.743
Tangential Flange	Sto	164.45	250.00	0.658	164.45	250.00	0.658
Combined	Sh/f + St	489.64	749.99	0.653	489.64	749.99	0.653
Combined	Sr + St	289.91	749.99	0.387	289.91	749.99	0.387
Combined	Sh + Sto	664.44	749.99	0.886	664.44	749.99	0.886
Radial Flange	Sr	11.44	499.99	0.023	11.44	499.99	0.023
Maximum Stress Ratio				1.000			1.000

Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)
 = max(1; 3.28/1.38)
 = 2.3678

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
 = max(0.25; 1.01/1.28) if h/ho < 0.35, else max (0.25; 0.58/0.74)
 = 0.7788

In this case, the hub ratio h/ho, 0.4172 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	599.3 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	9.0645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	222.2 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.591 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2554^2 - 2510^2)$$

$$= 1750.0 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 208 \cdot 1750 / (64 \cdot 9.06)$$

$$= 627.4 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(627; 508)$$

$$= 507.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(508; 290)$$

$$= 507.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(508; 599)$$

$$= 507.5 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 508 \geq 140 (1750 / (64 \cdot 9.06))$$

$$:: 507.5 \geq 422.3 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 508 \geq (97 \cdot 1750 + \pi/4 \cdot 545 \cdot 2510^2) / (0.7 \cdot 9.06 \cdot 64)$$

$$:: 507.5 \geq 484.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 508 \leq 276 (1750 / (64 \cdot 9.06))$$

$$:: 507.5 \leq 832.6 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 508 \leq 599 (1/0.59)$$



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:: 507.5 ≤ 1014.1 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi_b$$

$$= 508 \cdot 0.2 \cdot 9.06 \cdot 38.1$$

$$= 357.4 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 107 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 63956 / 1 \cdot 9807 \cdot 0.22 / (3.86 \cdot 201334 \cdot 13^2 \cdot 180 \cdot 0.3)$
 $= 1.000$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 21802 / 1 \cdot 9807 \cdot 0.22 / (3.86 \cdot 194922 \cdot 13^2 \cdot 180 \cdot 0.3)$
 $= 0.352$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 63956 / 1 \cdot 9807 \cdot 0.22 / (4.07 \cdot 201334 \cdot 13^2 \cdot 180 \cdot 0.3)$
 $= 0.949$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 21802 / 1 \cdot 9807 \cdot 0.22 / (4.07 \cdot 194922 \cdot 13^2 \cdot 180 \cdot 0.3)$
 $= 0.334$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.39, Temperature Reduction per Fig. UCS 66.1 = 55 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -31 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	45.51	207.00	130.95	207.00
Radial Flange	1.33	138.00	3.84	138.00
Tangential Flange	24.25	138.00	69.79	138.00
Maximum Average	34.88	138.00	100.37	138.00
Bolting	51.25	172.00	79.18	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 192.913 mm

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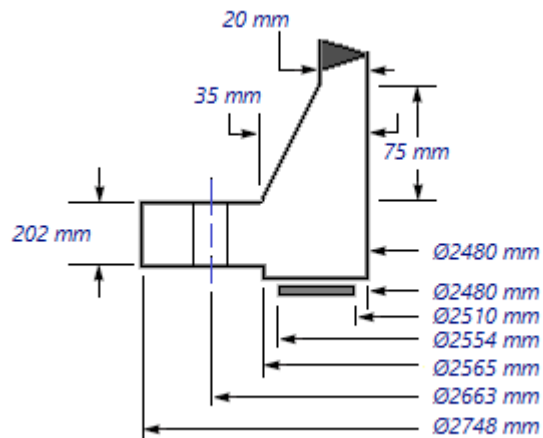
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Flange Input Data Values

Description: RHS Sh Fl 2 :

RHS Shell Flg-2

Flange Type	Integral Weld Neck		
Design Pressure	P	544.90	KPa.
Design Temperature		130	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2480.000	mm
Flange Outside Diameter	A	2748.000	mm
Flange Thickness	t	202.0000	mm
Thickness of Hub at Small End	go	20.0000	mm
Thickness of Hub at Large End	gl	35.0000	mm
Length of Hub	h	75.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2663.000	mm
Nominal Bolt Diameter	a	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		64	
Flange Face Outside Diameter	Fod	2565.000	mm
Flange Face Inside Diameter	Fid	2480.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2554.000	mm
Gasket Inside Diameter	Gi	2510.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (545 \cdot (2480/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 545) + Ca$$

$$= 7.9201 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 17) / (1243 + 0.6 \cdot 17)$$

$$= 1871.865 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 20) / (1240 + 0.6 \cdot 20)$$

$$= 2204.305 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	199.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2486.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	32.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	17.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	56.500 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2537.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2537^2 \cdot 545$$

$$= 280956.812 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2537 \cdot 3 \cdot 545$$

$$= 22210.883 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2486^2 \cdot 545$$

$$= 269713.844 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 280957 - 269714$$

$$= 11242.958 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o, 1 + W_o, 2 + H'p, 0)$$

$$= \max(280957 + 22211 + 0, 0)$$

$$= 303167.688 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2537 + 0 \cdot 0$$

$$= 468391.500 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + Fa + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((303168 + 0 + 4 \cdot 0 / 2537) / 172, 468392 / 172)$$

$$= 267.065 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 38.1 + 6 \cdot 199/(3 + 0.5)$$

$$= 417.343 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2663 \cdot \sin(\pi / 64)$$

$$= 130.667 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{131/(2 \cdot 38.1 + 199)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	267.065	580.128	
Radial Distance between Hub and Bolts:	50.800	56.500	
Radial Distance between Bolts and Edge:	38.100	42.500	
Circ. Spacing between the Bolts:	82.550	130.667	417.343

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 580 \cdot 172 / (68.9 \cdot \pi (2554 + 2510))$$

$$= 9.096 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 580$$

$$= 1017456.06 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 303168 - 280957$$

$$= 22210.89 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2663 - 2537) / 2$$

$$= 62.8575 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((2663 - 2486) / 2 + 62.9)$$

$$= 75.6788 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2663 - 2486 - 32) / 2$$

$$= 72.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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End Pressure, Md	269714.	72.5000	1.0000	19555.
Face Pressure, Mt	11243.	75.6788	1.0000	851.
Gasket Load, Mg	22211.	62.8575	1.0000	1396.
Gasket Seating, Matm	1017456.	62.8575	1.0000	63956.

Total Moment for Operation, Mo 21802. Kg-m.
 Total Moment for Gasket seating, Mg 63956. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 205.577 mm
 Hub Ratio, h/h0 = HL / H0 0.365
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.882

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.857
 Factor V per 4.16.5 0.288
 Factor f per 4.16.5 1.564

Stress factor from Table 4.16.4 [K]:

= A / B
 = 2748/2486
 = 1.105

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
 = (0.66845 + (5.7169(1.11² • log10(1.11))) / (1.11² - 1))/(1.11 - 1)
 = 19.341

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
 = (1.11²(1 + 8.55246 • log10(1.11)) - 1) / ((1.0472 + 1.9448 • 1.11²) • (1.11 - 1))
 = 1.875

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
 = (1.11²(1 + 8.55246 • log10(1.11)) - 1) / (1.36136(1.11² - 1) • (1.11 - 1))
 = 21.254

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
 = (1.11² + 1) / (1.11² - 1)
 = 10.014

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
 = 21.3 • 17² • 206 / 0.29
 = 4390807.500 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
 = 0.86/206
 = 0.004 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
 = (199 • 0.0042 + 1) / 1.88 + 199³/4390807
 = 2.770

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
 = (1.56•21802/2486)/(2.77•32²)
 = 47.42 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{1^2})$$

$$= (1.56 \cdot 63956 / 2486) / (2.77 \cdot 32^2)$$

$$= 139.11 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.11 \cdot 21802 / 2486) / (2.77 \cdot 199^2)$$

$$= 1.65 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.11 \cdot 63956 / 2486) / (2.77 \cdot 199^2)$$

$$= 4.84 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (19.3 \cdot 21802 / (199^2 \cdot 2486)) - 10 \cdot 1.65$$

$$= 25.47 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (19.3 \cdot 63956 / (199^2 \cdot 2486)) - 10 \cdot 4.84$$

$$= 74.72 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (47.4 + \max(1.65, 25.5)) / 2$$

$$= 36.45 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHA} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (139 + \max(4.84, 74.7)) / 2$$

$$= 106.92 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 303168 / 580$$

$$= 51.25 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (468392 / 580)$$

$$= 79.18 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	47.42	207.00	139.11	207.00
Radial Flange	1.65	138.00	4.84	138.00
Tangential Flange	25.47	138.00	74.72	138.00
Maximum Average	36.45	138.00	106.92	138.00
Bolting	51.25	172.00	79.18	172.00

Minimum Required Flange Thickness [Rigidity] 196.850 mm

Estimated M.A.W.P. (Operating) 1636 KPa.
 Estimated Finished Weight of Flange at given Thk. 1848.8 kg.
 Estimated Unfinished Weight of Forging at given Thk 2362.5 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):



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		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	267.95	374.99	0.715	267.95	374.99	0.715
Tangential Flange	Sto	160.58	250.00	0.642	160.58	250.00	0.642
Combined	Sh/f + St	586.85	749.99	0.782	586.85	749.99	0.782
Combined	Sr + St	285.32	749.99	0.380	285.32	749.99	0.380
Combined	Sh + Sto	660.57	749.99	0.881	660.57	749.99	0.881
Radial Flange	Sr	17.37	499.99	0.035	17.37	499.99	0.035
Maximum Stress Ratio				1.000			1.000

Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)
 = max(1; 2.09/1.33)
 = 1.5678

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)
 = max(0.25; 0.63/1.26) if $h/h0 < 0.35$, else max (0.25; 0.38/0.75)
 = 0.5035

In this case, the hub ratio $h/h0$, 0.3648 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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 Flg Calc [Int P]: RHS Sh Fl 2 Flng: 36 6:07pm Sep 8,2026

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	616.8 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	9.0645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	222.2 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.679 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2554^2 - 2510^2)$$

$$= 1750.0 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 208 \cdot 1750 / (64 \cdot 9.06)$$

$$= 627.4 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(627; 508)$$

$$= 507.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(508; 290)$$

$$= 507.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(508; 617)$$

$$= 507.5 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 508 \geq 140 (1750 / (64 \cdot 9.06))$$

$$:: 507.5 \geq 422.3 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 508 \geq (97 \cdot 1750 + \pi/4 \cdot 545 \cdot 2510^2) / (0.7 \cdot 9.06 \cdot 64)$$

$$:: 507.5 \geq 484.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 508 \leq 276 (1750 / (64 \cdot 9.06))$$

$$:: 507.5 \leq 832.6 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 508 \leq 617 (1/0.68)$$



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:: 507.5 ≤ 908.7 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= S_{bsel} \cdot K \cdot A_{bs} \cdot \phi_b$$

$$= 508 \cdot 0.2 \cdot 9.06 \cdot 38.1$$

$$= 357.4 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 107 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since $S_b \geq S_{bB8}$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 63956 / 1 \cdot 9807 \cdot 0.29 / (2.62 \cdot 201334 \cdot 17^2 \cdot 206 \cdot 0.3)$
 $= 0.999$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 21802 / 1 \cdot 9807 \cdot 0.29 / (2.62 \cdot 194922 \cdot 17^2 \cdot 206 \cdot 0.3)$
 $= 0.351$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 63956 / 1 \cdot 9807 \cdot 0.29 / (2.77 \cdot 201334 \cdot 17^2 \cdot 206 \cdot 0.3)$
 $= 0.946$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$
 $= 52.14 \cdot 21802 / 1 \cdot 9807 \cdot 0.29 / (2.77 \cdot 194922 \cdot 17^2 \cdot 206 \cdot 0.3)$
 $= 0.333$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.56, Temperature Reduction per Fig. UCS 66.1 = 25 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	44.15	207.00	127.04	207.00
Radial Flange	1.87	138.00	5.38	138.00
Tangential Flange	22.41	138.00	64.48	138.00
Maximum Average	33.28	138.00	95.76	138.00
Bolting	51.25	172.00	79.18	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 180.797 mm

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 Internal Pressure Calculations: Step: 9 6:07pm Sep 8,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Blind Flang		1027.9	144	3	2748	138	1
LHS CH flange-1		1024.9	220	3	2480	138	2
LHS Ch Shell		1024.1	20	3	2480	138	3
LHS Ch Flg-2		1024.9	220	3	2478	138	4
LHS Shell Flg-1		544.9	228	3	2480	138	5
Main Shell		544.32	28	3	2480	138	6
Main Shell		544.32	18	3	2480	138	7
Main Shell		544.14	28	3	2480	138	8
RHS Shell Flg-1		544.9	219	3	2480	138	9
RHS Shell Flg-2		544.9	210	3	2480	138	10
RHS		544.14	20	3	2480	138	11
RHS dish end		544.32	20	3	2480	138	12

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Blind Flang		1000	1088	1166	136	130.673	1
LHS CH flange-1		1000	1819	2057	212	207.95	2
LHS Ch Shell		1000	1848	2204	20	12.2667	3
LHS Ch Flg-2		1000	1806	2057	212	207.467	4
LHS Shell Flg-1		520	1409	1767	220	216.84	5
Main Shell		520	2718	3074	28	7.91482	6
Main Shell		520	1629	1986	18	7.91482	7
Main Shell		520	2718	3074	28	7.91322	8
RHS Shell Flg-1		520	1409	1767	211	206.375	9
RHS Shell Flg-2		520	1611	1829	202	196.85	10
RHS		520	1848	2204	20	7.91322	11
RHS dish end		520	1422	1778	16	7.88939	12

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 1057 KPa.
 Shell Side MAPnc = 1167 KPa.
 Channel Side MAWP = 1057 KPa.
 Channel Side MAPnc = 1166 KPa.

Elements Suitable for Design Internal Pressure.

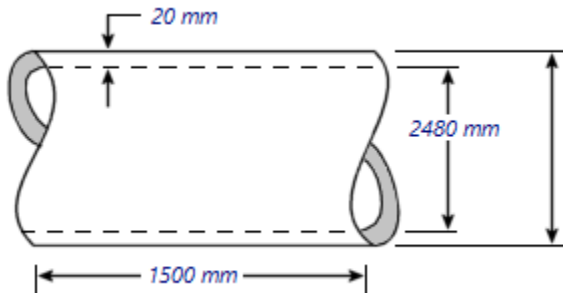
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 Element: LHS Ch Shell Step: 10 6:07pm Sep 8,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	1024.142 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	1500.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2480.000 mm	Actual Stress	75.5 N./mm ²
Spec. Min. Thk t,e, or sa	20.0000 mm	Spec. Nominal Thk.	20.000 mm
Surface Area	118752.2 cm ²	SG of Contents	1.000
Element Volume	7247067.0 cm ³	Weight of Contents	7233.8 Kg
Empty Weight	1826.2 Kg	Operating Weight	9059.9 Kg
		Comp. All. Ope. Stress	119.7 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1024 \cdot 1243) / (138 \cdot 1 - 0.6 \cdot 1024)$$

$$= 9.2667 + 3.0000 = 12.2667 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.142 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 17) / (1243 + 0.6 \cdot 17)$$

$$= 1872 - 24.1 = 1848 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 20) / (1240 + 0.6 \cdot 20)$$

$$= 2204 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1024 \cdot (1243 + 0.6 \cdot 17)) / (1 \cdot 17)$$

$$= 75.502 \text{ N./mm}^2$$



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 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 74 of 285
 Element: LHS Ch Shell Step: 10 6:07pm Sep 8,2026

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 0.800 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 20, tr = 9.56, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.56, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -33 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1056.657 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	20.000 mm	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.562	Temperature Difference	26 °C
UCS-66(c) red.	No	Computed Min. Temp.	-33 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: LHS Ch Shell

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.000	2520.00	1500.00	148.24	0.5952	0.0012866	92.67

$$MAEP = (4*B)/(3*(Do/t)) = (4*92.7)/(3*148) = 833 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.517	2520.00	1500.00	386.70	0.5952	0.0003000	30.00

$$MAEP = (4*B)/(3*(Do/t)) = (4*30)/(3*387) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.000	2520.00	14520.72	148.24	5.7622	0.0001151	11.50

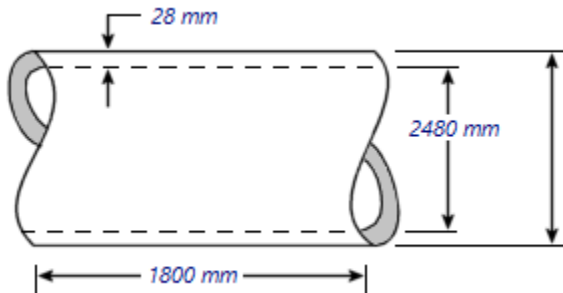
$$MAEP = (4*B)/(3*(Do/t)) = (4*11.5)/(3*148) = 103 \text{ KPa.}$$



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 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 75 of 285
 Element: Main Shell Step: 11 6:07pm Sep 8,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	544.319 KPa.	Int. Design Temperature	130 °C
Material	SA-516 70	Ext. Design Temperature	130 °C
Length	1800.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2480.000 mm	Actual Stress	27.4 N./mm ²
Spec. Min. Thk t,e, or sa	28.0000 mm	Spec. Nominal Thk.	28.000 mm
Surface Area	143407.4 cm ²	SG of Contents	1.000
Element Volume	5617681.0 cm ³	Weight of Contents	5614.3 Kg
Empty Weight	3077.8 Kg	Operating Weight	8692.0 Kg
		Comp. All. Ope. Stress	129.8 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (544 \cdot 1243) / (138 \cdot 1 - 0.6 \cdot 544) \\
 &= 4.9148 + 3.0000 = 7.9148 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.319 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 25) / (1243 + 0.6 \cdot 25) \\
 &= 2742 - 24.3 = 2718 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 28) / (1240 + 0.6 \cdot 28) \\
 &= 3074 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (544 \cdot (1243 + 0.6 \cdot 25)) / (1 \cdot 25) \\
 &= 27.392 \text{ N./mm}^2
 \end{aligned}$$



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 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 76 of 285
 Element: Main Shell Step: 11 6:07pm Sep 8,2026

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 1.116 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 28, tr = 9.56, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.38, Temp. Reduction = 55 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1056.657 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	28.000 mm	Basic MDMT	2 °C
Stress/Thk. Ratio	0.382	Temperature Difference	50 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 130 °C

Elastic Modulus from Chart: CS-2 at 130 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	2536.00	5600.00	101.44	2.2082	0.0005657	56.56

$$MAEP = (4*B)/(3*(Do/t)) = (4*56.6)/(3*101) = 743 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.291	2536.00	5600.00	224.61	2.2082	0.0001743	17.42

$$MAEP = (4*B)/(3*(Do/t)) = (4*17.4)/(3*225) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	2536.00	130606.08	101.44	50.0000	0.0001073	10.73

$$MAEP = (4*B)/(3*(Do/t)) = (4*10.7)/(3*101) = 141 \text{ KPa.}$$

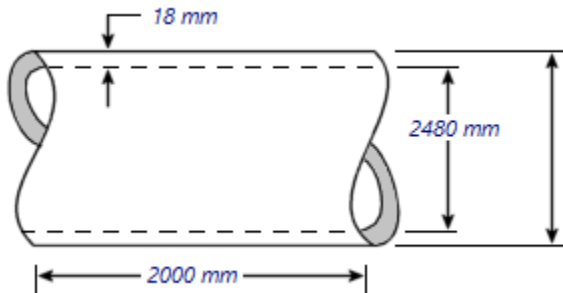
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 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 77 of 285
 Element: Main Shell Step: 12 6:07pm Sep 8,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	544.319 KPa.	Int. Design Temperature	130 °C
Material	SA-516 70	Ext. Design Temperature	130 °C
Length	2000.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2480.000 mm	Actual Stress	45.4 N./mm ²
Spec. Min. Thk t,e, or sa	18.0000 mm	Spec. Nominal Thk.	18.000 mm
Surface Area	158084.9 cm ²	SG of Contents	1.000
Element Volume	6241868.0 cm ³	Weight of Contents	6238.1 Kg
Empty Weight	2189.7 Kg	Operating Weight	8427.7 Kg
		Comp. All. Ope. Stress	115.9 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (544 \cdot 1243) / (138 \cdot 1 - 0.6 \cdot 544) \\
 &= 4.9148 + 3.0000 = 7.9148 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.319 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 15) / (1243 + 0.6 \cdot 15) \\
 &= 1653 - 24.3 = 1629 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 18) / (1240 + 0.6 \cdot 18) \\
 &= 1986 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (544 \cdot (1243 + 0.6 \cdot 15)) / (1 \cdot 15) \\
 &= 45.435 \text{ N./mm}^2
 \end{aligned}$$



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 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 78 of 285
 Element: Main Shell Step: 12 6:07pm Sep 8,2026

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 0.721 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 18, tr = 9.56, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.64, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1056.657 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.637	Temperature Difference	20 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 130 °C

Elastic Modulus from Chart: CS-2 at 130 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	2516.00	5600.00	167.73	2.2258	0.0002685	26.85

$$MAEP = (4*B)/(3*(Do/t)) = (4*26.8)/(3*168) = 213 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.239	2516.00	5600.00	223.86	2.2258	0.0001737	17.37

$$MAEP = (4*B)/(3*(Do/t)) = (4*17.4)/(3*224) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	2516.00	11109.44	167.73	4.4155	0.0001302	13.02

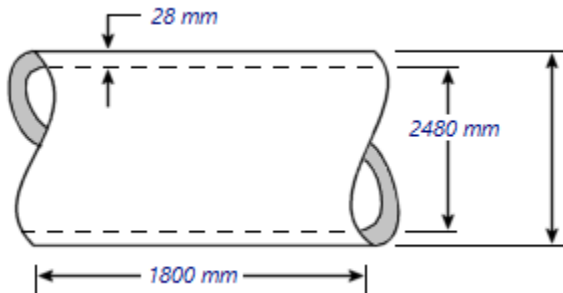
$$MAEP = (4*B)/(3*(Do/t)) = (4*13)/(3*168) = 103 \text{ KPa.}$$



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 Element: Main Shell Step: 13 6:07pm Sep 8,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	544.142 KPa.	Int. Design Temperature	130 °C
Material	SA-516 70	Ext. Design Temperature	130 °C
Length	1800.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2480.000 mm	Actual Stress	27.4 N./mm ²
Spec. Min. Thk t,e, or sa	28.0000 mm	Spec. Nominal Thk.	28.000 mm
Surface Area	143407.4 cm ²	SG of Contents	1.000
Element Volume	5634784.5 cm ³	Weight of Contents	5631.3 Kg
Empty Weight	3077.8 Kg	Operating Weight	8709.1 Kg
		Comp. All. Ope. Stress	129.8 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (544 \cdot 1243) / (138 \cdot 1 - 0.6 \cdot 544) \\
 &= 4.9132 + 3.0000 = 7.9132 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.142 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 25) / (1243 + 0.6 \cdot 25) \\
 &= 2742 - 24.1 = 2718 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 28) / (1240 + 0.6 \cdot 28) \\
 &= 3074 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (544 \cdot (1243 + 0.6 \cdot 25)) / (1 \cdot 25) \\
 &= 27.383 \text{ N./mm}^2
 \end{aligned}$$



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 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 80 of 285
 Element: Main Shell Step: 13 6:07pm Sep 8,2026

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 1.116 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 28, tr = 9.56, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.38, Temp. Reduction = 55 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1056.657 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	28.000 mm	Basic MDMT	2 °C
Stress/Thk. Ratio	0.382	Temperature Difference	50 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 130 °C

Elastic Modulus from Chart: CS-2 at 130 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	2536.00	5600.00	101.44	2.2082	0.0005657	56.56

$$MAEP = (4*B)/(3*(Do/t)) = (4*56.6)/(3*101) = 743 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.291	2536.00	5600.00	224.61	2.2082	0.0001743	17.42

$$MAEP = (4*B)/(3*(Do/t)) = (4*17.4)/(3*225) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	2536.00	130606.08	101.44	50.0000	0.0001073	10.73

$$MAEP = (4*B)/(3*(Do/t)) = (4*10.7)/(3*101) = 141 \text{ KPa.}$$

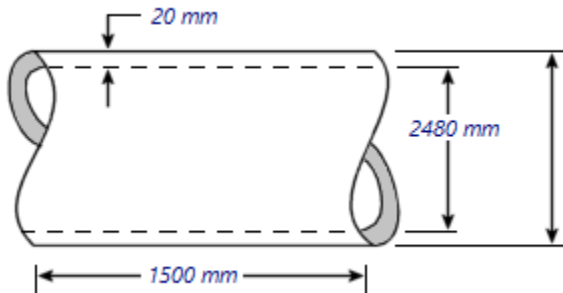
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 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 81 of 285
 Element: RHS Step: 14 6:07pm Sep 8,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: RHS

Cylindrical Shell Design Information:			
Design Pressure P	544.142 KPa.	Int. Design Temperature	130 °C
Material	SA-516 70	Ext. Design Temperature	130 °C
Length	1500.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2480.000 mm	Actual Stress	40.1 N./mm ²
Spec. Min. Thk t,e, or sa	20.0000 mm	Spec. Nominal Thk.	20.000 mm
Surface Area	118752.2 cm ²	SG of Contents	1.000
Element Volume	7247067.0 cm ³	Weight of Contents	7233.8 Kg
Empty Weight	1826.2 Kg	Operating Weight	9059.9 Kg
		Comp. All. Ope. Stress	119.7 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (544 \cdot 1243) / (138 \cdot 1 - 0.6 \cdot 544) \\
 &= 4.9132 + 3.0000 = 7.9132 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.142 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 17) / (1243 + 0.6 \cdot 17) \\
 &= 1872 - 24.1 = 1848 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 20) / (1240 + 0.6 \cdot 20) \\
 &= 2204 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (544 \cdot (1243 + 0.6 \cdot 17)) / (1 \cdot 17) \\
 &= 40.115 \text{ N./mm}^2
 \end{aligned}$$



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 Element: RHS Step: 14 6:07pm Sep 8,2026

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 0.800 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 20, tr = 9.56, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.56, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1056.657 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	20.000 mm	Basic MDMT	-24 °C
Stress/Thk. Ratio	0.562	Temperature Difference	24 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: RHS

Cylindrical Shell From 110 to 120 Ext. Chart: CS-2 at 130 °C

Elastic Modulus from Chart: CS-2 at 130 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.000	2520.00	1757.47	148.24	0.6974	0.0010867	87.81

MAEP = (4*B)/(3*(Do/t)) = (4*87.8)/(3*148) = 790 KPa.

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.935	2520.00	1757.47	363.36	0.6974	0.0002819	28.19

MAEP = (4*B)/(3*(Do/t)) = (4*28.2)/(3*363) = 103 KPa.

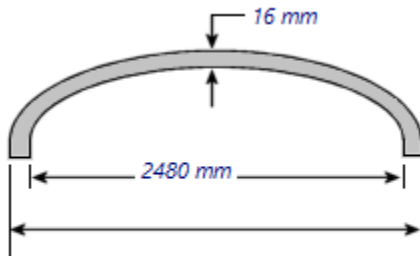
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.000	2520.00	14521.50	148.24	5.7625	0.0001151	11.50

MAEP = (4*B)/(3*(Do/t)) = (4*11.5)/(3*148) = 103 KPa.



Internal Pressure Calculation Results:

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Internal Pressure Results for: RHS dish end

Elliptical Head Design Information:			
Design Pressure P	544.319 KPa.	Int. Design Temperature	130 °C
Material	SA-516 70	Ext. Design Temperature	130 °C
Length	50.800 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2480.000 mm	Actual Stress	51.9 N./mm ²
Spec. Min. Thk t,e, or sa	16.0000 mm	Spec. Nominal Thk.	20.000 mm
Surface Area	72859.2 cm ²	SG of Contents	1.000
Element Volume	2242403.5 cm ³	Weight of Contents	2241.0 Kgf
Empty Weight	1221.4 Kgf	Operating Weight	3462.4 Kgf
		Comp. All. Ope. Stress	119.7 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.000 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.500 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (544 \cdot 2486 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 544)$$

$$= 4.8894 + 3.0000 = 7.8894 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 24.319 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 13) / (1 \cdot 2486 + 0.2 \cdot 13)$$

$$= 1446 - 24.3 = 1422 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 16) / (1 \cdot 2480 + 0.2 \cdot 16)$$

$$= 1778 \text{ KPa.}$$



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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (544 \cdot (1 \cdot 2486 + 0.2 \cdot 13)) / (2 \cdot 1 \cdot 13)$$

$$= 51.936 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (544 \cdot 1243) / (138 \cdot 1 - 0.6 \cdot 544) + 3$$

$$= 7.915 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 24.319 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 17) / (1243 + 0.6 \cdot 17)$$

$$= 1872 - 24.3 = 1848 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (2486 / (2 \cdot 623))^2) / 6$$

$$= 0.996794$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 3.475 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 16, tr = 9.5, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.73$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-29 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 20, tr = 9.56, c = 3 mm, E* = 1

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.56$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-33 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1056.657 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	20.000 mm	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.562	Temperature Difference	26 °C
UCS-66(c) red.	No	Computed Min. Temp.	-33 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 31.000 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 15.500 mm

EXTERNAL PRESSURE RESULTS for: RHS dish end

Elliptical Head From 120 to 130 Ext. Chart: CS-2 at 130 °C

Elastic Modulus from Chart: CS-2 at 130 °C: 199943392 KPa.



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Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
13.000	2512.00	193.23	0.0007188	71.86

$$\text{MAEP} = B / (K0 * \text{Do} / t) = 71.9 / (0.9 * 193) = 413 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
6.504	2512.00	386.22	0.0003596	35.95

$$\text{MAEP} = B / (K0 * \text{Do} / t) = 36 / (0.9 * 386) = 103 \text{ KPa.}$$

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 Shop Pressure Test Results: Step: 16 6:07pm Sep 8,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1373.654 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	676.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1489.714 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1162.323 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	743.600 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	743.600 KPa.
App 27-4	= M.A.W.P.	1056.657 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1373.654 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	676.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1489.379 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1162.323 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	743.600 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	743.600 KPa.
App 27-4	= M.A.W.P.	1056.657 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	111.6	234.0	0.477	1513.69
Main Shell	76.2	234.0	0.326	1514.02
Main Shell	126.4	234.0	0.540	1514.02
Main Shell	76.2	234.0	0.326	1514.02
RHS	111.6	234.0	0.477	1513.69
RHS dish end	144.4	234.0	0.617	1513.69

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3-Tube Inlet	Nozzle	138.00	138.00	1.000
N3-Tube Inlet	Pad	138.00	138.00	1.000
N1-Shell Inlet	Nozzle	118.00	118.00	1.000
N1-Shell Inlet	Pad	138.00	138.00	1.000
Drain	Nozzle	138.00	138.00	1.000
Vent	Nozzle	138.00	138.00	1.000
N2-Shell Outlet	Nozzle	118.00	118.00	1.000
N2-Shell Outlet	Pad	138.00	138.00	1.000
N4-Tube Outlet	Nozzle	138.00	138.00	1.000
N4-Tube Outlet	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind Flang	138.00	138.00	1.000
LHS CH flange-1	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000



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LHS Ch Flg-2	138.00	138.00	1.000
LHS Shell Flg-1	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
RHS Shell Flg-1	138.00	138.00	1.000
RHS Shell Flg-2	138.00	138.00	1.000
RHS	138.00	138.00	1.000
RHS dish end	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Floating Head Matl.	138.00	138.00	1.000
Backing Ring Matl.	137.90	137.90	1.000
Flange Material	138.00	138.00	1.000
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3-Tube Inlet	31.35	234.00	0.134
N1-Shell Inlet	21.67	216.00	0.100
Drain	7.85	179.40	0.044
Vent	7.85	179.40	0.044
N2-Shell Outlet	21.67	216.00	0.100
N4-Tube Outlet	31.35	234.00	0.134

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1373.654 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	676.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1489.714 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1162.323 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	743.600 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	743.600 KPa.
App 27-4	= M.A.W.P.	1056.657 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1373.654 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	676.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1489.379 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1162.323 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	743.600 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	743.600 KPa.
App 27-4	= M.A.W.P.	1056.657 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

$$= \text{Test Factor} \cdot \text{Design Pressure} \cdot \text{Stress Ratio}$$

$$= 1.3 \cdot 520 \cdot 1$$

$$= 676.000 \text{ KPa.}$$

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

$$= \text{Test Factor} \cdot \text{Design Pressure} \cdot \text{Stress Ratio}$$

$$= 1.3 \cdot 520 \cdot 1$$

$$= 676.000 \text{ KPa.}$$

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	51.6	234.0	0.221	700.31
Main Shell	35.2	234.0	0.151	700.31
Main Shell	58.5	234.0	0.250	700.31
Main Shell	35.2	234.0	0.151	700.31
RHS	51.6	234.0	0.221	700.31
RHS dish end	66.8	234.0	0.286	700.31

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3-Tube Inlet	Nozzle	138.00	138.00	1.000
N3-Tube Inlet	Pad	138.00	138.00	1.000
N1-Shell Inlet	Nozzle	118.00	118.00	1.000
N1-Shell Inlet	Pad	138.00	138.00	1.000
Drain	Nozzle	138.00	138.00	1.000
Vent	Nozzle	138.00	138.00	1.000
N2-Shell Outlet	Nozzle	118.00	118.00	1.000
N2-Shell Outlet	Pad	138.00	138.00	1.000
N4-Tube Outlet	Nozzle	138.00	138.00	1.000
N4-Tube Outlet	Pad	138.00	138.00	1.000



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Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind Flang	138.00	138.00	1.000
LHS CH flange-1	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000
LHS Ch Flg-2	138.00	138.00	1.000
LHS Shell Flg-1	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
RHS Shell Flg-1	138.00	138.00	1.000
RHS Shell Flg-2	138.00	138.00	1.000
RHS	138.00	138.00	1.000
RHS dish end	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Floating Head Matl.	138.00	138.00	1.000
Backing Ring Matl.	137.90	137.90	1.000
Flange Material	138.00	138.00	1.000
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3-Tube Inlet	14.50	234.00	0.062
N1-Shell Inlet	10.02	216.00	0.046
Drain	3.63	179.40	0.020
Vent	3.63	179.40	0.020
N2-Shell Outlet	10.02	216.00	0.046
N4-Tube Outlet	14.50	234.00	0.062

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description: Stationary Tube

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		676.00 KPa.
Shell Thickness	ts	28.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2480.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		676.00 KPa.
Channel Thickness	tc	20.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2480.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	6000
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	6000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5770.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	1130.510 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	32948.0 cm ²
Perimeter of Tube Layout	Cp	7006.7 mm



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Tubesheet Data:

Tubesheet TYPE: Stationary, Gasketed both Sides, Conf. d
 Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	115.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2562.000 mm
Dimension G for the Channel Side	Gc	2537.285 mm
Area of the Untubed Lanes	AL	0.0 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	635.910 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	2748.000 mm
Flange Inside Diameter	B	2480.000 mm
Flange Face Outside Diameter	Fod	2565.000 mm
Flange Face Inside Diameter	Fid	2480.000 mm
Gasket Outside Diameter	Go	2554.000 mm
Gasket Inside Diameter	Gi	2510.000 mm
Small end Hub thk.	g0	16.0000 mm
Large end Hub thk.	g1	35.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch	Code Sketch 1a	
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option	Program Selects	
Length of Partition Gasket	lp	5118.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	0.0000
Partition Gasket Design Seating Stress	yPart	0.00 N./mm ²
Partition Gasket Facing Sketch	Code Sketch 1a	
Partition Gasket Column for Gasket Seating	Code Column II	

Bolting Information:

Diameter of Bolt Circle	C	2663.000 mm
Nominal Bolt Diameter	dB	38.1000 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	72
Bolt Material		SA-193 B7
Bolt Material UNS Number		G41400
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Alternate Flange Operating Bolt Load, Wm1	570228.06 KgF
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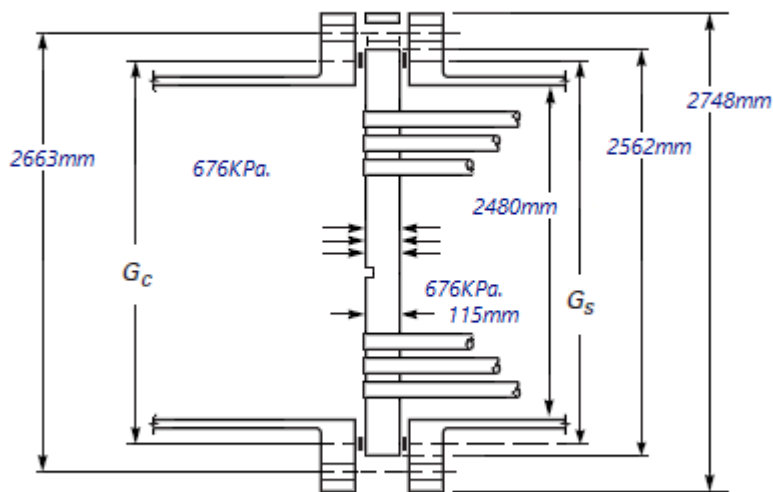
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Alternate Flange Seating Bolt Load, Wm2 468391.50 Kgf
 Alternate Flange Design Bolt Load, W 1144638.00 Kgf

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with None
 Tubesheet Extended as Flange No
 Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.



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Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature 21.1 °C 0.20132E+09 KPa.
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20132E+09 KPa.

Note:
 The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (676 \cdot (19/2 - 0)) / (92.4 \cdot 1 + 0.4 \cdot 676)$
 $= 0.0695 + 0.0000 = 0.0695 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	5770.00	9.03	50.0000	0.0134948	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 15019 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3171	19.05	5770.00	60.08	50.0000	0.0003047	30.46

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 30.5) / (3 \cdot 60.1) = 676 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3171 mm
Allowable Internal Pressure at Corroded thickness	22456.99 KPa.
Required Internal Design Pressure	676.00 KPa.
Allowable External Pressure at Corroded thickness	15019.20 KPa.
Required External Design Pressure	676.00 KPa.
Required Thickness due to Shell Side pressure	0.3171 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

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Gasket Contact Width,	$N = (Goc - Gic) / 2$	22.000 mm
Basic Gasket Width,	$b0 = N / 2$	11.000 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	8.355 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	2537.291 mm

Results for ASME Stationary Tubesheet Calculations for Configuration d:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio $[\rho]$:
 $= ltx / h$ (modified for corrosion if present)



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$$= 0/115$$

$$= 0.0000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (p); dt - 2 \cdot tt)$$

$$= \max(19 - 2 \cdot 2.11 (202701232/201322320) \cdot (92.4/138) \cdot (0); 19 - 2 \cdot 2.11)$$

$$= 19.0500 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 1131 + 19$$

$$= 2280.070 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (25.4 - 19) / 25.4$$

$$= 0.250$$

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 2280/2$$

$$= 1140.0350 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$$

$$= 25.4 / \sqrt{ 1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 2280 \cdot 25.4) / (\pi \cdot 2280^2) }$$

$$= 25.4000 \text{ mm}$$

Effective ligament efficiency for bending [μ*]:

$$= (p* - d*) / p*$$

$$= (25.4 - 19) / 25.4$$

$$= 0.2500$$

Ratio [ps]:

$$= as / ao$$

$$= 1269/1140$$

$$= 1.1128$$

Ratio [pc]:

$$= ac / ao$$

$$= 1269/1140$$

$$= 1.1128$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 6000 ((19 - 2 \cdot 2.11) / (2 \cdot 1140))^2$$

$$= 0.7462$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 6000 (19 / (2 \cdot 1140))^2$$

$$= 0.5812$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:
 β_s , k_s , λ_s , δ_s are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:



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β_C , k_c , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$h/p = 4.527559$; $\mu^* = 0.250000$; $E^* = 43608536$ KPa.
 $E^*/E = 0.216611$; $\nu^* = 0.396457$

Note: As h/p (4.528) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor $[X_a]$:
 $= ((24(1 - \nu^*)Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$
 $= ((24(1 - 0.4^2)6000 \cdot 202701232 \cdot 2.11$
 $(19 - 2.11)1140^2) / (43608536 \cdot 5770 \cdot 115^3))^{1/4}$
 $= 7.3924$

Values from Table 4.18.3

$Z_d = 0.003673$; $Z_v = 0.018388$; $Z_m = 0.193865$
 $Z_a = 0.485565E+03$; $Z_w = 0.018388$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio $[K]$:

$= A / Do$
 $= 2562/2280$
 $= 1.1236$

Compute coefficient $[F]$:

$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$
 $= (1 - 0.4) / 43608536 \cdot (0 + 0 +$
 $201322320 \cdot \ln(1.12))$
 $= 0.3248$

Compute parameter $[\phi]$:

$= (1 + \nu^*) F$
 $= (1 + 0.4) 0.32$
 $= 0.4536$

Compute parameter $[Q_1]$:

$= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$
 $= (1.11 - 1 - 0.45 \cdot 0.018) / (1 + 0.45 \cdot 0.19)$
 $= 0.096025$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor $[\gamma_b]$:

$= (G_c - G_s) / Do$ (config d)
 $= (2537 - 2537) / 2280 = 0$

Calculate parameter $[\omega_s]$:

$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$
 $= 1.11 \cdot 0 \cdot 0 \cdot 0.000 (1 + 115 \cdot 0)$
 $= 0.0000 \text{ mm}^2$

Calculate parameter $[\omega^*s]$:

$= ao^2(p_s^2 - 1) \cdot (p_s - 1) / 4 - \omega_s$
 $= 1140^2(1.11^2 - 1) \cdot (1.11 - 1) / 4 - 0$
 $= 8736.4102 \text{ mm}^2$

Parameter $[\omega_c]$:

$= p_c \cdot k_c [\delta_C (Cm_{\theta} \cdot \beta_C + Cq_{\Delta} \cdot h \cdot \beta_C^2) -$
 $\zeta_c (Cm_{\theta} + h \cdot Cq_{\theta} \cdot h \cdot \beta_C / 2))$
 $= 1.11 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 115) -$
 $0.000 (0/115 + 0 \cdot 0/2))$
 $= 0.0000 \text{ mm}^2$



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Parameter [$\omega \cdot c$]:

$$= a_o^2 [(p_c^2 + 1) \cdot (p_c - 1)/4 - (p_s - 1)/2] - \omega c$$

$$= 1140^2 [(1.11^2 + 1) \cdot (1.11 - 1)/4 - (1.11 - 1)/2] - 0$$

$$= 8736.4131 \text{ mm}^2$$

Pressure [$P \cdot S$]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [$P \cdot C$]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$P_e = P_s - P_t$$

$$= 676 - 676 = 0 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q_2]:

$$= [((\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) - (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)) +$$

$$W \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((8736 \cdot 676 - 8736 \cdot 676) -$$

$$(0 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 +$$

$$1144638 \cdot 0 / (2 \cdot \pi)] / (1 + 0.45 \cdot 0.19)$$

$$= -0.0002 \text{ Kg}$$

Tubesheet Bending Stress - original thickness [σ]:

$$= 6 \cdot Q_2 / (\mu \cdot (H - h'g)^2)$$

$$= 6 \cdot -0.0002 / (0.25 \cdot (115 - 0)^2)$$

$$= -0.0000 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot S_y = 1.5 \cdot 0.95 \cdot 250 = 356 \text{ N./mm}^2, 4.18.233$$

Could not converge on a min. required tubesheet thk. to satisfy the Bending Stress criteria.



Use the best Estimated Required Tubesheet Thickness value as design guidance & Compare the Actual Stresses versus Allowable Stresses.

$$\text{Tubesheet thickness} = 6.5134 \text{ mm}$$

$$\text{Tubesheet Bending Stress} = 0.0012 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.9.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot S_y)$$

$$= (2 \cdot 0.25 \cdot 115 / 1140) \cdot 190$$

$$= 9.5829 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ_{limit}

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p)$$

$$= (32948 \cdot 0) / (0.25 \cdot 6.51 \cdot 7007)$$

$$= 0.0000 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [H_{reqS}]:

$$= H + C_{ats} + C_{atc}$$

$$= 6.51 + 0 + 0$$



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= 6.5134 mm

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

= max(HreqB; HreqS)
 = max(6.51 ; 6.51)
 = 6.5134 mm

ASME Section VIII-2 Part 4.18.9.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

Ftmin = -7.66, Ftmax = 27.5

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

= ((Ps • xs - Pt • xt) - 2 • Q2 • Ftmin / ao²) / (Xt - Xs)
 = ((676 • 0.58 - 676 • 0.75) - 2 • -0.00045 • -7.66 /
 1140²) / (0.75 - 0.58)
 = -0.6760 N./mm²

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

= ((Ps • xs - Pt • xt) - 2 • Q2 • Ftmax / ao²) / (Xt - Xs)
 = ((676 • 0.58 - 676 • 0.75) - 2 • -0.00045 • 27.5 /
 1140²) / (0.75 - 0.58)
 = -0.6760 N./mm²

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

= max(| $\sigma_{t,1}$ | ; | $\sigma_{t,2}$ |) = 0.68 N./mm²

Allowable Tube Stress [$\sigma_{t,A}$]:

= $\beta_T \cdot S_{y,t} = 170.9969$ N./mm²

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

= 1.2500 (as Pe is 0.0)

Determine the factor [rt]:

= ($\sqrt{dt^2 + (dt - 2 \cdot tt)^2}$) / 4
 = ($\sqrt{19^2 + (19 - 2 \cdot 2.11)^2}$) / 4
 = 6.0355 mm

Determine the factor [Ct]:

= $\sqrt{(2 \cdot \pi^2 \cdot Et / Syt)}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 202701232 / 180)}$
 = 155.4270

Determine the factor [Ft]:

= k • L/r
 = 1 • 636 / 6.04
 = 105.3620

Buckling Allowable Stress [Stb]:

= (Sy,t / Fs) • (1 - Ft / (2 • Ct))
 = (180 / 1.25) • (1 - 105 / (2 • 155))
 = 170.997 N./mm² (Never greater than Sot)

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

= $\sigma_{t,max} \cdot \text{Tube Area}$
 = 0.68 • 1.12
 = 7.74 Kg

Tube Metal Area [At]:

= $\pi (do - t) t$
 = $\pi (19 - 2.11) 2.11$



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= 1.123 cm²

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (19 - 2.11) 171$$

$$= 1957.967 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (19 + 0.67 \cdot 2.11) 171$$

$$= 1957.9670 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 171/171$$

$$= 1.000$$

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 (19 + 0.67 \cdot 2.11) 171$$

$$= 1300.8848 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 2.11 \cdot (19 - 2.11) 1 \cdot 1)} - 0.75 \cdot 19 \right)$$

$$= 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 1958 is >= tube strength 1958 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Computations completed for ASME tubesheet configuration: d

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	0.0 <=	356.2 N./mm ²	Pass
Tubesheet Shear Stress	0.0 <=	190.0 N./mm ²	Pass
Maximum Stress in the Tubes	0.7 <=	171.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-0.7 <=	-171.0 N./mm ²	Pass
Maximum Force on any one Tube	7.7 <=	1958.0 Kgf	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	6.513	115.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

Stationary Tubesheet results per ASME Div. 2 Part 4.18.9 2025



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Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet		Stresses		Case Type	Pass/Fail	
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd			
D1uc	63.827	3.843	120	356	11	190	Ps+Pt-Th	D1	Ok
D2uc	63.827	3.843	120	356	11	190	Ps+Pt-Th	D2	Ok
D3uc	6.513	...	...	356	...	190	Ps+Pt-Th	D3	Ok
Max:	63.8266	3.843	mm	0.337		0.058	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Tube, Shell and Channel Stress Summary:

Case#	Ten	---- Tube Stresses		Tube Loads		Shell Stress		Channel Stress		Pass
		Allwd	Cmp	Allwd	Ld	Allwd	Allwd	Stress	Allwd	
D1uc	32	170	-8	-170	362	1957	...	...	...	Ok
D2uc	32	170	-32	-170	370	1957	...	...	...	Ok
D3uc	1	170	-1	-170	8	1957	...	...	...	Ok
Max	RATIO	0.189		0.189		0.189	...		...	

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	63.827	115.000	Ok
Tube thickness :	0.317	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Note: This is a full strength Tube to Tubesheet Joint.

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Input Echo, Tubesheet Number 1, Description: Floatating tube

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		676.00 KPa.
Shell Thickness	ts	28.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2480.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		676.00 KPa.
Channel Thickness	tc	20.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2480.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	6000
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	6000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5770.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	1130.510 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	32948.0 cm ²
Perimeter of Tube Layout	Cp	7006.7 mm



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Tubesheet Data:

Tubesheet TYPE: Floating, Tubesheet Gasketed, Conf. B
 Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	115.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2465.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? No

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 635.910 mm
 Tube End condition corresponding to Span (l) k 1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Junction Stress Reduction option Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	2465.000 mm
Flange Inside Diameter	B	2297.000 mm
Flange Face Outside Diameter	Fod	2326.000 mm
Flange Face Inside Diameter	Fid	2297.000 mm
Gasket Outside Diameter	Go	2323.000 mm
Gasket Inside Diameter	Gi	2297.000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch	Code	Sketch 1a
Gasket Thickness	tg	6.3500 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	2368.000 mm
Nominal Bolt Diameter	dB	38.1000 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	72
Bolt Material		SA-193 B7M
Bolt Material UNS Number		G41400
Bolt Allowable Stress At Temperature	Sb	138.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	138.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load Sa*Ab, was used for all load cases.

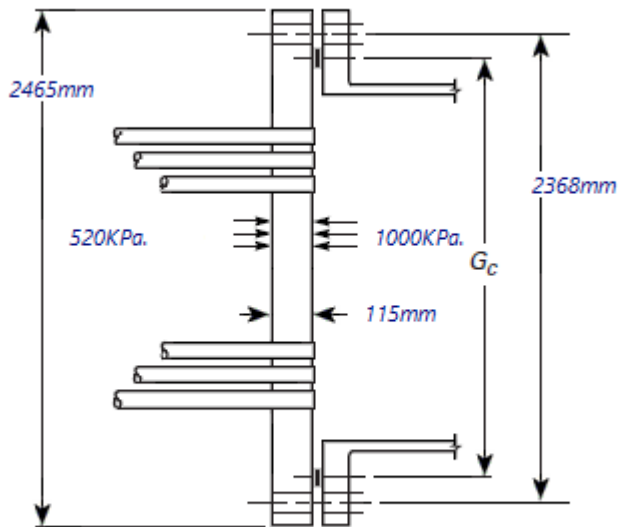
Tubesheet Extended as Flange Yes
 Thickness of Extended Portion of Tubesheet Tf 99.0000 mm
 Is Bolt Load Transferred to the Tubesheet Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



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Tubesheet Gasketed, Extended as a Flange



Configuration B:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20132E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.



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Note:
 The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (676 \cdot (19/2 - 0)) / (92.4 \cdot 1 + 0.4 \cdot 676)$
 $= 0.0695 + 0.0000 = 0.0695 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	5770.00	9.03	50.0000	0.0134948	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 15019 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3171	19.05	5770.00	60.08	50.0000	0.0003047	30.46

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 30.5) / (3 \cdot 60.1) = 676 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3171 mm
Allowable Internal Pressure at Corroded thickness	22456.99 KPa.
Required Internal Design Pressure	676.00 KPa.
Allowable External Pressure at Corroded thickness	15019.20 KPa.
Required External Design Pressure	676.00 KPa.
Required Thickness due to Shell Side pressure	0.3171 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

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Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	2310.156 mm

Results for ASME Floating Tubesheet Calculations for Configuration B:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio $[\rho]$:
 $= ltx / h$ (modified for corrosion if present)
 $= 0/115$
 $= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d*]$:
 $= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$
 $= \max(19 - 2 \cdot 2.11 (202701232/201322320) \cdot (92.4/138) \cdot (0); 19 - 2 \cdot 2.11)$
 $= 19.0500 \text{ mm}$



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Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 1131 + 19$$

$$= 2280.070 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (25.4 - 19) / 25.4$$

$$= 0.250$$

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 2280 / 2$$

$$= 1140.0350 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$$

$$= 25.4 / \sqrt{1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 2280 \cdot 25.4) / (\pi \cdot 2280^2) }$$

$$= 25.4000 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (25.4 - 19) / 25.4$$

$$= 0.2500$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 1155 / 1140$$

$$= 1.0132$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 1155 / 1140$$

$$= 1.0132$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 6000 ((19 - 2 \cdot 2.11) / (2 \cdot 1140))^2$$

$$= 0.7462$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot a_o))^2$$

$$= 1 - 6000 (19 / (2 \cdot 1140))^2$$

$$= 0.5812$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$h/p = 4.527559$; $\mu^* = 0.250000$; $E^* = 43608536 \text{ KPa}$.

$E^*/E = 0.216611$; $\nu^* = 0.396457$

Note: As h/p (4.528) is > 2 , data values for $h/p = 2$ were used.



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$$\begin{aligned} \text{Tube Bundle Stiffness Factor [Xa]:} \\ &= ((24(1 - \nu^2)Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E \cdot L \cdot H^3))^{1/4} \\ &= ((24(1 - 0.4^2)6000 \cdot 202701232 \cdot 2.11 \\ &\quad (19 - 2.11)1140^2) / (43608536 \cdot 5770 \cdot 115^3))^{1/4} \\ &= 7.3924 \end{aligned}$$

Values from Table 4.18.3

$$\begin{aligned} Z_d &= 0.003673 ; Z_v = 0.018388 ; Z_m = 0.193865 \\ Z_a &= 0.485565E+03 ; Z_w = 0.018388 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned} &= A / Do \\ &= 2465/2280 \\ &= 1.0811 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned} &= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.4) / 43608536 \cdot (0 + 0 + \\ &\quad 201322320 \cdot \ln(1.08)) \\ &= 0.2173 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned} &= (1 + \nu) F \\ &= (1 + 0.4) 0.22 \\ &= 0.3034 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned} &= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ &= (1.01 - 1 - 0.3 \cdot 0.018) / (1 + 0.3 \cdot 0.19) \\ &= 0.007192 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor [γ_b]:

$$\begin{aligned} &= (G_c - C) / Do \text{ (config b)} \\ &= (2310 - 2368) / 2280 = -0.025 \end{aligned}$$

Calculate parameter [ω_s]:

$$\begin{aligned} &= p_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s) \\ &= 1.01 \cdot 0 \cdot 0 \cdot 0.000 (1 + 115 \cdot 0) \\ &= 0.0000 \text{ mm}^2 \end{aligned}$$

Calculate parameter [ω_s^*]:

$$\begin{aligned} &= ao^2(p_s^2 - 1) \cdot (p_s - 1) / 4 - \omega_s \\ &= 1140^2(1.01^2 - 1) \cdot (1.01 - 1) / 4 - 0 \\ &= 113.8897 \text{ mm}^2 \end{aligned}$$

Parameter [ω_c]:

$$\begin{aligned} &= p_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \\ &\quad \zeta c (C_m \Theta + h \cdot C_q \Theta + h \cdot \beta C / 2)) \\ &= 1.01 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 115) - \\ &\quad 0.000 (0/115 + 0 \cdot 0/2)) \\ &= 0.0000 \text{ mm}^2 \end{aligned}$$

Parameter [ω_c^*]:

$$\begin{aligned} &= ao^2[(p_c^2 + 1) \cdot (p_c - 1) / 4 - (p_s - 1) / 2] - \omega_c \\ &= 1140^2[(1.01^2 + 1) \cdot (1.01 - 1) / 4 - (1.01 - 1) / 2] - 0 \\ &= 113.8891 \text{ mm}^2 \end{aligned}$$

Pressure [P^*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P^*C]:



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= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$P_e = P_s - P_t$$

$$= 676 - 676 = 0 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$= [((\omega * s * P_s - \omega * c * P_t) - (\omega * s * P * s - \omega * c * P * c)) +$$

$$W * \gamma_b / (2 * \pi)] / (1 + \phi * Z_m)$$

$$= [((114 * 676 - 114 * 676) -$$

$$(0 * 0 - 0 * 0)) * 0.0001 +$$

$$918372 * -0.025 / (2 * \pi)] / (1 + 0.3 * 0.19)$$

$$= -3502.0864 \text{ Kg}$$

Tubesheet Bending Stress - original thickness [σ]:

$$= 6 * Q2 / (\mu * (H - h'g)^2)$$

$$= 6 * -3502 / (0.25 * (115 - 0)^2)$$

$$= -62.3263 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 * \beta T * S_y = 1.5 * 0.95 * 250 = 356 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= 6 * Q2 / (\mu * (H - h'g)^2)$$

$$= 6 * -3598 / (0.25 * (48.8 - 0)^2)$$

$$= 356.2436 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 48.8 + 0 + 0$$

$$= 48.7526 \text{ mm}$$

$$H \text{ reqd.} = H_{ReqB} = 48.7526 \text{ mm}$$

Computations completed for ASME tubesheet configuration: B

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	62.3 <=	356.2 N./mm ²	Pass

Floating Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail	
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd			
D1uc	57.592	42.248	105	356	...	...	Ps+Pt-Th	D1	Ok
D2uc	31.361	...	56	356	...	...	Ps+Pt-Th	D2	Ok
D3uc	48.753	...	62	356	...	...	Ps+Pt-Th	D3	Ok
Max:	57.5921	42.248 mm		0.293		...	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the



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Load Case Tables

[(+)-Th]:
With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
With or Without Corrosion Allowance

[D1, D2, D3]:
Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
Operating Load Cases using the Maximum and Minimum Operating Pressures and
Operating Temperatures

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	57.592	115.000	Ok
Tubesheet thickness flanged extension :	42.248	99.000	Ok
Tube thickness :	0.317	2.110	Ok
Tube-tubesheet fillet weld leg :	0.000	2.110	Ok
Tube-tubesheet groove weld leg :	0.000	2.110	Ok

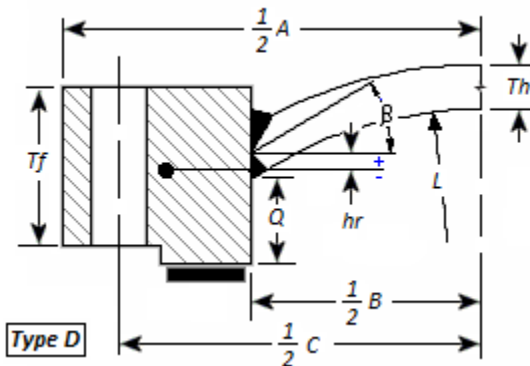
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 Floating Head: Operating Case: 1 6:07p Sep 8,2026

Input Echo, Floating Head Item 1, Description: Operating

Floating Head Type		D	
Tube Side (Internal) Design Pressure	Pts	1000.00	KPa.
Shell Side (External) Design Pressure	Pss	520.00	KPa.
Design Temperature for Spherical Head	Temp	130.00	°C
Head Material		SA-516 70	
Head Material UNS Number		K02700	
Head Allowable Stress at Temperature	Sho	138.00	N./mm ²
Head Allowable Stress at Ambient	Sha	138.00	N./mm ²
Crown Radius for Spherical Head	L	2067.000	mm
Head Thickness	Th	30.000	mm
Tube Side (Internal) Corrosion Allowance	Cats	0.0000	mm
Shell Side (External) Corrosion Allowance	Cass	0.0000	mm
Flange Material		SA-266	
Flange Allowable Stress at Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress at Ambient	Sfa	138.00	N./mm ²
Flange Outside Diameter	Fod	2465.000	mm
Flange Inside Diameter	Fid	2297.000	mm
Flange Thickness	Tf	170.0000	mm
Bolt Material		SA-193 B7M	
Bolt Material UNS Number		G41400	
Bolt Allowable Stress At Temperature	Sb	138.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	138.00	N./mm ²
Diameter of Bolt Circle	C	2368.000	mm
Nominal Bolt Diameter	dB	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		72	
Full face Gasket Flange Option	Program Selects		
Flange Face Outside Diameter	Ffod	2326.000	mm
Flange Face Inside Diameter	Ffid	2297.000	mm
Gasket Outside Diameter	Go	2323.000	mm
Gasket Inside Diameter	Gi	2297.000	mm
Gasket Factor,	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Flange Facing Sketch	Code Sketch 1a		
Column for Gasket Seating	Code Column I		
Gasket Thickness		6.3500	mm
Width of raised face		0.0000	mm
Dist. from Head Centerline to Flange Centroid hr		15.0566	mm
The Flange is not Slotted.			
This unit does not have a Backing Ring.			
Use Full Bolt Load in Flange Design		Yes	



Internal Pressure results for Spherical Heads:

ASME Code, Section VIII, Division 1, 2025, App. 1-6

Thickness Due to Internal Pressure[thr]:
 $= 5PL / 6Sho$
 $= (5 \cdot 1000 \cdot 2067) / (6 \cdot 138)$
 $= 12.4828 \text{ mm}$

Maximum Allowable Working Pressure at Given Thickness[Pa]:
 $= 6Sho \cdot (Th - Cass - Cats) / 5L$
 $= (6 \cdot 138 \cdot 30) / (5 \cdot 2067)$
 $= 2403.30 \text{ KPa.}$

Maximum Allowable Pressure, New and Cold[Pnc]:
 $= 6Sha / 5L$
 $= (6 \cdot 138 \cdot 30) / (5 \cdot 2067)$
 $= 2403.30 \text{ KPa.}$

Actual Stress at given Pressure and Thickness[Sact]:
 $= 5PL / 6(Th - Cass - Cats)$
 $= (5 \cdot 1000 \cdot 2067) / (6 \cdot 30)$
 $= 57. \text{ N./mm}^2$

Spherical head Percent Elongation $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$: 1.081 %

External Pressure results, Spherical Head ASME Code, Section VIII Division 1, 2025

External Pressure Chart CS-2 at 130.00 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Maximum Allowable External Pressure:

Corroded Thickness of Shell	TCA	30.0000	mm
Outside Crown Radius of Shell	OD/2	2096.9998	mm
Crown Radius / Thickness Ratio	$(OD/2)/T$	69.9000	
Geometry Factor, A	f(D/T, LD)	0.0017883	
Materials Factor, B	f(A, Chart)	101.1631	N./mm ²
Maximum Allowable Working Pressure		1447.17	KPa.
EMAWP = $B / ((OD/T)/2) = 101 / (140/2.0) = 1447$			

Results for Req'd Thickness for Ext. Pressure (Tca):

Corroded Thickness of Head	TCA	13.7952	mm
Outside Diameter of Shell	ODCA	4194.000	mm
Diameter / Thickness Ratio	(D/T)	304.0179	
Geometry Factor, A	f(DT, LD)	0.0008223	
Materials Factor, B	f(A, Chart)	79.0580	N./mm ²
Maximum Allowable Working Pressure		520.06	KPa.
EMAWP = $B / ((D/T)/2) = 79.1 / (304/2) = 520$			



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Intermediate Calculations for Flanged portion:

ASME Code, Section VIII, Div. 1, 2025 App. 2

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000 mm
Basic Gasket Width,	$b_0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b_0) * 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = G_o - 2.0 * b$	2310.156 mm

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsm_{ax}]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 * 38.1 + 6 * 170 / (3 + 0.5)$$

$$= 367.629 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C * \sin(\pi / n)$$

$$= 2368 * \sin(\pi / 72)$$

$$= 103.291 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{103 / (2 * 38.1 + 170)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

Total Area of Bolts 652.644 cm²

	Minimum	Actual	Maximum
Radial Distance between Bolts and Edge:	38.100	48.500	
Circ. Spacing between the Bolts:	82.550	103.291	367.629

Results for the Internal Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	427433.9	Kgf
Contact Load on Gasket Surfaces	Hp	28518.3	Kgf
Hydrostatic End Load at Flange ID	Hd	422579.5	Kgf
Pressure Force on Flange Face	Ht	4854.4	Kgf
Radial Component of Head Membrane Force	Hr	635376.6	Kgf
Operating Bolt Load:	Wm1	455952.2	Kgf
Gasket Seating Bolt Load	Wm2	327703.7	Kgf
Required Bolt Area	Am	324.024	cm ²

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calculation]:

$$= Ab * Sb / (y * PI * (Go + Gi))$$

$$= 653 * 138 / (68.9 * 3.14 * (2323 + 2297))$$

$$= 9.000 \text{ mm}$$

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :

= 38.100 mm [Note: Exceeds actual gasket width, 13.000]

Flange Design Bolt Load (Seating)	W	918372.3	Kgf
Gasket Seating Force (Operating)	Hg	28518.3	Kgf
Distance to Gasket Load Reaction	hg	28.9222	mm
Distance to Face Pressure Reaction	ht	32.2111	mm
Distance to End Pressure Reaction	hd	35.5000	mm

Summary of Moments for Internal Pressure:

Loading	Force	Distance	Bolt Corr	Moment
End Pressure,	Md	422580.	35.5000	1.0000 15002. Kg-m.



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Face Pressure,	Mt	4854.	32.2111	1.0000	156. Kg-m.
Gasket Load,	Mg	28518.	28.9222	1.0000	825. Kg-m.
Floating Hd. Load,	Mh	635377.	15.0566	1.0000	-9567. Kg-m.
Gasket Seating,	Ma	918372.	28.9222	1.0000	26562. Kg-m.

Total Moment for Operation (Internal Pressure) 6416.250 Kg-m.
 Total Moment for Gasket Seating (Int. Pressure) 26561.844 Kg-m.

Results for the External Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	222265.6	Kgf
Hydrostatic End Load at Flange ID	Hd	219741.4	Kgf
Pressure Force on Flange Face	Ht	2524.3	Kgf
Radial Component of Head Membrane Force	Hr	330395.8	Kgf
Gasket Seating Bolt Load	Wm2	327703.7	Kgf
Required Bolt Area	Am	232.884	cm ²

Flange Design Bolt Load (Seating)	W	918372.3	Kgf
Distance to Gasket Load Reaction	hg	28.9222	mm
Distance to Face Pressure Reaction	ht	32.2111	mm
Distance to End Pressure Reaction	hd	35.5000	mm

Summary of Moments for External Pressure:

Loading	Force	Distance	Bolt Corr	Moment	
End Pressure,	Md	219741.	6.5778	1.0000	1445. Kg-m.
Face Pressure,	Mt	2524.	3.2889	1.0000	8. Kg-m.
Floating Hd. Load,	Mh	330396.	15.0566	1.0000	-4975. Kg-m.
Gasket Seating,	Ma	918372.	28.9222	1.0000	26562. Kg-m.

Total Moment for Operation (External Pressure) 3520.991 Kg-m.
 Total Moment for Gasket Seating (Ext. Pressure) 26561.844 Kg-m.

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 1000 \cdot 2297 \cdot \sqrt{(4 \cdot 2067 \cdot 2067 - 2297 \cdot 2297) / (8 \cdot 138(2465 - 2297))}$$

$$= 42.5706 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 6416(2465 + 2297) / (138 \cdot 2297)(2465 - 2297)$$

$$= 5626.6411 \text{ mm}^2$$

Required thickness for Main Flange, Internal Operating Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 42.6 + \sqrt{(42.6 \cdot 42.6 + 5627)}$$

$$= 128.8196 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 26562(2465 + 2297) / (138 \cdot 2297)(2465 - 2297)$$

$$= 23293.0391 \text{ mm}^2$$

Required thickness for Main Flange, Internal Bolt-up Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 0 + \sqrt{(0 \cdot 0 + 23293)}$$

$$= 152.6206 \text{ mm}$$

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 520 \cdot 2297 \cdot \sqrt{(4 \cdot 2067 \cdot 2067 - 2297 \cdot 2297) / (8 \cdot 138(2465 - 2297))}$$



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= 22.1367 mm

Factor [J]:

= $Mo(A + B)/(S \cdot B)(A - B)$
 = $3521(2465+2297)/(138 \cdot 2297)(2465-2297)$
 = 3087.6841 mm²

Required thickness for Main Flange, External Operating Condition [T]:

= $F + \sqrt{(F \cdot F + J)}$ per 1-6(g)
 = $22.1 + \sqrt{(22.1 \cdot 22.1 + 3088)}$
 = 81.9507 mm

Factor [J]:

= $Mo(A + B)/(S \cdot B)(A - B)$
 = $26562(2465+2297)/(138 \cdot 2297)(2465-2297)$
 = 23293.0391 mm²

Required thickness for Main Flange, External Bolt-up Condition [T]:

= $F + \sqrt{(F \cdot F + J)}$ per 1-6(g)
 = $0 + \sqrt{(0 \cdot 0 + 23293)}$
 = 152.6206 mm

Summary of Required Thicknesses(mm):

	Head	Flange
Tubeside (Internal) Pressure	12.4828	128.8196
Shellside (External) Pressure	13.7952	81.9507
Tubeside Gasket Seating Load		152.6206
Shellside Gasket Seating Load		152.6206
Maximum + Corrosion Allowance	13.7952	152.6206
Actual Thickness as Given	30.0000	170.0000

Floating Head MAWP and MAPnc (KPa.):

Stress Condition	Tubeside		Shellside	
	MAWP	MAPnc	MAWP	MAPnc
Flange Stress	1440.87	1440.87	1412.60	1412.60
Floating head Stress	2403.20	2403.20	1447.07	1447.07
Bolt Stress	2014.09	2014.09	...	...
Minimum MAWP & MAPnc	1440.87	1440.87	1412.60	1412.60

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	133	No Calc	No Calc
20	30	No Calc	...	209	No Calc	No Calc
30	40	1500	2520	17	0.0012866	92.6695
40	50	No Calc	...	209	No Calc	No Calc
50	60	No Calc	...	217	No Calc	No Calc
60	70	5600	2536	25	0.00056568	56.5554
70	80	5600	2516	15	0.00026855	26.8488
80	90	5600	2536	25	0.00056568	56.5554
90	100	No Calc	...	208	No Calc	No Calc
100	110	No Calc	...	199	No Calc	No Calc
110	120	1757.47	2520	17	0.0010867	87.8149
120	130	No Calc	2512	13	0.00071877	71.861

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	136	64.897	103.419	No Calc
20	30	212	207.95	103.419	No Calc
30	40	20	9.51664	103.419	833
40	50	212	207.467	103.419	No Calc
50	60	220	216.84	103.419	No Calc
60	70	28	14.2907	103.419	743
70	80	18	14.2392	103.419	213
80	90	28	14.2907	103.419	743
90	100	211	206.375	103.419	No Calc
100	110	202	196.85	103.419	No Calc
110	120	20	9.93533	103.419	790
120	130	16	9.50406	103.419	413

Minimum

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External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	40	1500	14520.7	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	5600	130606	No Calc	No Calc
70	80	5600	11109.4	No Calc	No Calc
80	90	5600	130606	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc
100	110	No Calc	No Calc	No Calc	No Calc
110	120	1757.47	14521.5	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc



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Elements Suitable for External Pressure.

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 Element and Detail Weights: Step: 19 6:07pm Sep 8,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	6619.36	...	6619.36	...	661.936
20	30	2002.27	1430405	1948.77	1432161	200.227
30	40	1826.18	7247067	1554.11	7282176	182.618
40	50	2015.45	1428103	1961.99	1429858	201.545
50	60	2061.25	1469244	2006.29	1470999	206.125
60	70	3077.79	5617681	2751.31	5659812	307.779
70	80	2189.66	6241868	1826.9	6288681	218.966
80	90	3077.79	5634784	2751.31	5676916	307.779
90	100	1984.49	1425551	1931.17	1427306	198.449
100	110	1916.98	1381858	1865.29	1383613	191.698
110	120	1826.18	7247067	1554.11	7282176	182.618
120	130	1221.37	2242403	1038.17	2258122	122.137
Total		29818	41366032.00	27808	41591824.00	2981



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	...	68	-51	...	Liquid: 10
20	Liqd	1429.53	143.5	-30	...	Liquid: 20
30	Liqd	7233.76	750	9.99997	...	Liquid: 30
30	Nozl	220.451	750	1545	...	N3-Tube Inlet
40	Liqd	1427.23	143.5	-31	...	Liquid: 40
50	Liqd	1468.35	147.5	-30	...	Liquid: 50
50	Insl	52.4162	147.5	...	...	Insulation
50	Forc	...	295	...	...	TOP HE FORCES
50	Forc	...	295	...	...	TOP HE FORCES
50	Forc	...	295	...	...	TOP HE FORCES
50	Forc	...	295	...	...	TOP HE FORCES
60	Sadl	1535.91	1120	1571	...	Left Saddle
60	Liqd	5614.25	900	0.99994	...	Liquid: 60
60	Insl	525.314	900	...	...	IN
60	Nozl	125.647	824.229	1417.8	...	N1-Shell Inlet
60	Wght	500	1120	-1900	...	Fireproofing we
70	Liqd	6238.06	1000	0.99994	...	Liquid: 70
70	Insl	579.253	1000	...	...	iN
70	Nozl	8.16602	1448.81	-1264.5	...	Drain
70	Nozl	8.16602	1323.56	1264.5	...	Vent
70	Wght	79500	390.901	3899.39	-68.7519	TOP HE EMPTY
70	Wght	50000	390.901	3900	...	TOP HE OPERATIN
70	Wght	50000	390.901	3899.39	-68.7519	TOP HE TEST
80	Sadl	1535.91	680	1571	...	Sdl 2 Fr60
80	Liqd	5631.35	900	9.99997	...	Liquid: 60
80	Insl	525.314	900	...	...	Insulation
80	Nozl	125.647	1318.03	-1417.8	...	N2-Shell Outlet
80	Wght	500	680.001	-1900	...	Wght:[2 of 2]
90	Liqd	1424.68	143	-30	...	Liquid: 70
90	Insl	52.4781	156	...	...	Insulation
100	Liqd	1381.02	138.5	-30	...	Liquid: 80
100	Insl	52.8106	156	...	...	Insulation



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110	Liqd	7233.76	750	9.99997	...	Liquid: 90
110	Insl	435.104	750	...	...	Insulation
110	Nozl	203.095	750	-1545	...	N4-Tube Outlet
120	Liqd	2241.03	257.467	0.99994	...	Liquid: 100
120	Insl	363.227	25.25	...	...	Insulation
40	FTsh	3377.7	347.675	...	...	Stationary Tube
40	Tube	34464.4	3290.17	...	...	Tubes
40	FlTs	3002.22	6290.17	...	...	
40	FlHd	2066.42	6375.17	...	...	
50	Bafl	284.226	636	...	...	Baffles
50	Bafl	284.226	1272	...	...	Baffles
50	Bafl	284.226	1908	...	...	Baffles
50	Bafl	284.226	2544	...	...	Baffles
50	Bafl	284.226	3180	...	...	Baffles
50	Bafl	284.226	3816	...	...	Baffles
50	Bafl	284.226	4452	...	...	Baffles
50	Bafl	284.226	5088	...	...	Baffles
50	Bafl	284.863	5724	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	3071.8
Liquid	41323.0
Insulation	2585.9
Nozzles	691.2
Weights	180500.0
Exchanger Components	45469.4
Liquid in Tubes	6215.6

Sum of the Detail Weights 279857.0 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	32800.7	32800.7	32800.7	32800.7	32800.7	32800.7
Saddles	3071.8	3071.8	3071.8	3071.8	3071.8	3071.8
Nozzles	691.2	691.2	691.2	691.2	691.2	691.2
Wld Weights	79500.0	79500.0	79500.0	79500.0	79500.0	79500.0
Exchanger	45469.4	45469.4	45469.4	45469.4	45469.4	45469.4
Insulation	...	...	...	...	2585.9	2585.9 *
Test Weights	...	51000.0	...	...	...	...
Ope Weights	...	...	...	...	...	51000.0
Ope. Liquid	...	...	...	...	...	41323.0
Tube Ope Lqd	...	...	...	...	...	6215.6
Test Liquid	...	41340.8	...	...	...	...
Tube Tst Lqd	...	6215.7	...	...	...	...
Totals	161533.1	260089.5	161533.1	161533.1	164119.0	262657.7

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	161533.1 kg.
Shop Test Wt.	- Fabricated weight + water (full)	260089.5 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	161533.1 kg.



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Element and Detail Weights:

Step: 19

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Erected Wt.	- Fab. wt + or - loose items (trays, platforms etc.)	161533.1 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	164119.0 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	262657.7 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	260446.7 kg.
Field Test Wt.	- Empty weight + test medium (full)	260690.1 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	6219458.0 cm ³
Weight of Ope Liq in tubes :	6215.6 kg.
Weight of Water in tubes :	6215.7 kg.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	71741	...
20	30	36005.4	22983.9
30	40	118752	116867
40	50	36078.5	22965.4
50	60	36696	23607.2
60	70	143407	140241
70	80	158085	155823
80	90	143407	140241
90	100	35919	22906
100	110	35142.1	22204.8
110	120	118752	116867
120	130	72859.2	70627.3

Total	1006845.4	855333.5 cm ²
Total	1083.8	920.7 ft ²

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 Nozzle Flange MAWP: Step: 20 6:07pm Sep 8,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Flange Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N3-Tube Inlet	1820.4	1960.0	82	150	GR 1.1	...	...	...	...
N1-Shell Inlet	1656.0	1960.0	130	150	GR 1.1	2318.5	1324.8	1656.0	165.5
Drain	4574.0	5100.0	130	300	GR 1.1	...	...	...	...
Vent	4574.0	5100.0	130	300	GR 1.1	...	...	...	...
N2-Shell Outle	1656.0	1960.0	130	150	GR 1.1	2318.5	1324.8	1656.0	165.5
N4-Tube Outlet	1656.0	1960.0	130	150	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 1324.771 KPa.
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 1960.006 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 1656.044 KPa.
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 1960.006 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N1-Shell Inlet	389.001	1.20	2472	2677	1056.7	1324.8	Pass
N2-Shell Outlet	389.001	1.20	2472	2677	1056.7	1324.8	Pass

Evaluating Flange on Nozzle [N1-Shell Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(2472) + 4 \cdot \text{abs}(2677) \cdot 389.001 \leq \pi \cdot 389.001^3 [(1656.044 - 1056.657) + 1.200 \cdot 1656.044]$$

$$= 0.43724E+05 \leq 0.48780E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Shell Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1656.044, 1.200 \cdot 1656.044 + 1656.044 - 43723.9 / (\pi \cdot 389.001^3))$$

$$= \min(1656.044, 1324.771)$$

$$= 1324.771 \text{ KPa.}$$

Evaluating Flange on Nozzle [N2-Shell Outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(2472) + 4 \cdot \text{abs}(2677) \cdot 389.001 \leq \pi \cdot 389.001^3 [(1656.044 - 1056.657) + 1.200 \cdot 1656.044]$$

$$= 0.43724E+05 \leq 0.48780E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-Shell Outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1656.044, 1.200 \cdot 1656.044 + 1656.044 - 43723.9 / (\pi \cdot 389.001^3))$$

$$= \min(1656.044, 1324.771)$$

$$= 1324.771 \text{ KPa.}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition



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Nozzle Flange MAWP: Step: 20 6:07pm Sep 8, 2026

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	92.7 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.23 \cdot 0.97) / (1 + 1.7 \cdot 3.4 \cdot 0.23)))$
 $= \min(0.85, 0.91)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.554
Structure Height to Diameter ratio	4.244

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Zg = 274320 \text{ mm}$

Effective Height [z]:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1950 + 300 = 2250 \text{ mm}$
 $= 7.38 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z [Kz]:
 Because $z (7.38 \text{ ft.}) < 15 \text{ ft.}$
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900)^{2/9.5}$
 $= 0.85$

Type of Hill: No Hill



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 Wind Load Calculation: Step: 21 6:07pm Sep 8,2026

Wind Directionality Factor [Kd]:
 = 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:
 = (1 + K1 • K2 • K3)²
 = (1 + 0 • 0 • 0)²
 = 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 = max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)
 = max(16, 0.00256 • 0.85 • 1 • 1 • 1 • 57.6²)
 = 16 psf [78.1] Kgs/m²

Force on the first element [F]:
 = qz • G • Cf • WindArea
 = 16 • 0.85 • 0.55 • 4.83
 = 36.4 lbs. [16.5] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Blind Flang	2250.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS CH flange-1	2250.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS Ch Shell	2250.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS Ch Flg-2	2250.0	0.000	0.000	0.000	0.849	1.000	78.120
LHS Shell Flg-1	2250.0	0.000	0.000	0.000	0.849	1.000	78.120
Main Shell	2250.0	0.000	0.000	0.000	0.849	1.000	78.120
Main Shell	2250.0	0.000	0.000	0.000	0.849	1.000	78.120
Main Shell	2250.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS Shell Flg-1	2250.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS Shell Flg-2	2250.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS	2250.0	0.000	0.000	0.000	0.849	1.000	78.120
RHS dish end	2250.0	0.000	0.000	0.000	0.849	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Fireproofing	0.00	300.00	78.12	0.00
TOP HE EMPTY	363329.00	6149.39	78.12	2838.33
TOP HE OPERAT	1.00	6150.00	78.12	0.01
TOP HE TEST	-1.01	6149.39	78.12	-0.01
Wght:[2 of 2]	0.00	300.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	2250	3297.6	4484.74	78.12	18.1496
20	30	2250	2976	8541.12	78.12	34.5657
30	40	2250	3024	45360	78.12	183.571
40	50	2250	2973.6	8534.23	78.12	34.5379
50	60	2250	3216	9487.2	78.12	38.3945
60	70	2250	3283.2	59097.6	78.12	239.167
70	80	2250	3259.2	65184	78.12	3385.96
80	90	2250	3283.2	59097.6	78.12	239.167
90	100	2250	3216	9197.76	78.12	37.2231
100	110	2250	3216	8908.32	78.12	36.0518



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Wind Load Calculation:

Step: 21

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110	120	2250	3264	48960	78.12	198.14
120	130	2250	3252.98	17329.3	78.12	70.1314

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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 Earthquake Load Calculation: Step: 22 6:07pm Sep 8,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(10.90.75)$
 = 0.120 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.12$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.16 \cdot 259586 \cdot 1.25$
 = 15263.650 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.16 \cdot 259586$
 = 8140.6 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	1374	18541.8	1198.89



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Earthquake Load Calculation:

Step: 22

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20	30	1240	18541.8	1081.97
30	40	1240	18541.8	1081.97
40	50	1239	18541.8	1081.1
50	60	1240	18541.8	1081.97
60	70	1240	18541.8	1081.97
60	70	1240	18541.8	1081.97
70	80	1240	18541.8	1081.97
80	0	1240	18541.8	1081.97
80	90	1240	18541.8	1081.97
90	100	1240	18541.8	1081.97
100	110	1240	18541.8	1081.97
110	120	1240	18541.8	1081.97
120	130	1240	18541.8	1081.97

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 23 6:07pm Sep 8,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5429.524 mm
Center of Gravity of the Liquid	5494.887 mm
Center of Gravity of the Insulation	6875.757 mm
Center of Gravity of the Nozzles	5353.757 mm
Center of Gravity of the Added Weights (Operating)	4825.092 mm
Center of Gravity of the Added Weights (Empty)	4820.425 mm
Center of Gravity of the Tubesheet(s)	5070.231 mm
Center of Gravity of the Tubes	5216.350 mm
Center of Gravity of the Floating Head	8304.524 mm
Center of Gravity of the Baffles	5515.157 mm
Center of Gravity of Bare Shell New and Cold	4072.983 mm
Center of Gravity of Bare Shell Corroded	3952.387 mm
Vessel CG in the Operating Condition	4959.258 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4832.431 mm
Vessel CG in the Test Condition	4975.893 mm

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ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1254.00	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	6458.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1120.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	700.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1950.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		45500.00	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		57576.50	Kgf
Pressure used in Analysis	P	0.000	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-3.91	-210.96
Long. Stress at Top of Midspan		3.91	138.00
Long. Stress at Bottom of Midspan		3.91	138.00
Long. Stress at Top of Saddles		2.78	138.00
Long. Stress at Bottom of Saddles		-1.54	-234.00
Long. Stress at Bottom of Saddles		1.54	138.00
Tangential Shear in Shell		12.30	110.40
Circ. Stress at Horn of Saddle		29.59	207.00
Circ. Compressive Stress in Shell		1.29	138.00

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 45500 • 1.5 • 1950/3360
 = 39609.4 Kgf

Saddle Reaction Force due to Earthquake FI [Fsl]:
 = max(FI,eq; Sum of X Forces) • B / Ls
 = max(0, 68250) • 1950/3360



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 Saddle Calcs: Bundle Pull New: Step: 24 6:07pm Sep 8, 2026

= 7051.3 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:

= $F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 = $3 (0/2 + 8867) \cdot 1950/2231$
 = 0.0 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 17967 + max(39609, 0, 7051, 0)
 = 57576.5 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	59112.41	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	68250.00	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= $\min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot R_m)$
 = $\min(2(2480/2 + 28 + 20) \sin(120/2), 2 \cdot 1254)$
 = 2230.882 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0444	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0365			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 = $-57577 \cdot 1120 [1 - (1 - 1120/6458 + (1254^2 - 0^2)/(2 \cdot 1120 \cdot 6458))/(1 + (4 \cdot 0)/(3 \cdot 6458))]$
 = -4173.8 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 = $57577 \cdot 6458/4 ((1 + 2(1254^2 - 0^2)/(6458^2))/(1 + (4 \cdot 0)/(3 \cdot 6458)) - 4 \cdot 1120/6458)$
 = 35482.2 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$
 = $0 \cdot 1254/(2 \cdot 18) - 35482/(\pi \cdot 1254^2 \cdot 18)$
 = -3.91 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= $P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$
 = $0 \cdot 1254/(2 \cdot 18) + 35482/(\pi \cdot 1254^2 \cdot 18)$
 = 3.91 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

= $P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$
 = $0 \cdot 1254/(2 \cdot 28) - 4174/(0.11 \cdot \pi \cdot 1254^2 \cdot 28)$
 = 2.78 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

= $P \cdot R_m/(2t) + M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$
 = $0 \cdot 1254/(2 \cdot 28) + 4174/(0.19 \cdot \pi \cdot 1254^2 \cdot 28)$
 = -1.54 N./mm²



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 Saddle Calcs: Bundle Pull New: Step: 24 6:07pm Sep 8,2026

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L-2a)/(L+(4 \cdot hm/3))$
 $= 57577 (6458 - 2 \cdot 1120)/(6458 + (4 \cdot 0/3))$
 $= 37605.7 \text{ Kg}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ]:
 $= K2 \cdot T / (Rm \cdot t)$
 $= 1.17 \cdot 37606 / (1254 \cdot 28)$
 $= 12.30 \text{ N./mm}^2$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{(Rm \cdot t)}$
 $= 0.78 \cdot \sqrt{(1254 \cdot 28)}$
 $= 146.158 \text{ mm}$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$
 $= \min(400 + 1.56 \cdot \sqrt{(1254 \cdot 28)}, 2 \cdot 1120)$
 $= 692.32 \text{ mm}$

Wear Plate/Shell Stress ratio (4.15.26) [η]:
 $= 1.0000$ Materials are the same, bundle pull case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:
 $= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$
 $= -0.76 \cdot 57577 \cdot 0.1 / (692(28 + 1 \cdot 20))$
 $= -1.29 \text{ N./mm}^2$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r^*]:
 $= -Q/(4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$
 $= -57577 / (4(28 + 1 \cdot 20)692) -$
 $12 \cdot 0.044 \cdot 57577 \cdot 1254 / (6458(28 + 1 \cdot 20)^2)$
 $= -29.59 \text{ N./mm}^2$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	52.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	250.0	N./mm ²
Height of Web at Center	Hw, c	680.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(1240 \cdot 28)} + 700)28^3)/12$
 $= 181.228 \text{ cm}^4$

Resolved Inertia for the Shell [I, shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 181 + 277(268 - 14)^2$
 $= 178492.031 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	990.7	28.0	14.0	277.4	388346.4	0.178E+06
Wearplate	700.0	20.0	38.0	140.0	531999.9	0.738E+05



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Web	20.0	610.0	353.0	122.0	4306599.0	0.467E+05
BasePlate	425.0	52.0	684.0	221.0	15116398.0	0.384E+06
Totals	...	...	...	760.4	20343344.0	0.683E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 20343344 / 760$$

$$= 267.538 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 57577$$

$$= 11718.0664 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.3792 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 149.9973 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1950 - 1240 \cdot \sin(120/2) / 1.05$$

$$= 924.528 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 10833.8965 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 4.1626 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 166.6637 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(57577 + 1536) \cdot 425) / (4 \cdot 2350 \cdot 167)}$$

$$= 21.721 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1240 + 28 + 20)$$

$$= 2230.881 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2231 / (5 - 1)$$

$$= 557.720 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 558 \cdot 425 / 2$$

$$= 1185.156 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 59112 / (2350 \cdot 425)$$

$$= 5.919 \text{ Kg/cm}^2$$

Axial Load [P]:



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$$= A_p \cdot B_p$$

$$= 1185 \cdot 5.92$$

$$= 7014.510 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 76 + 55.8$$

$$= 131.772 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 7015/132$$

$$= 5.220 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	400.0	...	80.0	...	0.107E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 400 - 20$$

$$= 380.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/132$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(10667/132)}$$

$$= 89.971 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1950 - \cos(120/2) (1240 + 28 + 20) - 52$$

$$= 1254.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/250)}$$

$$= 125.651$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1254/90$$

$$= 13.938$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 68250 / (2 \cdot 2350) \cdot 558 \cdot 1254/2$$

$$= 5078.055 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($13.9 < 126$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (13.9)^2 / (2 \cdot 126^2)) 250 /$$



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$$\begin{aligned} & (5/3+3 \cdot (13.9)/(8 \cdot 126) - (13.9^3)/(8 \cdot 126^3)) \\ & = 145.5 \text{ N./mm}^2 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned} & = Sc/Sca + (Rm \cdot C1/I)/Sba \\ & = 5.22/145 + (5078 \cdot 200/106666664)/167 \\ & = 0.596 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	380.0	0.0	76.0	0.0	0.107E+05
Web	557.7	20.0	0.0	111.5	0.0	37.2
Totals	...	...	...	187.5	...	0.107E+05

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} & = AY / A \\ & = 0/188 \\ & = 0.000 \text{ mm} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} & = \text{Saddle Width} / 2 \\ & = 400/2 \\ & = 200.000 \text{ mm} \end{aligned}$$

Length of Inner Rib [L]:

$$\begin{aligned} & = \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk} \\ & = 1950 - \cos(1288/2)(52 + 0 + 0) - 0 \\ & = 610.000 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} & = \sqrt{(\text{Total Inertia} / \text{Total Area})} \\ & = \sqrt{(10703/188)} \\ & = 75.542 \text{ mm} \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} & = KL/r \\ & = 1 \cdot 610/75.5 \\ & = 8.075 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned} & = FI / (2 \cdot \text{Baseplate Length}) \\ & = 68250/(2 \cdot 2350) \\ & = 14.521 \text{ Kg/mm} \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned} & = \text{Unit Force} \cdot e \cdot L \\ & = 14.5 \cdot 558 \cdot 610 \\ & = 4940.372 \text{ Kg-m.} \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned} & = \text{Bending Moment} / \text{Section Modulus} \\ & = 4940/535126 \\ & = 90.537 \text{ N./mm}^2 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($8.07 < 126$) per AISC E2-1 [Sca]:

$$\begin{aligned} & = (1 - (KL/r)^2/(2 \cdot Cc^2))Fy/(5/3+3 \cdot (KL/r)/(8 \cdot Cc) - (KL/r^3)/(8 \cdot Cc^3)) \\ & = (1 - (8.07)^2/(2 \cdot 126^2))250/ \\ & \quad (5/3+3 \cdot (8.07)/(8 \cdot 126) - (8.07^3)/(8 \cdot 126^3)) \\ & = 147.6 \text{ N./mm}^2 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I)/Sba$$



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$$= 7.34/148 + (4940 \cdot 200/10703)/167$$

$$= 0.593$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	9.7967 cm ²
Saddle Load Q0 (Weight)	Q0	52061.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	68250.0 Kg
Maximum Transverse Force	Ft	0.0 Kg
Maximum Longitudinal Force	Fl	68250.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 $= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 68250 / (9.8 \cdot 8)$
 $= 85.4 \text{ N./mm}^2$. Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 0 / (9.8 \cdot 8)$
 $= 0.0 \text{ N./mm}^2$. Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= (QL - Q0) / (Stba \cdot Rnbolts)$
 $= (68250 - 52061) / (172 \cdot 8)$
 $= 1.1538 \text{ cm}^2$

Bolt Area due to Shear Load [Bltarears]:
 $= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 $= 68250 / (103 \cdot 8)$
 $= 8.1072 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 1950 \cdot 0 + 0$
 $= 0.00 \text{ Kg-m.}$

Eccentricity (e):
 $= Rmom / Q0$
 $= 0/52061$
 $= 0.00 \text{ mm} < Bplen/6 \rightarrow \text{No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
 $= 0 \text{ (No Uplift)}$

Required Area of a Single Bolt [Bltarear]:
 $= \max[Bltarearl, Bltarears, Bltareart]$
 $= \max[1.15, 8.11, 0]$
 $= 8.1072 \text{ cm}^2$

Review notes about nozzle loadings and supports in the Warnings report.



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ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1255.50	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	6458.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1120.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	700.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1950.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		45500.00	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		56471.00	Kgf
Pressure used in Analysis	P	0.000	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-4.60	-147.17
Long. Stress at Top of Midspan		4.60	138.00
Long. Stress at Bottom of Midspan		4.60	138.00
Long. Stress at Top of Saddles		3.03	138.00
Long. Stress at Bottom of Saddles		-1.68	-234.00
Long. Stress at Bottom of Saddles		1.68	138.00
Tangential Shear in Shell		13.49	110.40
Circ. Stress at Horn of Saddle		32.81	207.00
Circ. Compressive Stress in Shell		1.38	138.00

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 45500 • 1.5 • 1950/3360
 = 39609.4 Kgf

Saddle Reaction Force due to Earthquake FI [Fsl]:
 = max(FI,eq; Sum of X Forces) • B / Ls
 = max(0, 68250) • 1950/3360



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= 7051.3 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:

= $F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 = $3 (0/2 + 8867) \cdot 1950/2231$
 = 0.0 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 16862 + max(39609, 0, 7051, 0)
 = 56471.0 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	58006.90	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	68250.00	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot R_m)$
 = $\min(2(2480/2 + 28 + 20) \sin(120/2), 2 \cdot 1256)$
 = 2230.882 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0443	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0364			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 = $-56471 \cdot 1120 [1 - (1 - 1120/6458 + (1255^2 - 0^2)/(2 \cdot 1120 \cdot 6458))/(1 + (4 \cdot 0)/(3 \cdot 6458))]$
 = -4077.2 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 = $56471 \cdot 6458/4 ((1 + 2(1255^2 - 0^2)/(6458^2))/(1 + (4 \cdot 0)/(3 \cdot 6458)) - 4 \cdot 1120/6458)$
 = 34817.4 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$
 = $0 \cdot 1256/(2 \cdot 15) - 34817/(\pi \cdot 1256^2 \cdot 15)$
 = -4.60 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= $P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$
 = $0 \cdot 1256/(2 \cdot 15) + 34817/(\pi \cdot 1256^2 \cdot 15)$
 = 4.60 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

= $P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$
 = $0 \cdot 1256/(2 \cdot 25) - 4077/(0.11 \cdot \pi \cdot 1256^2 \cdot 25)$
 = 3.03 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

= $P \cdot R_m/(2t) + M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$
 = $0 \cdot 1256/(2 \cdot 25) + 4077/(0.19 \cdot \pi \cdot 1256^2 \cdot 25)$
 = -1.68 N./mm²



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Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L-2a)/(L+(4 \cdot hm/3))$
 $= 56471 (6458 - 2 \cdot 1120)/(6458 + (4 \cdot 0/3))$
 $= 36883.7 \text{ Kg}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ]:
 $= K2 \cdot T / (Rm \cdot t)$
 $= 1.17 \cdot 36884 / (1256 \cdot 25)$
 $= 13.49 \text{ N./mm}^2$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{(Rm \cdot t)}$
 $= 0.78 \cdot \sqrt{(1256 \cdot 25)}$
 $= 138.189 \text{ mm}$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$
 $= \min(400 + 1.56 \cdot \sqrt{(1256 \cdot 25)}, 2 \cdot 1120)$
 $= 676.38 \text{ mm}$

Wear Plate/Shell Stress ratio (4.15.26) [η]:
 $= 1.0000$ Materials are the same, bundle pull case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:
 $= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$
 $= -0.76 \cdot 56471 \cdot 0.1 / (676(25 + 1 \cdot 20))$
 $= -1.38 \text{ N./mm}^2$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r^*]:
 $= -Q/(4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$
 $= -56471 / (4(25 + 1 \cdot 20)676) -$
 $12 \cdot 0.044 \cdot 56471 \cdot 1256 / (6458(25 + 1 \cdot 20)^2)$
 $= -32.81 \text{ N./mm}^2$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	52.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	250.0	N./mm ²
Height of Web at Center	Hw, c	680.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(1243 \cdot 25)} + 700)25^3)/12$
 $= 126.953 \text{ cm}^4$

Resolved Inertia for the Shell [I, shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 127 + 244(277 - 12.5)^2$
 $= 170367.719 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	975.0	25.0	12.5	243.7	304687.0	0.170E+06
Wearplate	700.0	20.0	35.0	140.0	489999.9	0.819E+05



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Web	20.0	610.0	350.0	122.0	4269999.5	0.444E+05
BasePlate	425.0	52.0	681.0	221.0	15050100.0	0.362E+06
Totals	...	...	...	726.7	20114788.0	0.658E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 20114788/727$$

$$= 276.777 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 56471$$

$$= 11493.0723 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 2.3336 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 149.9973 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1950 - 1243 \cdot \sin(120/2) / 1.05$$

$$= 922.047 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 10597.3652 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 4.3699 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 166.6637 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(56471 + 1536) \cdot 425) / (4 \cdot 2350 \cdot 167)}$$

$$= 21.517 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1240 + 28 + 20)$$

$$= 2230.881 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2231 / (5 - 1)$$

$$= 557.720 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 558 \cdot 425/2$$

$$= 1185.156 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 58007 / (2350 \cdot 425)$$

$$= 5.808 \text{ Kgf/cm}^2$$

Axial Load [P]:



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$$= A_p \cdot B_p$$

$$= 1185 \cdot 5.81$$

$$= 6883.326 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 76 + 55.8$$

$$= 131.772 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 6883/132$$

$$= 5.123 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	400.0	...	80.0	...	0.107E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 400 - 20$$

$$= 380.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/132$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(10667/132)}$$

$$= 89.971 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1950 - \cos(120/2) (1240 + 28 + 20) - 52$$

$$= 1254.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/250)}$$

$$= 125.651$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1254/90$$

$$= 13.938$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 68250 / (2 \cdot 2350) \cdot 558 \cdot 1254/2$$

$$= 5078.055 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($13.9 < 126$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (13.9)^2 / (2 \cdot 126^2)) 250 /$$



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$$\begin{aligned} & (5/3+3 \cdot (13.9)/(8 \cdot 126) - (13.9^3)/(8 \cdot 126^3)) \\ & = 145.5 \text{ N./mm}^2 \end{aligned}$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$\begin{aligned} & = Sc/Sca + (Rm \cdot C1/I)/Sba \\ & = 5.12/145 + (5078 \cdot 200/106666664)/167 \\ & = 0.595 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	380.0	0.0	76.0	0.0	0.107E+05
Web	557.7	20.0	0.0	111.5	0.0	37.2
Totals	...	...	...	187.5	...	0.107E+05

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} & = AY / A \\ & = 0/188 \\ & = 0.000 \text{ mm} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} & = \text{Saddle Width} / 2 \\ & = 400/2 \\ & = 200.000 \text{ mm} \end{aligned}$$

Length of Inner Rib [L]:

$$\begin{aligned} & = \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk} \\ & = 1950 - \cos(1288/2)(52 + 0 + 0) - 0 \\ & = 610.000 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} & = \sqrt{(\text{Total Inertia} / \text{Total Area})} \\ & = \sqrt{(10703/188)} \\ & = 75.542 \text{ mm} \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} & = KL/r \\ & = 1 \cdot 610/75.5 \\ & = 8.075 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned} & = FI / (2 \cdot \text{Baseplate Length}) \\ & = 68250/(2 \cdot 2350) \\ & = 14.521 \text{ Kg/mm} \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned} & = \text{Unit Force} \cdot e \cdot L \\ & = 14.5 \cdot 558 \cdot 610 \\ & = 4940.372 \text{ Kg-m.} \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned} & = \text{Bending Moment} / \text{Section Modulus} \\ & = 4940/535126 \\ & = 90.537 \text{ N./mm}^2 \end{aligned}$$

Compressive Allowable, $KL/r < Cc$ ($8.07 < 126$) per AISC E2-1 [Sca]:

$$\begin{aligned} & = (1 - (KL/r)^2/(2 \cdot Cc^2))Fy/(5/3+3 \cdot (KL/r)/(8 \cdot Cc) - (KL/r^3)/(8 \cdot Cc^3)) \\ & = (1 - (8.07)^2/(2 \cdot 126^2))/250/ \\ & \quad (5/3+3 \cdot (8.07)/(8 \cdot 126) - (8.07^3)/(8 \cdot 126^3)) \\ & = 147.6 \text{ N./mm}^2 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I)/Sba$$



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$$= 7.2/148 + (4940 \cdot 200/10703)/167$$

$$= 0.592$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	9.7967 cm ²
Saddle Load Q0 (Weight)	Q0	50955.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	68250.0 Kg
Maximum Transverse Force	Ft	0.0 Kg
Maximum Longitudinal Force	F1	68250.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 $= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 68250 / (9.8 \cdot 8)$
 $= 85.4 \text{ N./mm}^2$. Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 0 / (9.8 \cdot 8)$
 $= 0.0 \text{ N./mm}^2$. Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= (QL - Q0) / (Stba \cdot Rnbolts)$
 $= (68250 - 50956) / (172 \cdot 8)$
 $= 1.2326 \text{ cm}^2$

Bolt Area due to Shear Load [Bltarears]:
 $= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 $= 68250 / (103 \cdot 8)$
 $= 8.1072 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 1950 \cdot 0 + 0$
 $= 0.00 \text{ Kg-m.}$

Eccentricity (e):
 $= Rmom / Q0$
 $= 0/50956$
 $= 0.00 \text{ mm} < Bplen/6 \rightarrow \text{No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
 $= 0 \text{ (No Uplift)}$

Required Area of a Single Bolt [Bltarear]:
 $= \max[Bltarearl, Bltarears, Bltareart]$
 $= \max[1.23, 8.11, 0]$
 $= 8.1072 \text{ cm}^2$

Review notes about nozzle loadings and supports in the Warnings report.



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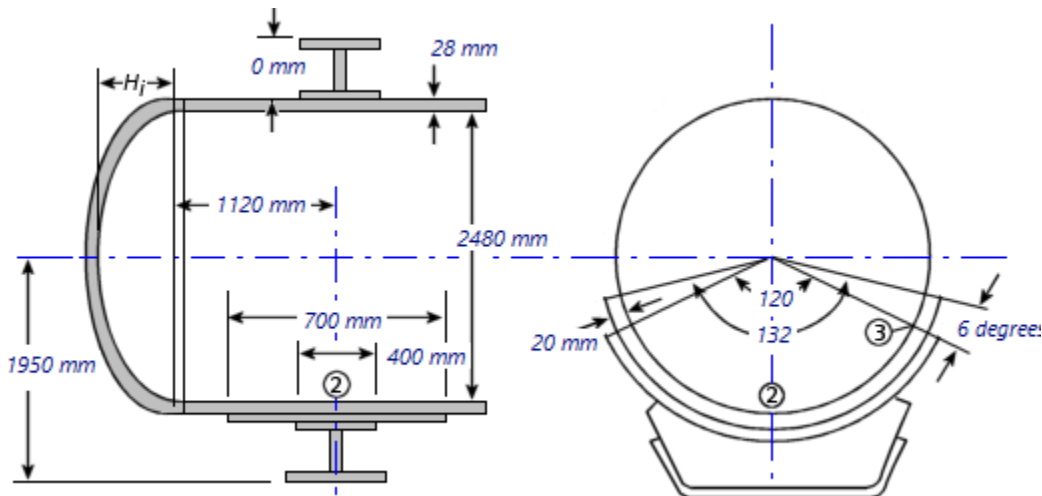
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	1254.00	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	6458.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1120.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	700.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1950.00	mm
Coefficient of Friction	mu	0.30	

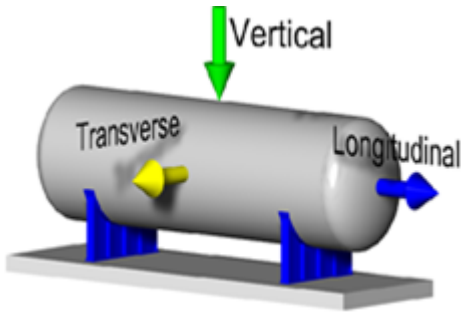


Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		122675.21	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-8.34	-210.96
Long. Stress at Top of Midspan	8.34	138.00
Long. Stress at Bottom of Midspan	8.34	138.00
Long. Stress at Top of Saddles	5.91	138.00
Long. Stress at Bottom of Saddles	-3.28	-234.00
Long. Stress at Bottom of Saddles	3.28	138.00
Tangential Shear in Shell	26.20	110.40
Circ. Stress at Horn of Saddle	63.05	207.00
Circ. Compressive Stress in Shell	2.75	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:
 $= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf$
 $= 370 \cdot 0.00256 \cdot 57.6^2 \cdot 1$
 $= 3146.9 \text{ lbf}$
 $= 1427.436 \text{ Kgf}$

Longitudinal Force due to Wind during Transport [Flw]:
 $= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult}$
 $= \pi \cdot 8.32^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2$
 $= 554.2 \text{ lbf}$
 $= 251.4 \text{ Kgf}$

Transverse Force due to Lateral Acceleration [Ft]:
 $= gz \cdot \text{Shipping Weight} \cdot gw + Ftw$
 $= 0.1 \cdot 158461 \cdot 1 + 1427$
 $= 17273.6 \text{ Kgf}$

Longitudinal Force due to Axial Acceleration [FI]:
 $= gx \cdot \text{Shipping Weight} \cdot gw + Flw$
 $= 0.1 \cdot 158461 \cdot 1 + 251$
 $= 16097.5 \text{ Kgf}$

Saddle Reaction Force due to Transverse Load [Fwt]:
 $= Ft/\text{Num of Saddles} \cdot B / E$
 $= 17274/2 \cdot 1950/2231$
 $= 7549.4 \text{ Kgf}$

Saddle Reaction Force due to Longitudinal Load [Fwl]:
 $= FI \cdot B / Ls$
 $= 16098 \cdot 1950/3360$
 $= 9342.3 \text{ Kgf}$

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:
 $= \text{Saddle Load} \cdot gy$
 $= 107936 \cdot 0.05$



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= 5396.805 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)
 = 107936 + 5397 + max(9342, 7549)
 = 122675.2 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	124211.12	Kgf
Transverse Shear Load	8636.78	Kgf
Longitudinal Shear Load	16097.51	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)
 = min(2(2480/2 + 28 + 20)sin(120/2), 2•1254)
 = 2230.882 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0444	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0365			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -122675•1120[1-(1-1120/6458+(1254²-0²)/
 (2•1120•6458))/(1+(4•0)/(3•6458))]
 = -8892.9 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)
 = 122675•6458/4((1+2(1254²-0²)/(6458²))/(1+(4•0)/
 (3•6458))-4•1120/6458)
 = 75600.0 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)
 = 0 • 1254/(2•18) - 75600/(π•1254²•18)
 = -8.34 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 0 • 1254/(2 • 18) + 75600/(π • 1254² • 18)
 = 8.34 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)
 = 0•1254/(2•28)--8893/(0.11•π•1254²•28)
 = 5.91 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 0•1254/(2•28)+-8893/(0.19•π•1254²•28)
 = -3.28 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))
 = 122675 (6458 - 2 • 1120)/(6458 + (4 • 0/3))
 = 80124.5 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 80125 / (1254 \cdot 28)$$

$$= 26.20 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1254 \cdot 28)}$$

$$= 146.158 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{(1254 \cdot 28)}, 2 \cdot 1120)$$

$$= 692.32 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 122675 \cdot 0.1 / (692(28 + 1 \cdot 20))$$

$$= -2.75 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -122675 / (4(28 + 1 \cdot 20)692) -$$

$$12 \cdot 0.044 \cdot 122675 \cdot 1254 / (6458(28 + 1 \cdot 20)^2)$$

$$= -63.05 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	52.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	250.0	N./mm ²
Height of Web at Center	Hw, c	680.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1240 \cdot 28)} + 700)28^3) / 12$$

$$= 181.228 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 181 + 277(268 - 14)^2$$

$$= 178492.031 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	990.7	28.0	14.0	277.4	388346.4	0.178E+06
Wearplate	700.0	20.0	38.0	140.0	531999.9	0.738E+05
Web	20.0	610.0	353.0	122.0	4306599.0	0.467E+05
BasePlate	425.0	52.0	684.0	221.0	15116398.0	0.384E+06
Totals	...	...	...	760.4	20343344.0	0.683E+06

Distance to Centroid [C1]:

$$= AY / A$$



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$$= 20343344/760$$

$$= 267.538 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 122675$$

$$= 24967.0645 \text{ Kg}$$

Tension Stress, $St = (Fh/As) = 5.0693 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 149.9973 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1950 - 1240 \cdot \sin(120/2) / 1.05$$

$$= 924.528 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 23083.2109 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 8.8691 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 166.6637 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(122675 + 1536) \cdot 425) / (4 \cdot 2350 \cdot 167)}$$

$$= 31.486 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1240 + 28 + 20)$$

$$= 2230.881 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2231 / (5 - 1)$$

$$= 557.720 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 558 \cdot 425/2$$

$$= 1185.156 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 124211 / (2350 \cdot 425)$$

$$= 12.437 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1185 \cdot 12.4$$

$$= 14739.376 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



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$$= 76 + 55.8$$

$$= 131.772 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 14739/132$$

$$= 10.969 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	400.0	...	80.0	...	0.107E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 400 - 20$$

$$= 380.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/132$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(10667/132)}$$

$$= 89.971 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1950 - \cos(120/2) (1240 + 28 + 20) - 52$$

$$= 1254.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/250)}$$

$$= 125.651$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1254/90$$

$$= 13.938$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 16098 / (2 \cdot 2350) \cdot 558 \cdot 1254/2$$

$$= 1197.715 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($13.9 < 126$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (13.9)^2 / (2 \cdot 126^2)) 250 / (5/3 + 3 \cdot (13.9) / (8 \cdot 126) - (13.9^3) / (8 \cdot 126^3))$$

$$= 145.5 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 11/145 + (1198 \cdot 200/106666664) / 167$$



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= 0.208

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	380.0	0.0	76.0	0.0	0.107E+05
Web	557.7	20.0	0.0	111.5	0.0	37.2
Totals	...	...	...	187.5	...	0.107E+05

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/188
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 400/2
 = 200.000 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = 1950 - cos(1288/2)(52 + 0 + 0) - 0
 = 610.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{(10703/188)}$
 = 75.542 mm

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 610/75.5
 = 8.075

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 16098/(2 • 2350)
 = 3.425 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 3.43 • 558 • 610
 = 1165.241 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σB,base,c]:

= Bending Moment / Section Modulus
 = 1165/535126
 = 21.354 N./mm²

Compressive Allowable, KL/r < Cc (8.07 < 126) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (8.07)^2 / (2 \cdot 126^2)) 250 /$
 $(5/3 + 3 \cdot (8.07) / (8 \cdot 126) - (8.07^3) / (8 \cdot 126^3))$
 = 147.6 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
 = 15.4/148 + (1165 • 200/10703)/167
 = 0.233

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



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Vessel Mean Radius	Rm	1254.00	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	6458.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1120.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	700.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1950.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		62393.73	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-4.24	-210.96
Long. Stress at Top of Midspan		4.24	138.00
Long. Stress at Bottom of Midspan		4.24	138.00
Long. Stress at Top of Saddles		3.01	138.00
Long. Stress at Bottom of Saddles		-1.67	-234.00
Long. Stress at Bottom of Saddles		1.67	138.00
Tangential Shear in Shell		13.32	110.40
Circ. Stress at Horn of Saddle		32.07	207.00
Circ. Compressive Stress in Shell		1.40	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 370 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 3146.9 \text{ lbf} \\
 &= 1427.436 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 8.32^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 554.2 \text{ lbf} \\
 &= 251.4 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 158461 \cdot 1 + 1427
 \end{aligned}$$



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= 17273.6 Kgf

Longitudinal Force due to Axial Acceleration [FI]:

= $g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw}$
 = $0.1 \cdot 158461 \cdot 1 + 251$
 = 16097.5 Kgf

Saddle Reaction Force due to Transverse Load [Fwt]:

= $F_t / \text{Num of Saddles} \cdot B / E$
 = $17274 / 2 \cdot 1950 / 2231$
 = 7549.4 Kgf

Saddle Reaction Force due to Longitudinal Load [Fwl]:

= $F_l \cdot B / L_s$
 = $16098 \cdot 1950 / 3360$
 = 9342.3 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

= Saddle Load $\cdot g_y$
 = $50525 \cdot 0.05$
 = 2526.258 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)
 = $50525 + 2526 + \max(9342, 7549)$
 = 62393.7 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	63929.63	Kgf
Transverse Shear Load	8636.78	Kgf
Longitudinal Shear Load	16097.51	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(\text{ShellID} / 2 + t + \text{WearPadThickness}) \sin(\theta / 2), 2 \cdot R_m)$
 = $\min(2(2480 / 2 + 28 + 20) \sin(120 / 2), 2 \cdot 1254)$
 = 2230.882 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0444	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0365			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m / 2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a / L + (R_m^2 - h_m^2) / (2a \cdot L)) / (1 + (4h_m) / 3L)]$
 = $-62394 \cdot 1120 [1 - (1 - 1120 / 6458 + (1254^2 - 0^2) / (2 \cdot 1120 \cdot 6458)) / (1 + (4 \cdot 0) / (3 \cdot 6458))]$
 = -4523.0 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2) / (L^2)) / (1 + (4h_m) / 3L) - 4a / L)$
 = $62394 \cdot 6458 / 4 ((1 + 2(1254^2 - 0^2) / (6458^2)) / (1 + (4 \cdot 0) / (3 \cdot 6458)) - 4 \cdot 1120 / 6458)$
 = 38450.9 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$
 = $0 \cdot 1254 / (2 \cdot 18) - 38451 / (\pi \cdot 1254^2 \cdot 18)$
 = -4.24 N./mm²



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Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1254 / (2 \cdot 18) + 38451 / (\pi \cdot 1254^2 \cdot 18)$$

$$= 4.24 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1254 / (2 \cdot 28) - 4523 / (0.11 \cdot \pi \cdot 1254^2 \cdot 28)$$

$$= 3.01 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1254 / (2 \cdot 28) + 4523 / (0.19 \cdot \pi \cdot 1254^2 \cdot 28)$$

$$= -1.67 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 62394 (6458 - 2 \cdot 1120) / (6458 + (4 \cdot 0 / 3))$$

$$= 40752.1 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 40752 / (1254 \cdot 28)$$

$$= 13.32 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1254 \cdot 28}$$

$$= 146.158 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{1254 \cdot 28}, 2 \cdot 1120)$$

$$= 692.32 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 62394 \cdot 0.1 / (692(28 + 1 \cdot 20))$$

$$= -1.40 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -62394 / (4(28 + 1 \cdot 20)692) - 12 \cdot 0.044 \cdot 62394 \cdot 1254 / (6458(28 + 1 \cdot 20)^2)$$

$$= -32.07 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	52.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	250.0	N./mm ²
Height of Web at Center	Hw,c	680.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



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Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1240 \cdot 28)} + 700)^3) / 12$$

$$= 181.228 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 181 + 277 (268 - 14)^2$$

$$= 178492.031 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	990.7	28.0	14.0	277.4	388346.4	0.178E+06
Wearplate	700.0	20.0	38.0	140.0	531999.9	0.738E+05
Web	20.0	610.0	353.0	122.0	4306599.0	0.467E+05
BasePlate	425.0	52.0	684.0	221.0	15116398.0	0.384E+06
Totals	...	...	...	760.4	20343344.0	0.683E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 20343344 / 760$$

$$= 267.538 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 62394$$

$$= 12698.4766 \text{ Kg}$$

Tension Stress, $St = (Fh / As) = 2.5783 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 149.9973 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1950 - 1240 \cdot \sin(120/2) / 1.05$$

$$= 924.528 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 11740.3311 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 4.5109 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 166.6637 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(62394 + 1536) 425 / (4 \cdot 2350 \cdot 167))}$$

$$= 22.588 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1240 + 28 + 20)$$

$$= 2230.881 \text{ mm}$$

Distance between Ribs [e]:



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$$\begin{aligned}
 &= \text{Web Length} / (\text{Nribs} - 1) \\
 &= 2231 / (5 - 1) \\
 &= 557.720 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 558 \cdot 425 / 2 \\
 &= 1185.156 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 63930 / (2350 \cdot 425) \\
 &= 6.401 \text{ Kg/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= \text{Ap} \cdot \text{Bp} \\
 &= 1185 \cdot 6.4 \\
 &= 7586.140 \text{ Kgf}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 76 + 55.8 \\
 &= 131.772 \text{ cm}^2
 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned}
 &= P / \text{Ar} \\
 &= 7586 / 132 \\
 &= 5.646 \text{ N./mm}^2
 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	400.0	...	80.0	...	0.107E+05

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 400 - 20 \\
 &= 380.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= \text{AY} / \text{A} \\
 &= 0 / 132 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 400 / 2 \\
 &= 200.000 \text{ mm}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{ \text{Total Inertia} / \text{Total Area} } \\
 &= \sqrt{ 10667 / 132 } \\
 &= 89.971 \text{ mm}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1950 - \cos(120 / 2) (1240 + 28 + 20) - 52 \\
 &= 1254.000 \text{ mm}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{ 2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress} } \\
 &= \sqrt{ 2 \cdot \pi^2 \cdot 199943008 / 250 }
 \end{aligned}$$



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$$= 125.651$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1254/90$$

$$= 13.938$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 16098 / (2 \cdot 2350) \cdot 558 \cdot 1254/2$$

$$= 1197.715 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($13.9 < 126$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (13.9)^2 / (2 \cdot 126^2)) 250 /$$

$$(5/3 + 3 \cdot (13.9) / (8 \cdot 126) - (13.9^3) / (8 \cdot 126^3))$$

$$= 145.5 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 5.65/145 + (1198 \cdot 200/106666664) / 167$$

$$= 0.171$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	380.0	0.0	76.0	0.0	0.107E+05
Web	557.7	20.0	0.0	111.5	0.0	37.2
Totals	...	...	...	187.5	...	0.107E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/188$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1950 - \cos(1288/2) (52 + 0 + 0) - 0$$

$$= 610.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (10703/188) }$$

$$= 75.542 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 610/75.5$$

$$= 8.075$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 16098 / (2 \cdot 2350)$$

$$= 3.425 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 3.43 \cdot 558 \cdot 610$$



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= 1165.241 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 1165/535126

= 21.354 N./mm²

Compressive Allowable, $KL/r < C_c$ (8.07 < 126) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)

= (1-(8.07)²/(2 • 126²))250/

(5/3+3•(8.07)/(8• 126)-(8.07³)/(8•126³)

= 147.6 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C1/I)/S_{ba}$

= 7.94/148 + (1165 • 200/10703)/167

= 0.182

Review notes about nozzle loadings and supports in the Warnings report.

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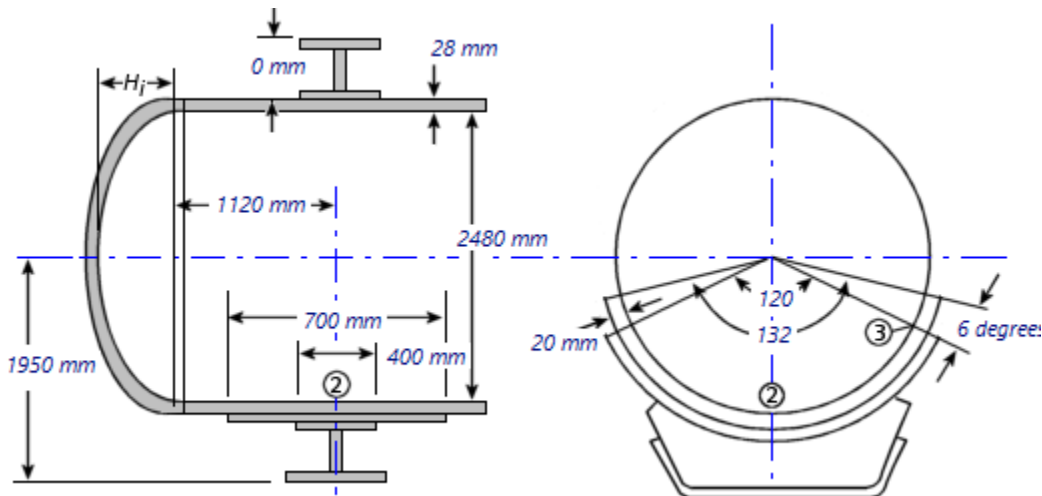
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1255.50	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	6458.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1120.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	700.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1950.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	283126.53	Kgf
Pressure used in Analysis	P	532.135 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-0.77	-176.60
Long. Stress at Top of Midspan	0.77	138.00
Long. Stress at Bottom of Midspan	45.32	138.00
Long. Stress at Top of Saddles	28.55	138.00
Long. Stress at Bottom of Saddles	4.94	138.00
Tangential Shear in Shell	67.64	110.40
Circ. Stress at Horn of Saddle	164.50	172.50



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Circ. Compressive Stress in Shell	6.94	138.00
-----------------------------------	------	--------

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (4515/2 + 35467) \cdot 1950/2231$
 $= 98923.5 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(625, 12150) \cdot 1950/3360$
 $= 7051.3 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(15264, 12150) \cdot 1950/3360$
 $= 8858.4 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (15264/2 + 35467) \cdot 1950/2231$
 $= 113016.4 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 170110 + \max(7051, 98923, 8858, 113016)$
 $= 283126.5 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4 (2.54 \cdot 1.2)^2)$
 $= 6129.442 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	284662.47	Kgf
Transverse Shear Load	43098.49	Kgf
Longitudinal Shear Load	15263.65	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2 (2480/2 + 28 + 20) \sin(120/2), 2 \cdot 1256)$
 $= 2230.882 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0443	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0364			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2) / (2a \cdot L)) / (1 + (4\text{hm}) / (3L))]$
 $= -283127 \cdot 1120 [1 - (1 - 1120/6458 + (1255^2 - 0^2) / (2 \cdot 1120 \cdot 6458)) / (1 + (4 \cdot 0) / (3 \cdot 6458))]$
 $= -20441.8 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2 (\text{Rm}^2 - \text{hm}^2) / (L^2)) / (1 + (4\text{hm}^2) / (3L)) - 4a/L)$



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$$= 283127 \cdot 6458 / 4 \left((1 + 2(1255^2 - 0^2) / (6458^2)) / (1 + (4 \cdot 0) / (3 \cdot 6458)) - 4 \cdot 1120 / 6458 \right)$$

$$= 174562.5 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 532 \cdot 1256 / (2 \cdot 15) - 174562 / (\pi \cdot 1256^2 \cdot 15)$$

$$= -0.77 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 532 \cdot 1256 / (2 \cdot 15) + 174562 / (\pi \cdot 1256^2 \cdot 15)$$

$$= 45.32 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 532 \cdot 1256 / (2 \cdot 25) - 20442 / (0.11 \cdot \pi \cdot 1256^2 \cdot 25)$$

$$= 28.55 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 532 \cdot 1256 / (2 \cdot 25) + 20442 / (0.19 \cdot \pi \cdot 1256^2 \cdot 25)$$

$$= 4.94 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 283127 (6458 - 2 \cdot 1120) / (6458 + (4 \cdot 0 / 3))$$

$$= 184922.2 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 184922 / (1256 \cdot 25)$$

$$= 67.64 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1256 \cdot 25)}$$

$$= 138.189 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{(1256 \cdot 25)}, 2 \cdot 1120)$$

$$= 676.38 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(138 / 138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 283127 \cdot 0.1 / (676(25 + 1 \cdot 20))$$

$$= -6.94 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -283127 / (4(25 + 1 \cdot 20)676) -$$

$$12 \cdot 0.044 \cdot 283127 \cdot 1256 / (6458(25 + 1 \cdot 20)^2)$$

$$= -164.50 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 3360.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$



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$$= 0.12297\text{E-}04 \cdot 3360 (130 - 21.1)$$

$$= 4.499 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	52.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	222.2	N./mm ²
Height of Web at Center	Hw,c	680.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1243 \cdot 25)} + 700)25^3)/12$$

$$= 126.953 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 127 + 244 (277 - 12.5)^2$$

$$= 170367.719 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	975.0	25.0	12.5	243.7	304687.0	0.170E+06
Wearplate	700.0	20.0	35.0	140.0	489999.9	0.819E+05
Web	20.0	610.0	350.0	122.0	4269999.5	0.444E+05
BasePlate	425.0	52.0	681.0	221.0	15050100.0	0.362E+06
Totals	...	...	...	726.7	20114788.0	0.658E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 20114788/727$$

$$= 276.777 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 283127$$

$$= 57622.3906 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 11.6996 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 133.3187 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1950 - 1243 \cdot \sin(120/2) / 1.05$$

$$= 922.047 \text{ mm}$$



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Bending Moment, $M = F_h \cdot d = 53131.6133 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 21.9093 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 148.1319 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(283127 + 1536) \cdot 425) / (4 \cdot 2350 \cdot 148)}$
 $= 50.559 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (1240 + 28 + 20)$
 $= 2230.881 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 2231 / (5 - 1)$
 $= 557.720 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 558 \cdot 425 / 2$
 $= 1185.156 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 284662 / (2350 \cdot 425)$
 $= 28.502 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 1185 \cdot 28.5$
 $= 33779.160 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 76 + 55.8$
 $= 131.772 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 33779 / 132$
 $= 25.139 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	400.0	...	80.0	...	0.107E+05

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 400 - 20$
 $= 380.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 132$
 $= 0.000 \text{ mm}$



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Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{10667/132}$$

$$= 89.971 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1950 - \cos(120/2) (1240 + 28 + 20) - 52$$

$$= 1254.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/222)}$$

$$= 133.279$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1254/90$$

$$= 13.938$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 15264 / (2 \cdot 2350) \cdot 558 \cdot 1254/2$$

$$= 1135.673 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($13.9 < 133$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (13.9)^2 / (2 \cdot 133^2)) 222 / (5/3 + 3 \cdot (13.9) / (8 \cdot 133) - (13.9)^3 / (8 \cdot 133^3))$$

$$= 129.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 25.1/130 + (1136 \cdot 200/106666664) / 148$$

$$= 0.335$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	380.0	0.0	76.0	0.0	0.107E+05
Web	557.7	20.0	0.0	111.5	0.0	37.2
Totals	...	...	...	187.5	...	0.107E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/188$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1950 - \cos(1288/2) (52 + 0 + 0) - 0$$

$$= 610.000 \text{ mm}$$



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Radius of Gyration [r]:
 $= \sqrt{(\text{Total Inertia} / \text{Total Area})}$
 $= \sqrt{(10703/188)}$
 $= 75.542 \text{ mm}$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 610/75.5$
 $= 8.075$

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 15264 / (2 \cdot 2350)$
 $= 3.248 \text{ Kgf/mm}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 3.25 \cdot 558 \cdot 610$
 $= 1104.881 \text{ Kg-m.}$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 1105/535126$
 $= 20.248 \text{ N./mm}^2$

Compressive Allowable, $KL/r < C_c$ ($8.07 < 133$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (8.07)^2 / (2 \cdot 133^2)) 222 /$
 $(5/3 + 3 \cdot (8.07) / (8 \cdot 133) - (8.07^3) / (8 \cdot 133^3))$
 $= 131.3 \text{ N./mm}^2$

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
 $= 35.3/131 + (1105 \cdot 200/10703) / 148$
 $= 0.406$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.38 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	9.7967 cm ²
Saddle Load Q0 (Weight)	Q0	171646.1 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	8858.4 Kgf
Maximum Transverse Force	Ft	43098.5 Kgf
Maximum Longitudinal Force	F1	15263.7 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 $= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 15264 / (9.8 \cdot 8)$
 $= 19.1 \text{ N./mm}^2$. Must be less than 103.4 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 43098 / (9.8 \cdot 8)$
 $= 53.9 \text{ N./mm}^2$. Must be less than 103.4 N./mm².

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $FI / (BoltShearAllowable \cdot Nbolts)$
 = $15264 / (103 \cdot 8)$
 = 1.8091 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + \text{Sum of X Moments}$
 = $1950 \cdot 43098 + 0$
 = 84043.72 Kg-m.

Eccentricity (e):
 = $Rmom / QO$
 = $84044 / 171646$
 = 489.62 mm > Bplen/6 --> Uplift in Transverse direction

$f = Bplen / 2 - Edgedis$
 = $2350 / 2 - 50.8$
 = 1124.20 mm

Modular Ratio Of Steel/Concrete (n1):
 = ES / EC
 = $203402 / 21526$
 = 9.45

$K1 = 3 (e - 0.5 \cdot Bplen)$
 = $3 (490 - 0.5 \cdot 2350)$
 = -2056.13 mm

$K2 = 6 \cdot n1 \cdot At / Bpwid \cdot (f + e)$
 = $6 \cdot 9.45 \cdot 39.2 / 425 \cdot (1124 + 490)$
 = 843614.55 mm²

$K3 = -K2 \cdot (0.5 \cdot Bplen + f)$
 = $-843615 \cdot (0.5 \cdot 2350 + 1124)$
 = -1939638454.47 mm³

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -2056 \cdot Y^2 + 843615 \cdot Y + -1939638400 = 0$
 $Y = 2095.31 \text{ mm}$

$Num = (Bplen / 2 - Y / 3 - e)$
 = $(2350 / 2 - 2095 / 3 - 490)$
 = -13.06

$Denom = (Bplen / 2 - Y / 3 + f)$
 = $(2350 / 2 - 2095 / 3 + 1124)$
 = 1600.76

Total Bolt Tension Force [Tforce]:
 = $-QO \cdot Num / Denom$
 = $-171646 \cdot -13.1 / 1601$
 = 1400.35 Kgf

Bolt Area Required due to Transverse Load [Bltareart]:
 = $Tforce / (Stba \cdot Nbt)$
 = $1400 / (172 \cdot 4)$
 = 0.1992 cm²



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Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 1.81, 0.2]
 = 1.8091 cm²

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)
 = 2(QO + Tforce) / (Y • Bpwid)
 = 2(171646 + 1400)/(2095 • 425)
 = 3811.18 KPa.

Distance from Baseplate Edge to the Web [ADIST]:
 = (Bplen - Weblngh) / 2
 = (2350 - 2299)/2
 = 25.4000 mm

Overturning Moment due To Bolt Tension [Mt]:
 = Tforce • Adist
 = 1400 • 25.4
 = 35.57 Kg-m.

Equivalent Bearing Pressure (f1):
 = fc • (Y - Adist) / Y
 = 3811 • (2095 - 25.4)/2095
 = 3764.98 KPa.

Overturning Moment due to Bearing Pressure [Mc]:
 = (Adist² • Bpwid / 6) • (f1 + 2 • fc)
 = (25.4² • 425/6) • (3765 + 2 • 3811)
 = 53.07 Kg-m.

Baseplate Required Thickness [Treq]:
 = $\sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$
 = $\sqrt{(6 \cdot \max(35.6, 53.1) / (425 \cdot 207))}$
 = 5.9577 mm

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 20, c = 3 mm, E* = 1
 Thickness Ratio = tr • E*/(tg - c) = 0.38, Temp. Reduction = 55 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1255.50	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	6458.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1120.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	700.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²



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Shell Allowable Stress used in Calculation	138.00	N./mm ²
Head Allowable Stress used in Calculation	138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle	1.00	
Circumferential Efficiency at Mid-Span	1.00	
Distance from Saddle Base to Centerline	B	1950.00 mm
Coefficient of Friction	mu	0.00

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	140880.00	Kgf
Pressure used in Analysis	P	531.959 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	10.80	138.00
Long. Stress at Bottom of Midspan	33.73	138.00
Long. Stress at Top of Saddles	20.92	138.00
Long. Stress at Bottom of Saddles	9.17	138.00
Tangential Shear in Shell	33.66	110.40
Circ. Stress at Horn of Saddle	81.85	172.50
Circ. Compressive Stress in Shell	3.45	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (4515/2 + 8867) \cdot 1950/2231$
 $= 29170.8\ Kg$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(625, 12150) \cdot 1950/3360$
 $= 7051.3\ Kg$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(15264, 12150) \cdot 1950/3360$
 $= 8858.4\ Kg$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (15264/2 + 8867) \cdot 1950/2231$
 $= 43263.7\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 97616 + \max(7051, 29171, 8858, 43264)$
 $= 140880.0\ Kg$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(2.54 \cdot 1.2)^2)$
 $= 6129.442\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	142415.91	Kgf
Transverse Shear Load	16498.49	Kgf
Longitudinal Shear Load	15263.65	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1



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Saddle Dimension [E]:

$$= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2480/2 + 28 + 20) \sin(120/2), 2 \cdot 1256)$$

$$= 2230.882 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0443	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0364			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -140880 \cdot 1120 [1 - (1 - 1120/6458 + (1256^2 - 0^2)/(2 \cdot 1120 \cdot 6458)) / (1 + (4 \cdot 0)/(3 \cdot 6458))]$$

$$= -10171.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 140880 \cdot 6458/4 ((1 + 2(1256^2 - 0^2)/(6458^2)) / (1 + (4 \cdot 0)/(3 \cdot 6458)) - 4 \cdot 1120/6458)$$

$$= 86860.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 532 \cdot 1256/(2 \cdot 15) - 86860/(\pi \cdot 1256^2 \cdot 15)$$

$$= 10.80 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 532 \cdot 1256/(2 \cdot 15) + 86860/(\pi \cdot 1256^2 \cdot 15)$$

$$= 33.73 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 532 \cdot 1256/(2 \cdot 25) - 10172/(0.11 \cdot \pi \cdot 1256^2 \cdot 25)$$

$$= 20.92 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 532 \cdot 1256/(2 \cdot 25) + 10172/(0.19 \cdot \pi \cdot 1256^2 \cdot 25)$$

$$= 9.17 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 140880 (6458 - 2 \cdot 1120)/(6458 + (4 \cdot 0/3))$$

$$= 92014.9 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 92015/(1256 \cdot 25)$$

$$= 33.66 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1256 \cdot 25)}$$

$$= 138.189 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$



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$$= \min(400 + 1.56 \cdot \sqrt{1256 \cdot 25}, 2 \cdot 1120)$$

$$= 676.38 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r/S, 1)$$

$$= \min(138/138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 140880 \cdot 0.1 / (676(25 + 1 \cdot 20))$$

$$= -3.45 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -140880 / (4(25 + 1 \cdot 20)676) - 12 \cdot 0.044 \cdot 140880 \cdot 1256 / (6458(25 + 1 \cdot 20)^2)$$

$$= -81.85 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	52.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	222.2	N./mm ²
Height of Web at Center	Hw, c	680.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{1243 \cdot 25} + 700)25^3) / 12$$

$$= 126.953 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 127 + 244(277 - 12.5)^2$$

$$= 170367.719 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	975.0	25.0	12.5	243.7	304687.0	0.170E+06
Wearplate	700.0	20.0	35.0	140.0	489999.9	0.819E+05
Web	20.0	610.0	350.0	122.0	4269999.5	0.444E+05
BasePlate	425.0	52.0	681.0	221.0	15050100.0	0.362E+06
Totals	...	...	...	726.7	20114788.0	0.658E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 20114788 / 727$$

$$= 276.777 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:



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$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 140880$$

$$= 28672.1367 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 5.8216 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 133.3187 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1950 - 1243 \cdot \sin(120/2) / 1.05$$

$$= 922.047 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 26437.5859 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 10.9018 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 148.1319 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(140880 + 1536) \cdot 425) / (4 \cdot 2350 \cdot 148)}$$

$$= 35.761 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1240 + 28 + 20)$$

$$= 2230.881 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2231 / (5 - 1)$$

$$= 557.720 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 558 \cdot 425 / 2$$

$$= 1185.156 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 142416 / (2350 \cdot 425)$$

$$= 14.259 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1185 \cdot 14.3$$

$$= 16899.627 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 76 + 55.8$$

$$= 131.772 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 16900 / 132$$

$$= 12.577 \text{ N./mm}^2$$

Check of Outside Ribs:



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Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	400.0	...	80.0	...	0.107E+05

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 400 - 20
 = 380.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/132
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 400/2
 = 200.000 mm

Radius of Gyration [r]:

$$= \sqrt{\frac{\text{Total Inertia}}{\text{Total Area}}}$$

$$= \sqrt{\frac{10667}{132}}$$
 = 89.971 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1950 - cos(120/2)(1240 + 28 + 20) - 52
 = 1254.000 mm

Intermediate Term [Cc]:

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot \text{Elastic Modulus}}{\text{Yield Stress}}}$$

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot 199943008}{222}}$$
 = 133.279

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1254/90
 = 13.938

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 15264 / (2 • 2350) • 558 • 1254/2
 = 1135.673 Kg-m.

Compressive Allowable, KL/r < Cc (13.9 < 133) per AISC E2-1 [Sca]:

$$= \left(1 - \frac{(KL/r)^2}{(2 \cdot Cc^2)} \right) F_y / \left(\frac{5}{3} + 3 \cdot \frac{(KL/r)}{(8 \cdot Cc)} - \frac{(KL/r)^3}{(8 \cdot Cc^3)} \right)$$

$$= \left(1 - \frac{(13.9)^2}{(2 \cdot 133^2)} \right) 222 / \left(\frac{5}{3} + 3 \cdot \frac{(13.9)}{(8 \cdot 133)} - \frac{(13.9)^3}{(8 \cdot 133^3)} \right)$$
 = 129.6 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I) / Sba
 = 12.6/130 + (1136 • 200/106666664) / 148
 = 0.238

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	380.0	0.0	76.0	0.0	0.107E+05
Web	557.7	20.0	0.0	111.5	0.0	37.2



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Totals	...	...	...	187.5	...	0.107E+05
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Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/188$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1950 - \cos(1288/2)(52 + 0 + 0) - 0$$

$$= 610.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{10703/188}$$

$$= 75.542 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 610/75.5$$

$$= 8.075$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 15264 / (2 \cdot 2350)$$

$$= 3.248 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 3.25 \cdot 558 \cdot 610$$

$$= 1104.881 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1105/535126$$

$$= 20.248 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.07 < 133$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.07)^2 / (2 \cdot 133^2)) 222 / (5/3 + 3 \cdot (8.07) / (8 \cdot 133) - (8.07^3) / (8 \cdot 133^3))$$

$$= 131.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 17.7/131 + (1105 \cdot 200/10703) / 148$$

$$= 0.271$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.38 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	9.7967 cm ²



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Saddle Load Q0 (Weight)	Q0	99152.2 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	8858.4 Kgf
Maximum Transverse Force	Ft	16498.5 Kgf
Maximum Longitudinal Force	F1	15263.7 Kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
 = 51493.8 Kgf

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 = $\max(Ff, F1) / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $51494 / (9.8 \cdot 8)$
 = 64.4 N./mm². Must be less than 103.4 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 = $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $16498 / (9.8 \cdot 8)$
 = 20.6 N./mm². Must be less than 103.4 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 = $15264 / (103 \cdot 8)$
 = 1.8091 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot Ft + \text{Sum of X Moments}$
 = $1950 \cdot 16498 + 0$
 = 32172.69 Kg-m.

Eccentricity (e):
 = $Rmom / Q0$
 = $32173 / 99152$
 = 324.47 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0, 1.81, 0]$
 = 1.8091 cm²

MDMT Calculations for: Sdl 2 Fr60, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 20, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.38$, Temp. Reduction = 55 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Review notes about nozzle loadings and supports in the Warnings report.

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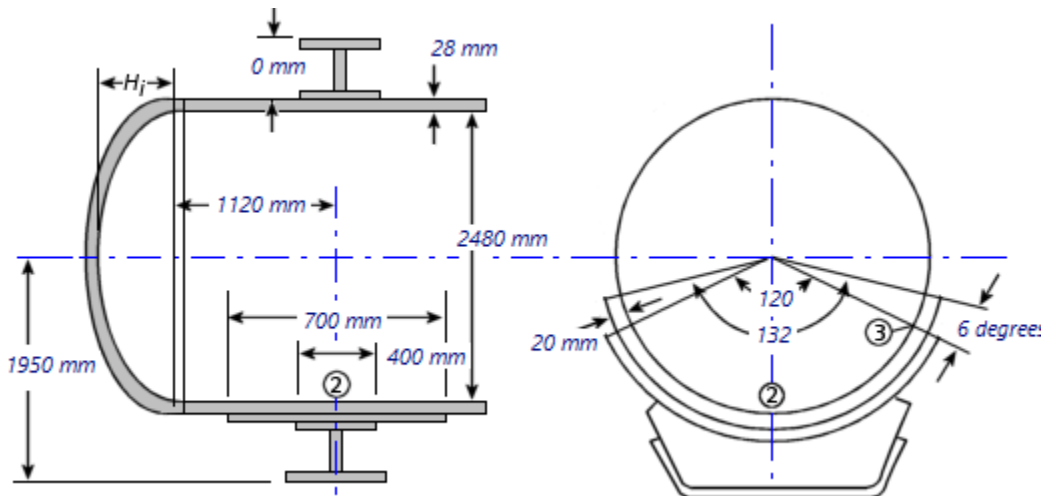
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1255.50	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	6458.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1120.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	700.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1950.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 168761.91 Kgf
 Pressure used in Analysis P 688.306 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	15.07	234.00
Long. Stress at Bottom of Midspan	42.54	234.00
Long. Stress at Top of Saddles	26.34	234.00
Long. Stress at Bottom of Saddles	12.27	234.00
Tangential Shear in Shell	40.32	138.58
Circ. Stress at Horn of Saddle	98.05	350.99
Circ. Compressive Stress in Shell	4.13	234.00



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (2258/2 + 0) \cdot 1950/2231$$

$$= 2959.9 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(313, 0) \cdot 1950/3360$$

$$= 181.4 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 165802 + \max(181, 2960, 0, 0)$$

$$= 168761.9 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.54 \cdot 1.2)^2)$$

$$= 6129.442 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	170297.81	Kgf
Transverse Shear Load	1128.77	Kgf
Longitudinal Shear Load	312.52	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(2480/2 + 28 + 20) \sin(120/2), 2 \cdot 1256)$$

$$= 2230.882 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0443	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0364			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -168762 \cdot 1120 [1 - (1 - 1120/6458 + (1255^2 - 0^2)/(2 \cdot 1120 \cdot 6458)) / (1 + (4 \cdot 0)/(3 \cdot 6458))]$$

$$= -12184.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 168762 \cdot 6458/4 ((1 + 2(1255^2 - 0^2)/(6458^2)) / (1 + (4 \cdot 0)/(3 \cdot 6458)) - 4 \cdot 1120/6458)$$

$$= 104050.7 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 688 \cdot 1256/(2 \cdot 15) - 104051/(\pi \cdot 1256^2 \cdot 15)$$

$$= 15.07 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$



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$$= 688 \cdot 1256 / (2 \cdot 15) + 104051 / (\pi \cdot 1256^2 \cdot 15)$$

$$= 42.54 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 688 \cdot 1256 / (2 \cdot 25) - 12185 / (0.11 \cdot \pi \cdot 1256^2 \cdot 25)$$

$$= 26.34 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 688 \cdot 1256 / (2 \cdot 25) + 12185 / (0.19 \cdot \pi \cdot 1256^2 \cdot 25)$$

$$= 12.27 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 168762 (6458 - 2 \cdot 1120) / (6458 + (4 \cdot 0 / 3))$$

$$= 110225.7 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 110226 / (1256 \cdot 25)$$

$$= 40.32 \text{ N./mm}^2$$

Decay Length (4.15.20) [$x1, x2$]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1256 \cdot 25)}$$

$$= 138.189 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{(1256 \cdot 25)}, 2 \cdot 1120)$$

$$= 676.38 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma6, r$]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 168762 \cdot 0.1 / (676(25 + 1 \cdot 20))$$

$$= -4.13 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [$\sigma7, r^*$]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -168762 / (4(25 + 1 \cdot 20)676) -$$

$$12 \cdot 0.044 \cdot 168762 \cdot 1256 / (6458(25 + 1 \cdot 20)^2)$$

$$= -98.05 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	52.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	250.0	N./mm ²
Height of Web at Center	Hw,c	680.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1243 \cdot 25)} + 700)^3) / 12$$

$$= 126.953 \text{ cm}^4$$



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Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 127 + 244 (277 - 12.5)^2$$

$$= 170367.719 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	975.0	25.0	12.5	243.7	304687.0	0.170E+06
Wearplate	700.0	20.0	35.0	140.0	489999.9	0.819E+05
Web	20.0	610.0	350.0	122.0	4269999.5	0.444E+05
BasePlate	425.0	52.0	681.0	221.0	15050100.0	0.362E+06
Totals	...	...	...	726.7	20114788.0	0.658E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 20114788/727$$

$$= 276.777 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 168762$$

$$= 34346.7070 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 6.9738 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 149.9973 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1950 - 1243 \cdot \sin(120/2) / 1.05$$

$$= 922.047 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 31669.9121 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 13.0593 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 166.6637 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(168762 + 1536)425 / (4 \cdot 2350 \cdot 167))}$$

$$= 36.867 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1240 + 28 + 20)$$

$$= 2230.881 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2231 / (5 - 1)$$

$$= 557.720 \text{ mm}$$



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 558 \cdot 425 / 2$$

$$= 1185.156 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 170298 / (2350 \cdot 425)$$

$$= 17.051 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1185 \cdot 17.1$$

$$= 20208.203 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 76 + 55.8$$

$$= 131.772 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 20208 / 132$$

$$= 15.039 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	400.0	...	80.0	...	0.107E+05

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 400 - 20$$

$$= 380.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 132$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (10667 / 132) }$$

$$= 89.971 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1950 - \cos(120 / 2) (1240 + 28 + 20) - 52$$

$$= 1254.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199943008 / 250) }$$

$$= 125.651$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 1254/90$$

$$= 13.938$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 313 / (2 \cdot 2350) \cdot 558 \cdot 1254 / 2$$

$$= 23.253 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($13.9 < 126$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (13.9)^2 / (2 \cdot 126^2)) 250 / (5/3 + 3 \cdot (13.9) / (8 \cdot 126) - (13.9^3) / (8 \cdot 126^3))$$

$$= 145.5 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 15/145 + (23.3 \cdot 200 / 106666664) / 167$$

$$= 0.106$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	380.0	0.0	76.0	0.0	0.107E+05
Web	557.7	20.0	0.0	111.5	0.0	37.2
Totals	...	...	...	187.5	...	0.107E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/188$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 1950 - \cos(1288/2) (52 + 0 + 0) - 0$$

$$= 610.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (10703/188) }$$

$$= 75.542 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 610/75.5$$

$$= 8.075$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 313 / (2 \cdot 2350)$$

$$= 0.066 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.066 \cdot 558 \cdot 610$$

$$= 22.622 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$



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$$= 22.6/535126$$

$$= 0.415 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.07 < 126$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.07)^2 / (2 \cdot 126^2)) 250 / (5/3 + 3 \cdot (8.07) / (8 \cdot 126) - (8.07^3) / (8 \cdot 126^3))$$

$$= 147.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 21.1/148 + (22.6 \cdot 200/10703) / 167$$

$$= 0.146$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	9.7967 cm ²
Saddle Load QO (Weight)	QO	167337.9 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	181.4 Kgf
Maximum Transverse Force	Ft	1128.8 Kgf
Maximum Longitudinal Force	F1	312.5 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F_1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 313 / (9.8 \cdot 8)$$

$$= 0.4 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1129 / (9.8 \cdot 8)$$

$$= 1.4 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 313 / (103 \cdot 8)$$

$$= 0.0371 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 1950 \cdot 1129 + 0$$

$$= 2201.14 \text{ Kg-m.}$$

Eccentricity (e):

$$= R_{mom} / QO$$

$$= 2201/167338$$

$$= 13.15 \text{ mm} < B_{plen}/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$



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Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.037, 0]
 = 0.0371 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1255.50	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	6458.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1120.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	700.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1950.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		94175.70	Kgf
Pressure used in Analysis	P	688.306	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	21.14	234.00
Long. Stress at Bottom of Midspan	36.47	234.00
Long. Stress at Top of Saddles	22.34	234.00
Long. Stress at Bottom of Saddles	14.48	234.00
Tangential Shear in Shell	22.50	138.58
Circ. Stress at Horn of Saddle	54.72	350.99
Circ. Compressive Stress in Shell	2.31	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = Ftr(Ft/Num of Saddles + Z Force Load) • B / E
 = 3 (2258/2 + 0) • 1950/2231
 = 2959.9 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
 = max(Fl,wind; Sum of X Forces) • B / Ls
 = max(313, 0) • 1950/3360
 = 181.4 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 91216 + max(181, 2960, 0, 0)



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= 94175.7 Kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 766 (0 + $\pi/4$ (2.54 • 1.2)²)
 = 6129.442 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	95711.59	Kgf
Transverse Shear Load	1128.77	Kgf
Longitudinal Shear Load	312.52	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(2480/2 + 28 + 20)sin(120/2), 2•1256)
 = 2230.882 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0443	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0364			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -94176•1120[1 - (1 - 1120/6458 + (1255²-0²)/
 (2•1120•6458))/(1+(4•0)/(3•6458))]
 = -6799.5 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L)) - 4a/L)
 = 94176•6458/4((1+2(1255²-0²)/(6458²))/(1+(4•0)/
 (3•6458)) - 4•1120/6458)
 = 58064.3 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)
 = 688 • 1256/(2•15) - 58064/(π •1256²•15)
 = 21.14 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 688 • 1256/(2 • 15) + 58064/(π • 1256² • 15)
 = 36.47 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 688•1256/(2•25)--6800/(0.11• π •1256²•25)
 = 22.34 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 688•1256/(2•25)+-6800/(0.19• π •1256²•25)
 = 14.48 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))
 = 94176 (6458 - 2 • 1120)/(6458 + (4 • 0/3))
 = 61510.2 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 61510 / (1256 \cdot 25)$$

$$= 22.50 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1256 \cdot 25)}$$

$$= 138.189 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(400 + 1.56 \cdot \sqrt{(1256 \cdot 25)}, 2 \cdot 1120)$$

$$= 676.38 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 94176 \cdot 0.1 / (676(25 + 1 \cdot 20))$$

$$= -2.31 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -94176 / (4(25 + 1 \cdot 20)676) - 12 \cdot 0.044 \cdot 94176 \cdot 1256 / (6458(25 + 1 \cdot 20)^2)$$

$$= -54.72 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	52.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	250.0	N./mm ²
Height of Web at Center	Hw,c	680.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1243 \cdot 25)} + 700)25^3) / 12$$

$$= 126.953 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 127 + 244(277 - 12.5)^2$$

$$= 170367.719 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	975.0	25.0	12.5	243.7	304687.0	0.170E+06
Wearplate	700.0	20.0	35.0	140.0	489999.9	0.819E+05
Web	20.0	610.0	350.0	122.0	4269999.5	0.444E+05
BasePlate	425.0	52.0	681.0	221.0	15050100.0	0.362E+06
Totals	...	...	...	726.7	20114788.0	0.658E+06

Distance to Centroid [C1]:



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$$= AY / A$$

$$= 20114788/727$$

$$= 276.777 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 94176$$

$$= 19166.7949 \text{ Kgf}$$

Tension Stress, $St = (Fh/As) = 3.8916 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 149.9973 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1950 - 1243 \cdot \sin(120/2) / 1.05$$

$$= 922.047 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 17673.0391 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 7.2876 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 166.6637 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(94176 + 1536)425 / (4 \cdot 2350 \cdot 167)}$$

$$= 27.639 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1240 + 28 + 20)$$

$$= 2230.881 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2231 / (5 - 1)$$

$$= 557.720 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{\text{pwid}} / 2$$

$$= 558 \cdot 425/2$$

$$= 1185.156 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 95712 / (2350 \cdot 425)$$

$$= 9.583 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1185 \cdot 9.58$$

$$= 11357.512 \text{ Kgf}$$

Area of the Rib and Web [Ar]:



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= Rib Area + Web Area
 = 76 + 55.8
 = 131.772 cm²

Compressive Stress [Sc]:
 = P/Ar
 = 11358/132
 = 8.453 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	400.0	...	80.0	...	0.107E+05

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 400 - 20
 = 380.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/132
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 400/2
 = 200.000 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{10667/132}$
 = 89.971 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1950 - cos(120/2)(1240 + 28 + 20) - 52
 = 1254.000 mm

Intermediate Term [Cc]:
 = $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 199943008/250)}$
 = 125.651

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1254/90
 = 13.938

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 313 / (2 • 2350) • 558 • 1254/2
 = 23.253 Kg-m.

Compressive Allowable, KL/r < Cc (13.9 < 126) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (13.9)^2 / (2 \cdot 126^2)) 250 /$
 $(5/3 + 3 \cdot (13.9) / (8 \cdot 126) - (13.9^3) / (8 \cdot 126^3))$
 = 145.5 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I) / Sba



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$$= 8.45/145 + (23.3 \cdot 200/106666664)/167$$

$$= 0.061$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	380.0	0.0	76.0	0.0	0.107E+05
Web	557.7	20.0	0.0	111.5	0.0	37.2
Totals	...	...	...	187.5	...	0.107E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/188$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1950 - \cos(1288/2)(52 + 0 + 0) - 0$$

$$= 610.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(10703/188)}$$

$$= 75.542 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 610/75.5$$

$$= 8.075$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 313/(2 \cdot 2350)$$

$$= 0.066 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.066 \cdot 558 \cdot 610$$

$$= 22.622 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 22.6/535126$$

$$= 0.415 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.07 < 126$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2/(2 \cdot C_c^2))F_y / (5/3 + 3 \cdot (KL/r)/(8 \cdot C_c) - (KL/r)^3/(8 \cdot C_c^3))$$

$$= (1 - (8.07)^2/(2 \cdot 126^2))250 / (5/3 + 3 \cdot (8.07)/(8 \cdot 126) - (8.07)^3/(8 \cdot 126^3))$$

$$= 147.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 11.9/148 + (22.6 \cdot 200/10703)/167$$

$$= 0.083$$

Input Data for Base Plate Bolting Calculations:



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Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	9.7967 cm ²
Saddle Load Q0 (Weight)	Q0	92751.7 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	181.4 Kg
Maximum Transverse Force	Ft	1128.8 Kg
Maximum Longitudinal Force	F1	312.5 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 313 / (9.8 \cdot 8)$$

$$= 0.4 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1129 / (9.8 \cdot 8)$$

$$= 1.4 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (Q0 > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 313 / (103 \cdot 8)$$

$$= 0.0371 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1950 \cdot 1129 + 0$$

$$= 2201.14 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 2201 / 92752$$

$$= 23.73 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$$

$$= \max[0, 0.037, 0]$$

$$= 0.0371 \text{ cm}^2$$

Review notes about nozzle loadings and supports in the Warnings report.

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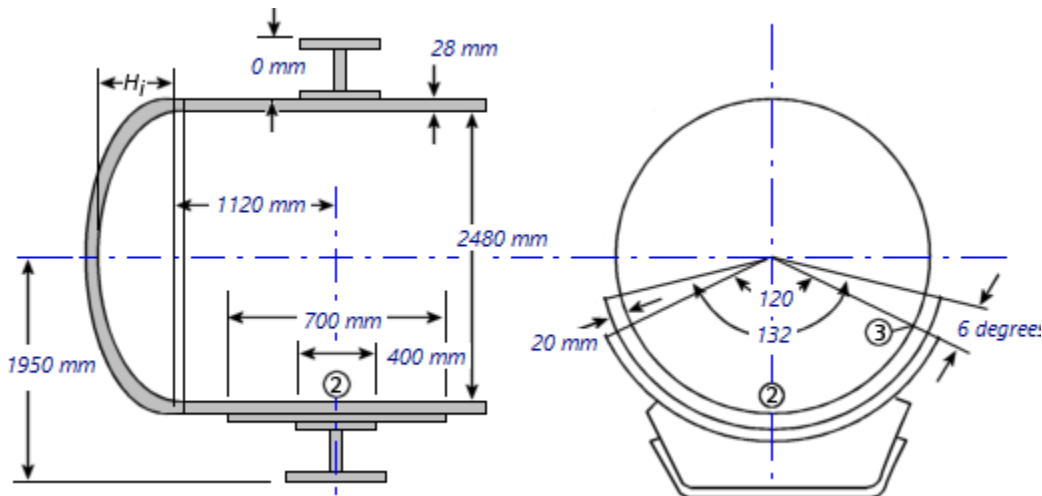
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	1255.50	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	6458.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1120.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	700.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1950.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	283126.53	Kgf
Pressure used in Analysis	P	-103.419 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-27.37	-176.60
Long. Stress at Top of Midspan	27.37	138.00
Long. Stress at Bottom of Midspan	18.72	138.00
Long. Stress at Top of Saddles	12.59	138.00
Long. Stress at Bottom of Saddles	-11.02	-210.96
Long. Stress at Bottom of Saddles	11.02	138.00
Tangential Shear in Shell	67.64	110.40



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Circ. Stress at Horn of Saddle	164.50	172.50
Circ. Compressive Stress in Shell	6.94	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (4515/2 + 35467) \cdot 1950/2231$
 $= 98923.5 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl, wind; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(625, 12150) \cdot 1950/3360$
 $= 7051.3 \text{ Kg}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl, eq; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(15264, 12150) \cdot 1950/3360$
 $= 8858.4 \text{ Kg}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (15264/2 + 35467) \cdot 1950/2231$
 $= 113016.4 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$
 $= 170110 + \max(7051, 98923, 8858, 113016)$
 $= 283126.5 \text{ Kg}$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform \text{ Area} + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(2.54 \cdot 1.2)^2)$
 $= 6129.442 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	284662.47	Kgf
Transverse Shear Load	43098.49	Kgf
Longitudinal Shear Load	15263.65	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(2480/2 + 28 + 20) \sin(120/2), 2 \cdot 1256)$
 $= 2230.882 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0443	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0364			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -283127 \cdot 1120 [1 - (1 - 1120/6458 + (1255^2 - 0^2) / (2 \cdot 1120 \cdot 6458)) / (1 + (4 \cdot 0) / (3 \cdot 6458))]$
 $= -20441.8 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:



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$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L) \\
 &= 283127 \cdot 6458 / 4 \cdot ((1 + 2(1255^2 - 0^2) / (6458^2)) / (1 + (4 \cdot 0) / (3 \cdot 6458)) - 4 \cdot 1120 / 6458) \\
 &= 174562.5 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= -103 \cdot 1256 / (2 \cdot 15) - 174562 / (\pi \cdot 1256^2 \cdot 15) \\
 &= -27.37 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= -103 \cdot 1256 / (2 \cdot 15) + 174562 / (\pi \cdot 1256^2 \cdot 15) \\
 &= 18.72 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -103 \cdot 1256 / (2 \cdot 25) - 20442 / (0.11 \cdot \pi \cdot 1256^2 \cdot 25) \\
 &= 12.59 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -103 \cdot 1256 / (2 \cdot 25) + 20442 / (0.19 \cdot \pi \cdot 1256^2 \cdot 25) \\
 &= -11.02 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 283127 (6458 - 2 \cdot 1120) / (6458 + (4 \cdot 0 / 3)) \\
 &= 184922.2 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.17 \cdot 184922 / (1256 \cdot 25) \\
 &= 67.64 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(1256 \cdot 25)} \\
 &= 138.189 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(400 + 1.56 \cdot \sqrt{(1256 \cdot 25)}, 2 \cdot 1120) \\
 &= 676.38 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$\begin{aligned}
 &= \min(Sr / S, 1) \\
 &= \min(138 / 138, 1) \\
 &= 1.0000
 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.76 \cdot 283127 \cdot 0.1 / (676(25 + 1 \cdot 20)) \\
 &= -6.94 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r_{*}]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2) \\
 &= -283127 / (4(25 + 1 \cdot 20)676) - 12 \cdot 0.044 \cdot 283127 \cdot 1256 / (6458(25 + 1 \cdot 20)^2) \\
 &= -164.50 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	52.0000	mm
Baseplate Width	Bpwid	425.0000	mm



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Number of Ribs (inc. outside ribs)	Nribs	5
Rib Thickness	Ribtk	20.0000 mm
Web Thickness	Webtk	20.0000 mm
Web Location	Webloc	Center
Saddle Design Temperature		21.1 °C
Saddle Yield Stress	Sy	222.2 N./mm ²
Height of Web at Center	Hw,c	680.0 mm
Friction Coefficient	mu	0.300

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(1243 \cdot 25)} + 700)25^3)/12$
 $= 126.953 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 127 + 244(277 - 12.5)^2$
 $= 170367.719 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	975.0	25.0	12.5	243.7	304687.0	0.170E+06
Wearplate	700.0	20.0	35.0	140.0	489999.9	0.819E+05
Web	20.0	610.0	350.0	122.0	4269999.5	0.444E+05
BasePlate	425.0	52.0	681.0	221.0	15050100.0	0.362E+06
Totals	...	...	...	726.7	20114788.0	0.658E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 20114788/727$
 $= 276.777 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 120/2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$
 $= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.2 \cdot 283127$
 $= 57622.3906 \text{ Kg}$

Tension Stress, St = (Fh/As) = 11.6996 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 133.3187 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1950 - 1243 \cdot \sin(120/2) / 1.05$
 $= 922.047 \text{ mm}$

Bending Moment, M = Fh * d = 53131.6133 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 21.9093 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 148.1319 N./mm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + \text{SaddleWt})\text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$



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$$= \sqrt{3(283127 + 1536)425 / (4 \cdot 2350 \cdot 148)}$$

$$= 50.559 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1240 + 28 + 20)$$

$$= 2230.881 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2231 / (5 - 1)$$

$$= 557.720 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 558 \cdot 425 / 2$$

$$= 1185.156 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 284662 / (2350 \cdot 425)$$

$$= 28.502 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 1185 \cdot 28.5$$

$$= 33779.160 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 76 + 55.8$$

$$= 131.772 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 33779 / 132$$

$$= 25.139 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	400.0	...	80.0	...	0.107E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 400 - 20$$

$$= 380.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 132$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400 / 2$$

$$= 200.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$



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$$= \sqrt{(10667/132)}$$

$$= 89.971 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1950 - \cos(120/2) (1240 + 28 + 20) - 52$$

$$= 1254.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 222)}$$

$$= 133.279$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1254/90$$

$$= 13.938$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 15264 / (2 \cdot 2350) \cdot 558 \cdot 1254 / 2$$

$$= 1135.673 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($13.9 < 133$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (13.9)^2 / (2 \cdot 133^2)) 222 / (5/3 + 3 \cdot (13.9) / (8 \cdot 133) - (13.9)^3 / (8 \cdot 133^3))$$

$$= 129.6 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 25.1/130 + (1136 \cdot 200/106666664) / 148$$

$$= 0.335$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	380.0	0.0	76.0	0.0	0.107E+05
Web	557.7	20.0	0.0	111.5	0.0	37.2
Totals	...	...	...	187.5	...	0.107E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/188$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 400/2$$

$$= 200.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1950 - \cos(1288/2) (52 + 0 + 0) - 0$$

$$= 610.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(10703/188)}$$

$$= 75.542 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$



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$$= 1 \cdot 610/75.5$$

$$= 8.075$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 15264 / (2 \cdot 2350)$$

$$= 3.248 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 3.25 \cdot 558 \cdot 610$$

$$= 1104.881 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1105/535126$$

$$= 20.248 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.07 < 133$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.07)^2 / (2 \cdot 133^2)) 222 / (5/3 + 3 \cdot (8.07) / (8 \cdot 133) - (8.07^3) / (8 \cdot 133^3))$$

$$= 131.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 35.3/131 + (1105 \cdot 200/10703) / 148$$

$$= 0.406$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1255.50	mm
Shell Thickness used in this Case	t	28.000	mm
Stiffened Vessel Length per 4.15.6	L	6458.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1120.00	mm
Saddle Width	b1 or b	400.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	700.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1950.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		140880.00	Kgf
Pressure used in Analysis	P	-103.419	KPa.

Horizontal Vessel Analysis Results:		Actual	Allowable
		N./mm ²	N./mm ²
Long. Stress at Top	of Midspan	-15.80	-176.60
Long. Stress at Top	of Midspan	15.80	138.00
Long. Stress at Bottom	of Midspan	7.14	138.00



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Long. Stress at Top of Saddles	4.96	138.00
Long. Stress at Bottom of Saddles	-6.79	-210.96
Long. Stress at Bottom of Saddles	6.79	138.00
Tangential Shear in Shell	33.66	110.40
Circ. Stress at Horn of Saddle	81.85	172.50
Circ. Compressive Stress in Shell	3.45	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (4515/2 + 8867) \cdot 1950/2231$
 $= 29170.8 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(625, 12150) \cdot 1950/3360$
 $= 7051.3 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(15264, 12150) \cdot 1950/3360$
 $= 8858.4 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (15264/2 + 8867) \cdot 1950/2231$
 $= 43263.7 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 97616 + \max(7051, 29171, 8858, 43264)$
 $= 140880.0 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(2.54 \cdot 1.2)^2)$
 $= 6129.442 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	142415.91	Kgf
Transverse Shear Load	16498.49	Kgf
Longitudinal Shear Load	15263.65	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(2480/2 + 28 + 20) \sin(120/2), 2 \cdot 1256)$
 $= 2230.882 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0443	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0364			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h m^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$



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$$= -140880 \cdot 1120 [1 - (1 - 1120/6458 + (1255^2 - 0^2) / (2 \cdot 1120 \cdot 6458)) / (1 + (4 \cdot 0) / (3 \cdot 6458))] \\ = -10171.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L) \\ = 140880 \cdot 6458 / 4 \cdot ((1 + 2(1255^2 - 0^2) / (6458^2)) / (1 + (4 \cdot 0) / (3 \cdot 6458)) - 4 \cdot 1120 / 6458) \\ = 86860.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\ = -103 \cdot 1256 / (2 \cdot 15) - 86860 / (\pi \cdot 1256^2 \cdot 15) \\ = -15.80 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\ = -103 \cdot 1256 / (2 \cdot 15) + 86860 / (\pi \cdot 1256^2 \cdot 15) \\ = 7.14 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = -103 \cdot 1256 / (2 \cdot 25) - 10172 / (0.11 \cdot \pi \cdot 1256^2 \cdot 25) \\ = 4.96 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = -103 \cdot 1256 / (2 \cdot 25) + 10172 / (0.19 \cdot \pi \cdot 1256^2 \cdot 25) \\ = -6.79 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\ = 140880 (6458 - 2 \cdot 1120) / (6458 + (4 \cdot 0 / 3)) \\ = 92014.9 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t) \\ = 1.17 \cdot 92015 / (1256 \cdot 25) \\ = 33.66 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)} \\ = 0.78 \cdot \sqrt{(1256 \cdot 25)} \\ = 138.189 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\ = \min(400 + 1.56 \cdot \sqrt{(1256 \cdot 25)}, 2 \cdot 1120) \\ = 676.38 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1) \\ = \min(138/138, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\ = -0.76 \cdot 140880 \cdot 0.1 / (676(25 + 1 \cdot 20)) \\ = -3.45 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2) \\ = -140880 / (4(25 + 1 \cdot 20)676) - 12 \cdot 0.044 \cdot 140880 \cdot 1256 / (6458(25 + 1 \cdot 20)^2) \\ = -81.85 \text{ N./mm}^2$$



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Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	52.0000	mm
Baseplate Width	Bpwid	425.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	222.2	N./mm ²
Height of Web at Center	Hw,c	680.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(1243 \cdot 25)} + 700)^3) / 12$
 $= 126.953 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 127 + 244 (277 - 12.5)^2$
 $= 170367.719 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	975.0	25.0	12.5	243.7	304687.0	0.170E+06
Wearplate	700.0	20.0	35.0	140.0	489999.9	0.819E+05
Web	20.0	610.0	350.0	122.0	4269999.5	0.444E+05
BasePlate	425.0	52.0	681.0	221.0	15050100.0	0.362E+06
Totals	...	...	...	726.7	20114788.0	0.658E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 20114788 / 727$
 $= 276.777 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.2 \cdot 140880$
 $= 28672.1367 \text{ Kg}$

Tension Stress, St = (Fh/As) = 5.8216 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 133.3187 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$
 $= 1950 - 1243 \cdot \sin(120 / 2) / 1.05$
 $= 922.047 \text{ mm}$

Bending Moment, M = Fh * d = 26437.5859 Kg-m.



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Bending Stress, $S_b = (M * C_1 / I) = 10.9018 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 * \text{Yield Str} = 148.1319 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{ (3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{ (3(140880 + 1536) 425 / (4 * 2350 * 148))}$
 $= 35.761 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2)(\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120/2)(1240 + 28 + 20)$
 $= 2230.881 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 2231 / (5 - 1)$
 $= 557.720 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 558 * 425 / 2$
 $= 1185.156 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 142416 / (2350 * 425)$
 $= 14.259 \text{ Kgf/cm}^2$

Axial Load [P]:
 $= A_p * B_p$
 $= 1185 * 14.3$
 $= 16899.627 \text{ Kgf}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 76 + 55.8$
 $= 131.772 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 16900 / 132$
 $= 12.577 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	400.0	...	80.0	...	0.107E+05

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 400 - 20$
 $= 380.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 132$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



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= Saddle Width / 2
 = 400/2
 = 200.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{10667/132}$
 = 89.971 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1950 - cos(120/2)(1240 + 28 + 20) - 52
 = 1254.000 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 199943008/222}$
 = 133.279

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1254/90
 = 13.938

Bending Moment [Rm]:
 $= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$
 $= 15264 / (2 \cdot 2350) \cdot 558 \cdot 1254/2$
 = 1135.673 Kg-m.

Compressive Allowable, KL/r < Cc (13.9 < 133) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (13.9)^2 / (2 \cdot 133^2)) 222 / (5/3 + 3 \cdot (13.9) / (8 \cdot 133) - (13.9)^3 / (8 \cdot 133^3))$
 = 129.6 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= Sc/Sca + (Rm \cdot C1 / I) / Sba$
 $= 12.6/130 + (1136 \cdot 200/106666664) / 148$
 = 0.238

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	380.0	0.0	76.0	0.0	0.107E+05
Web	557.7	20.0	0.0	111.5	0.0	37.2
Totals	...	...	...	187.5	...	0.107E+05

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0/188$
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 400/2
 = 200.000 mm

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = 1950 - cos(1288/2)(52 + 0 + 0) - 0
 = 610.000 mm

Radius of Gyration [r]:



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$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{10703/188}$$

$$= 75.542 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 610/75.5$$

$$= 8.075$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 15264 / (2 \cdot 2350)$$

$$= 3.248 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 3.25 \cdot 558 \cdot 610$$

$$= 1104.881 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1105/535126$$

$$= 20.248 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($8.07 < 133$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.07)^2 / (2 \cdot 133^2)) 222 / (5/3 + 3 \cdot (8.07) / (8 \cdot 133) - (8.07^3) / (8 \cdot 133^3))$$

$$= 131.3 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 17.7/131 + (1105 \cdot 200/10703) / 148$$

$$= 0.271$$

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N3-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	1056.657	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2480.00	mm
Design Length of Section	L	1500.0000	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1176.17	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

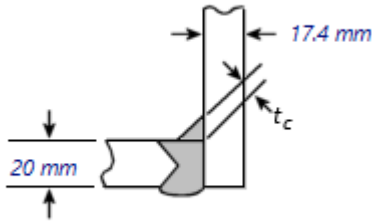
Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		24.0157	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	17.4498	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	16.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1020.0001	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		205.0006	mm
ASME Code Weld Type per UW-16		e	
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



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UW-16 (e)

Reinforcement CALCULATION, Description: N3-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.016 in.
Actual Thickness Used in Calculation	0.687 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1057 \cdot 1243 / (138 \cdot 1 - 0.6 \cdot 1057)$
 $= 9.5622 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1057 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 1057)$
 $= 2.3284 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.6380 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1162.1984	mm
Parallel to Vessel Wall, opening length	d	581.0992	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		42.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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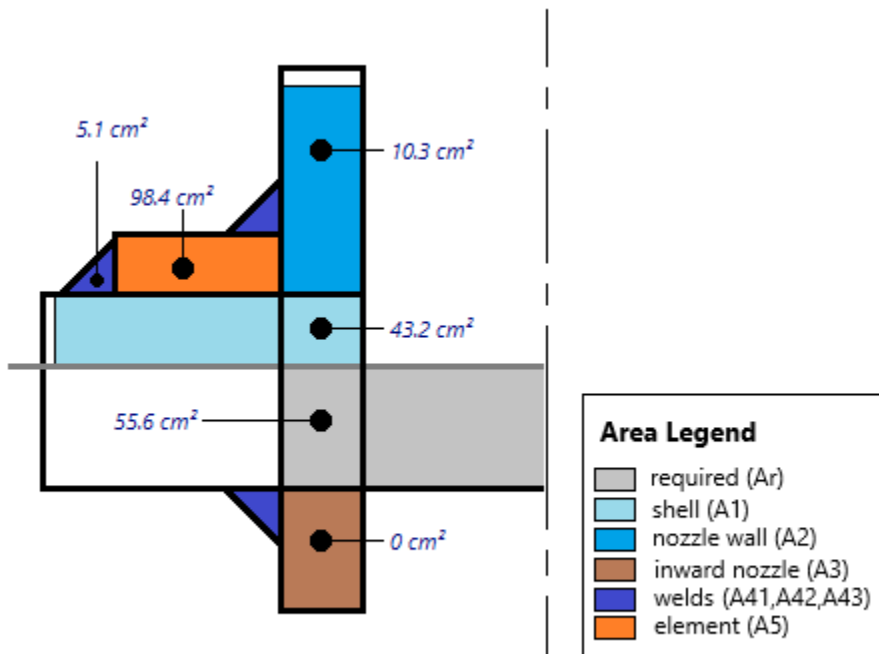


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	55.566	18.934	NA
Area in Shell	A1	43.221	60.919	NA
Area in Nozzle Wall	A2	10.303	10.890	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	5.120	5.120	NA
Area in Element	A5	98.400	98.400	NA
TOTAL AREA AVAILABLE	Atot	157.044	175.329	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 609.9988 24.0000 mm
 Based on given Pad Diameter: 1020.0001 0.0000 mm
 Based on Shell or Nozzle Thickness: 609.9988 17.4498 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (581 \cdot 9.56 \cdot 1 + 2 \cdot 14.4 \cdot 9.56 \cdot 1 \cdot (1 - 1))$
 $= 55.566 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 581(1 \cdot 17 - 1 \cdot 9.56) - 2 \cdot 14.4$
 $(1 \cdot 17 - 1 \cdot 9.56) \cdot (1 - 1)$



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$$= 43.221 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 42.5) \cdot (14.4 - 2.33) \cdot 1$$

$$= 10.303 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 16^2 \cdot 1 + (0)^2 \cdot 1 + 16^2 \cdot 1$$

$$= 5.120 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (1020 - 610) \cdot 24 \cdot 1$$

$$= 98.400 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.3284 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 12.5622 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 12.5622 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.9320 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(12.6, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.33, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 17.4498 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 26.8,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 318.2 N./mm ²	Passed
Occasional	: 10.4,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 12.6,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 17.4, tr = 2.33, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.16, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 17.4, tr = 2.33, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.16, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
--	--------



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Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 17.4, tr = 2.33, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E^* / (tg - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 20, tr = 9.56, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E^* / (tg - c) = 0.56$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -33 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 17.4, tr = 2.33, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E^* / (tg - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -33 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -33 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -46 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1056.66 / 1960.01 = 0.539$

Weld Size Calculations, Description: N3-Tube Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 14.4498 mm
 Intermediate Calc. for pad/shell Welds TminPad 17.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$10.1149 = 0.7 * t_{min}$	$11.3120 = 0.7 * W_o$ mm
Pad Weld	$8.5000 = 0.5 * t_{minPad}$	$11.3120 = 0.7 * W_p$ mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1848 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 38.0952 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 458.0952 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 * t_{nom} / R_f * (1 - R_f / R_o)$) 2.945 %



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This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 581.10 mm

Input Echo, WRC297 Item 1, Description: N3-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2480.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	609.999	mm
Nozzle Thickness	17.4498	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	1020.000	mm
Thickness of Reinforcing Pad	24.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1056.66	KPa.
Radial Load	P	-4589.03	Kgf
Circumferential Shear	Vc	3059.35	Kgf
Longitudinal Shear	Vl	3059.35	Kgf
Circumferential Moment	Mc	5137.73	Kg-m.
Longitudinal Moment	Ml	5137.73	Kg-m.
Torsional Moment	Mt	7339.62	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 610.0 + 2 \cdot 1.65 \cdot \sqrt{1263.5(20.0 - 3.0)}$
 $= 1093.643 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2527.000	mm
Nozzle Outside Diameter	(d) =	609.999	mm
Vessel Thickness used	(T) =	41.000	mm
Nozzle Thickness used	(t) =	14.450	mm
T / t	=	2.837	



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$$d / t = 42.215$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.895$$

Nr/P	=	0.058
Mr/P	=	0.073
MØ/P	=	0.057
NØ/P	=	0.186
MrD/Mc	=	0.157
NrDL/Mc	=	0.094
MØd/Mc	=	0.150
NØDL/Mc	=	0.242
MrD/Ml	=	0.093
NrDL/Ml	=	0.062
MØD/Ml	=	0.081
NØDL/Ml	=	0.357

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-17	-17	17	17
Outplane Bending (ML)	-23	23	23	-23
Normal Pressure Stress	31	32	31	32
Outplane Stress Summary	4	34	87	22

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	11	-11	11	-11
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-27	27	27	-27
Inplane Pressure Stress	15	15	15	15
Inplane Stress Summary	-1	29	59	-18

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	9	36	87	40



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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-44	44	44	-44
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	15	15	15	15
Outplane Stress Summary	-26	43	85	-20

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	11	-11	11	-11
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-46	46	46	-46
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	31	32	31	32
Inplane Stress Summary	-5	63	95	-19

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	2	2	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	26	64	96	23

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-17	-17	17	17
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	24	26	24	26
Hoop Stress Summary	12	13	47	48

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



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(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	57	-57	57	-57
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-12	-12	12	12
Axial Bending (ML)	-146	146	146	-146
Axial Pressure Stress	12	12	12	12
Axial Stress Summary	-87	90	230	-176

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8
Shear Stress Summary	10	10	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	102	91	230	225

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	24	26	24	26
Hoop Stress Summary	17	19	41	42

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	57	-57	57	-57
Axial Membrane (Mc)	-12	-12	12	12
Axial Bending (Mc)	-261	261	261	-261
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	12	12	12	12
Axial Stress Summary	-202	205	345	-291

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2



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Shear due to Torsion	8	8	8	8
Shear Stress Summary	6	6	10	10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	220	205	345	335

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 610.0 + 2 \cdot 1.65 \cdot \sqrt{1251.5(20.0 - 3.0)}$$

$$= 1091.341 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2503.000 mm
Nozzle Outside Diameter	(d) =	1020.000 mm
Vessel Thickness used	(T) =	17.000 mm
Nozzle Thickness used	(t) =	219.450 mm
T / t	=	0.077
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.945$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.046
Mr/P	=	0.071
M0/P	=	0.019
N0/P	=	0.027
MrD/Mc	=	0.241
NrDL/Mc	=	0.100
M0d/Mc	=	0.072
N0DL/Mc	=	0.052
MrD/Ml	=	0.050
NrDL/Ml	=	0.049
M0D/Ml	=	0.015
N0DL/Ml	=	0.026

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-15	15	15	-15
Normal Pressure Stress	76	77	76	77
Outplane Stress Summary	78	75	119	52

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7



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Inplane Bending (P)	65	-65	65	-65
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-51	51	51	-51
Inplane Pressure Stress	38	38	38	38
Inplane Stress Summary	51	22	170	-63

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane	Shear	(Vc)	1	1	-1	-1
Outplane	Shear	(Vl)	0	0	0	0
Outplane	Shear	(Mt)	2	2	2	2
Shear Stress Summary			3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	79	75	170	115

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-73	73	73	-73
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	38	38	38	38
Outplane Stress Summary	-21	89	143	-40

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)		7	7	7	7
Inplane Bending (P)		65	-65	65	-65
Inplane Membrane (Mc)		-17	-17	17	17
Inplane Bending (Mc)		-246	246	246	-246
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		76	77	76	77
Inplane Stress Summary		-113	248	413	-210

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			2	2	2	2



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Shear Stress Summary 1 1 3 3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	114	248	413	210

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	31	32	31	32	31	32	31	32
Circ. Pl (SUS)	-13	-13	21	21	-3	-3	5	5
Circ. Q (SUS)	-14	14	32	-32	-35	35	57	-57
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	-2	-2	4	4	-7	-7	15	15
Long. Q (SUS)	-16	16	38	-38	-35	35	53	-53
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	31.0	32.0	31.0	32.0	31.0	32.0	31.0	32.0
Pm+Pl (SUS)	18.0	19.0	52.0	53.0	28.0	29.0	36.0	37.0
Pm+Pl+Q (Total)	8.1	33.8	84.1	40.2	27.2	64.2	93.4	24.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	32.00	138.00	Passed
Pm+Pl (SUS)	53.00	207.00	Passed
Pm+Pl+Q (TOTAL)	93.39	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	76	77	76	77	76	77	76	77
Circ. Pl (SUS)	0	0	8	8	-10	-10	24	24
Circ. Q (SUS)	3	-3	33	-33	-181	181	311	-311
Long. Pm (SUS)	38	38	38	38	38	38	38	38
Long. Pl (SUS)	-1	-1	15	15	-4	-4	12	12
Long. Q (SUS)	14	-14	116	-116	-55	55	91	-91
Shear Pm (SUS)	0	0	0	0	0	0	0	0



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Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	76.0	77.0	76.0	77.0	76.0	77.0	76.0	77.0
Pm+P1 (SUS)	76.0	77.0	84.0	85.0	66.0	67.0	100.0	101.0
Pm+P1+Q (Total)	79.3	74.2	169.0	115.0	115.0	248.0	411.0	210.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	77.00	138.00	Passed
Pm+P1 (SUS)	101.02	207.00	Passed
Pm+P1+Q (TOTAL)	411.03	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	24	26	24	26	24	26	24	26
Circ. P1 (SUS)	-13	-13	21	21	-7	-7	15	15
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. P1 (SUS)	-11	-11	13	13	-11	-11	13	13
Long. Q (SUS)	-89	89	203	-203	-204	204	318	-318
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	24.0	26.0	24.0	26.0	24.0	26.0	24.0	26.0
Pm+P1 (SUS)	11.4	13.3	45.2	47.2	17.2	19.2	39.3	41.2
Pm+P1+Q (Total)	101.0	91.3	228.2	225.3	220.3	205.2	343.3	334.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	26.00	138.00	Passed
Pm+P1 (SUS)	47.18	207.00	Passed
Pm+P1+Q (TOTAL)	343.33	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



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Input, Nozzle Desc: N1-Shell Inlet

From: 60

Pressure for Reinforcement Calculations	P	1056.657	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2480.00	mm
Design Length of Section	L	5600.0000	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3453.75	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

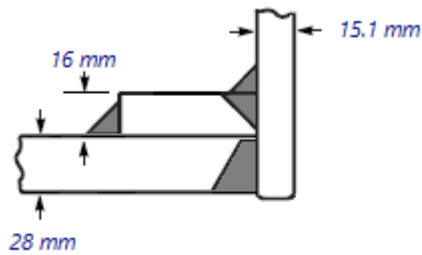
Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		60	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	595.6000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		120.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



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Reinforcement CALCULATION, Description: N1-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 14.000 in.
 Actual Thickness Used in Calculation 0.594 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1057 \cdot 1243 / (138 \cdot 1 - 0.6 \cdot 1057)$
 $= 9.5622 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1057 \cdot 178 / (118 \cdot 1 + 0.4 \cdot 1057)$
 $= 1.5866 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0591 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	662.8496	mm
Parallel to Vessel Wall, opening length	d	331.4248	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		46.2190	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

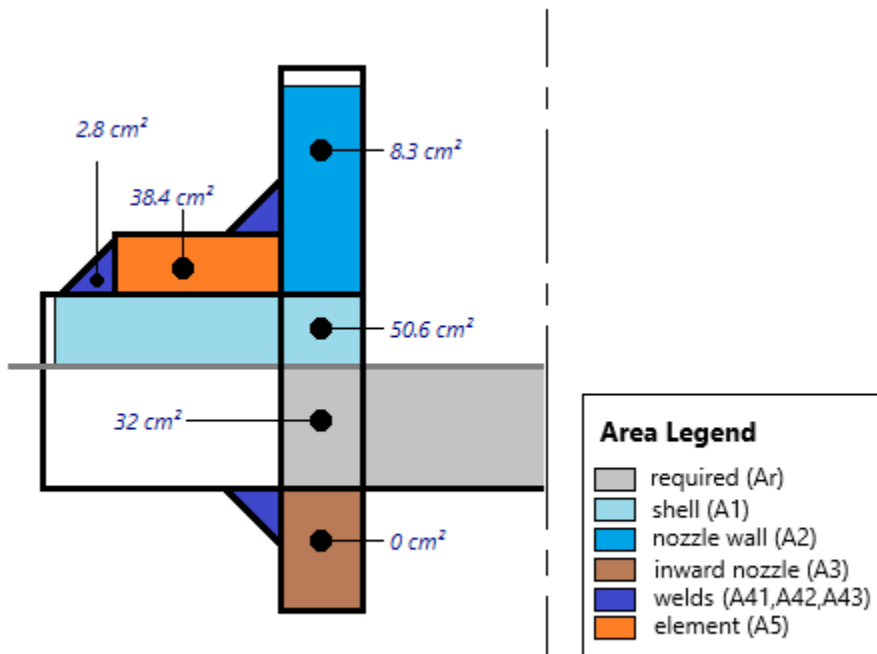


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	32.027	18.908	NA
Area in Shell	A1	50.624	44.956	NA
Area in Nozzle Wall	A2	8.300	8.717	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.815	2.815	NA
Area in Element	A5	38.400	38.400	NA
TOTAL AREA AVAILABLE	Atot	100.139	94.888	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (331 \cdot 9.56 \cdot 1 + 2 \cdot 12.1 \cdot 9.56 \cdot 1 \cdot (1 - 0.86))$$

$$= 32.027 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 331(1 \cdot 25 - 1 \cdot 9.56) - 2 \cdot 12.1$$

$$(1 \cdot 25 - 1 \cdot 9.56) \cdot (1 - 0.86)$$

$$= 50.624 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 46.2) \cdot (12.1 - 1.59) \cdot 0.86$$



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= 8.300 cm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 14^2 \cdot 1$
 = 2.815 cm²

Area Available in Element [A5]:
 = $(\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot fr_4$
 = $(596 - 356) \cdot 16 \cdot 1$
 = 38.400 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.5866 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 12.5622 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 12.5622 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.9320 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[11.3, max(12.6, 4.5)]
 = 11.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(4.59, 11.3)
 = 11.3312 mm

Available Nozzle Neck Thickness = 0.875 * 15.088 = 13.202 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	27.4,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	267.6 N./mm ²	Passed
Occasional	:	7.0,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	15.7,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 13.2, tr = 1.59, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 13.2, tr = 1.59, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Govrn. thk, $t_g = 13.2$, $t_r = 1.59$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 16$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.38$, Temp. Reduction = 55 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 13.2$, $t_r = 1.59$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -48 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-46 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1056.66 / 1960.01 = 0.539$

Weld Size Calculations, Description: N1-Shell Inlet

Intermediate Calc. for nozzle/shell Welds	T_{min}	12.0876 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	16.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$8.0000 = 0.5 * T_{minPad}$	$9.8980 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - t_r)) \cdot Sv)$
 $= \max(0, (32 - 50.6 + 2 \cdot 12.1 \cdot 0.86 \cdot (1 \cdot 25 - 9.56)) \cdot 138)$
 $= \max(0, -21678.49) \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (8.3 + 38.4 + 2.82 - 0 \cdot 0.86) \cdot 138$
 $= 69675.66 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$



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$$= (8.3 + 0 + 0.86 + (5.17)) \cdot 138$$

$$= 20154.83 \text{ Kg}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (8.3 + 0 + 2.82 + 38.4 + (5.17)) \cdot 138$$

$$= 76947.69 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 356 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 32932. \text{ Kg}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 596 \cdot 14 \cdot 0.49 \cdot 138$$

$$= 90311. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 172) \cdot (15.1 - 3) \cdot 0.7 \cdot 118$$

$$= 54935. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 356 \cdot 16 \cdot 0.74 \cdot 138$$

$$= 93063. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 356 \cdot (16 - 0) \cdot 0.74 \cdot 138$$

$$= 93063. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (90311 + 54935) = 145246 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (32932 + 93063 + 93063 + 0) = 219058 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (90311 + 93063 + 0) = 183374 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 145245 Kg, must exceed W = 0 Kg	or W1 = 69675 Kg
Path 2-2 = 219057 Kg, must exceed W = 0 Kg	or W2 = 20154 Kg
Path 3-3 = 183373 Kg, must exceed W = 0 Kg	or W3 = 76947 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2160 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 12.8134 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 340.8134 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center



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of the nozzle to the edge of the diameter limit is: 331.42 mm

Input Echo, WRC297 Item 1, Description: N1-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	3.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	2480.000	mm	
Shell Thickness	28.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	355.600	mm	
Nozzle Thickness	15.0876	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-106 B		
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	595.600	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1056.66	KPa.
Radial Load	P	-2676.93	Kgf
Circumferential Shear	Vc	1784.62	Kgf
Longitudinal Shear	Vl	1784.62	Kgf
Circumferential Moment	Mc	1748.26	Kg-m.
Longitudinal Moment	Ml	1748.26	Kg-m.
Torsional Moment	Mt	2497.50	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 355.6 + 2 \cdot 1.65 \cdot \sqrt{(1263.5(28.0 - 3.0))}$$

$$= 942.105 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2527.000	mm
Nozzle Outside Diameter	(d) =	355.600	mm
Vessel Thickness used	(T) =	41.000	mm
Nozzle Thickness used	(t) =	12.088	mm
T / t	=	3.392	
d / t	=	29.419	



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$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.105$$

Nr/P	=	0.077
Mr/P	=	0.088
M0/P	=	0.090
N0/P	=	0.235
MrD/Mc	=	0.155
NrDL/Mc	=	0.114
M0d/Mc	=	0.202
N0DL/Mc	=	0.173
MrD/M1	=	0.123
NrDL/M1	=	0.097
M0D/M1	=	0.147
N0DL/M1	=	0.346

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-25	25	25	-25
Normal Pressure Stress	31	32	31	32
Outplane Stress Summary	8	43	78	12

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-21	21	21	-21
Inplane Pressure Stress	15	15	15	15
Inplane Stress Summary	1	27	49	-9

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	3	3	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	10	43	78	22

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-34	34	34	-34
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	15	15	15	15

Outplane Stress Summary	-11	40	67	-18
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-26	26	26	-26
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	31	32	31	32

Inplane Stress Summary	11	48	70	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	23	49	73	22

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	18	16	18

Hoop Stress Summary	10	11	30	31
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	2	2	2	2
Axial Bending (P)	53	-53	53	-53
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-15	-15	15	15
Axial Bending (ML)	-147	147	147	-147
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-99	88	227	-174
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	12	12	7	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	112	90	227	206

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	18	16	18
Hoop Stress Summary	15	16	25	26

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	53	-53	53	-53
Axial Membrane (Mc)	-15	-15	15	15
Axial Bending (Mc)	-193	193	193	-193
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-145	134	272	-220

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10



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Shear Stress Summary	7	7	12	12
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	161	134	273	248

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{\text{Rmean}(t - ca)}$
 $= 355.6 + 2 \cdot 1.65 \cdot \sqrt{1255.5(28.0 - 3.0)}$
 $= 940.245 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2511.000 mm
Nozzle Outside Diameter	(d) =	595.600 mm
Vessel Thickness used	(T) =	25.000 mm
Nozzle Thickness used	(t) =	132.088 mm
T / t	=	0.189
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.377$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.065
Mr/P	=	0.128
M0/P	=	0.038
N0/P	=	0.069
MrD/Mc	=	0.281
NrDL/Mc	=	0.093
M0d/Mc	=	0.085
N0DL/Mc	=	0.081
MrD/Ml	=	0.098
NrDL/Ml	=	0.082
M0D/Ml	=	0.029
N0DL/Ml	=	0.090

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	52	53	52	53
Outplane Stress Summary	52	50	76	42

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	32	-32	32	-32
Inplane Membrane (Mc)	0	0	0	0



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Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-27	27	27	-27
Inplane Pressure Stress	26	26	26	26

Inplane Stress Summary	30	19	91	-26
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	52	50	91	69
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-23	23	23	-23
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	26	26	26	26

Outplane Stress Summary	11	39	65	0
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	32	-32	32	-32
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-77	77	77	-77
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	52	53	52	53

Inplane Stress Summary	5	96	168	-49
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
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PV Elite 28 Licensee: SL User
 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 224 of 285
 Nozzle Calcs.: N1-Shell Inlet Nozl: 14 6:07pm Sep 8,2026

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	11	96	168	49

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	31	32	31	32	31	32	31	32
Circ. Pl (SUS)	-6	-6	12	12	-2	-2	4	4
Circ. Q (SUS)	-17	17	33	-33	-18	18	34	-34
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	-1	-1	3	3	-1	-1	7	7
Long. Q (SUS)	-13	13	29	-29	-26	26	42	-42
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	31.0	32.0	31.0	32.0	31.0	32.0	31.0	32.0
Pm+Pl (SUS)	25.0	26.0	43.0	44.0	29.0	30.0	35.0	36.0
Pm+Pl+Q (Total)	9.2	43.5	76.3	22.8	23.8	49.0	70.4	22.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	32.00	138.00	Passed
Pm+Pl (SUS)	44.00	207.00	Passed
Pm+Pl+Q (TOTAL)	76.31	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	52	53	52	53	52	53	52	53
Circ. Pl (SUS)	-2	-2	6	6	-2	-2	6	6
Circ. Q (SUS)	1	-1	17	-17	-45	45	109	-109
Long. Pm (SUS)	26	26	26	26	26	26	26	26
Long. Pl (SUS)	-1	-1	5	5	-1	-1	5	5
Long. Q (SUS)	5	-5	59	-59	-14	14	32	-32
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1



PV Elite 28 Licensee: SL User
 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 225 of 285
 Nozzle Calcs.: N1-Shell Inlet Nozl: 14 6:07pm Sep 8,2026

Pm (SUS)	52.0	53.0	52.0	53.0	52.0	53.0	52.0	53.0
Pm+Pl (SUS)	50.0	51.0	58.0	59.0	50.0	51.0	58.0	59.0
Pm+Pl+Q (Total)	51.0	50.0	90.1	70.0	11.2	96.0	167.0	50.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	53.00	138.00	Passed
Pm+Pl (SUS)	59.00	207.00	Passed
Pm+Pl+Q (TOTAL)	167.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	16	18	16	18	16	18	16	18
Circ. Pl (SUS)	-6	-6	12	12	-1	-1	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-13	-13	17	17	-13	-13	17	17
Long. Q (SUS)	-94	94	200	-200	-140	140	246	-246
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	16.0	18.0	16.0	18.0	16.0	18.0	16.0	18.0
Pm+Pl (SUS)	15.5	17.5	29.0	30.7	20.4	22.4	26.2	27.0
Pm+Pl+Q (Total)	111.6	90.8	225.3	205.6	160.8	135.5	271.6	247.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	18.00	118.00	Passed
Pm+Pl (SUS)	30.70	177.00	Passed
Pm+Pl+Q (TOTAL)	271.58	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



PV Elite 28 Licensee: SL User
 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 226 of 285
 Nozzle Calcs.: Drain Nozl: 15 6:07pm Sep 8,2026

Input, Nozzle Desc: Drain

From: 70

Pressure for Reinforcement Calculations	P	1080.946	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2480.00	mm
Design Length of Section	L	5600.0000	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5878.33	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

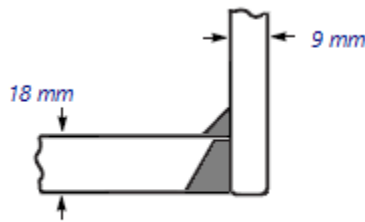
Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		1.9291	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	9.0000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



PV Elite 28 Licensee: SL User
 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 227 of 285
 Nozzle Calcs.: Drain Nozl: 15 6:07pm Sep 8,2026



Reinforcement CALCULATION, Description: Drain

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 1.929 in.
 Actual Thickness Used in Calculation 0.354 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1081 \cdot 1243 / (138 \cdot 1 - 0.6 \cdot 1081)$
 $= 9.7831 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1081 \cdot 27.5 / (138 \cdot 1 - 0.6 \cdot 1081)$
 $= 0.2164 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4019 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	109.9998	mm
Parallel to Vessel Wall, opening length	d	54.9999	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.0000	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Drain.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.4019 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 12.7831 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 12.7831 mm
Wall Thickness, shell/head, external pressure	trb2	= 3.9320 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 7.5200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.52, \max(12.8, 4.5)]$
 $= 7.5200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.4, 7.52)$
 $= 7.5200 \text{ mm}$

Available Nozzle Neck Thickness = 9.0000 mm -> OK



PV Elite 28 Licensee: SL User
 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 228 of 285
 Nozzle Calcs.: Drain Nozl: 15 6:07pm Sep 8,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9, tr = 0.22, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E^* / (tg - c) = 0.036$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1080.95/5100.00 = 0.212$

Weld Size Calculations, Description: Drain

Intermediate Calc. for nozzle/shell Welds Tmin 6.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.2000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4526 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.4525 mm

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PV Elite 28 Licensee: SL User
 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 229 of 285
 Nozzle Calcs.: Vent Nozl: 16 6:07pm Sep 8,2026

Input, Nozzle Desc: Vent

From: 70

Pressure for Reinforcement Calculations	P	1056.657	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2480.00	mm
Design Length of Section	L	5600.0000	mm
Shell Finished (Minimum) Thickness	t	18.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5753.08	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

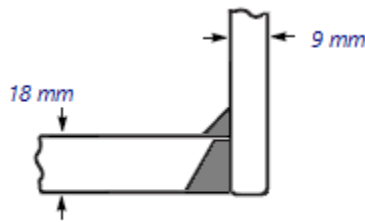
Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		1.9291	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	9.0000	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



PV Elite 28 Licensee: SL User
 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 230 of 285
 Nozzle Calcs.: Vent Nozl: 16 6:07pm Sep 8,2026



Reinforcement CALCULATION, Description: Vent

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.929 in.
Actual Thickness Used in Calculation	0.354 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1057 \cdot 1243 / (138 \cdot 1 - 0.6 \cdot 1057)$
 $= 9.5622 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1057 \cdot 27.5 / (138 \cdot 1 - 0.6 \cdot 1057)$
 $= 0.2116 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4019 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	109.9998 mm
Parallel to Vessel Wall, opening length	d	54.9999 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	15.0000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: Vent.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.4019 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 12.5622 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 12.5622 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.9320 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.5200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.52, \max(12.6, 4.5)]$
 $= 7.5200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.4, 7.52)$
 $= 7.5200 \text{ mm}$

Available Nozzle Neck Thickness = 9.0000 mm -> OK



PV Elite 28 Licensee: SL User
 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 231 of 285
 Nozzle Calcs.: Vent Nozl: 16 6:07pm Sep 8,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9, tr = 0.21, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1056.66 / 5100.00 = 0.207$

Weld Size Calculations, Description: Vent

Intermediate Calc. for nozzle/shell Welds Tmin 6.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.2000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4526 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 318.4525 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 55.00 mm

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PV Elite 28 Licensee: SL User
 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 232 of 285
 Nozzle Calcs.: N2-Shell Outlet Nozl: 17 6:07pm Sep 8,2026

Input, Nozzle Desc: N2-Shell Outlet

From: 80

Pressure for Reinforcement Calculations	P	1080.770	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2480.00	mm
Design Length of Section	L	5600.0000	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7747.55	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

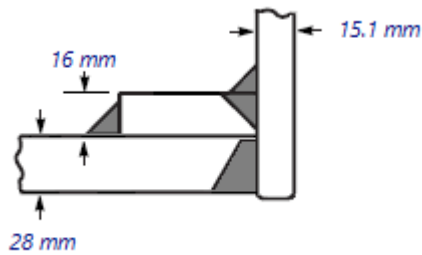
Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		60	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	595.6000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		120.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



PV Elite 28 Licensee: SL User
 FileName : 108-CA-CB_(Stack)-2026y-4m-20d-51s-9h-32m Page 233 of 285
 Nozzle Calcs.: N2-Shell Outlet Nozl: 17 6:07pm Sep 8,2026



Reinforcement CALCULATION, Description: N2-Shell Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	14.000 in.
Actual Thickness Used in Calculation	0.594 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1081 \cdot 1243 / (138 \cdot 1 - 0.6 \cdot 1081)$
 $= 9.7815 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1081 \cdot 178 / (118 \cdot 1 + 0.4 \cdot 1081)$
 $= 1.6227 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0591 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	662.8496 mm
Parallel to Vessel Wall, opening length	d	331.4248 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		46.2190 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



PV Elite 28 Licensee: SL User
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 Nozzle Calcs.: N2-Shell Outlet Nozl: 17 6:07pm Sep 8,2026

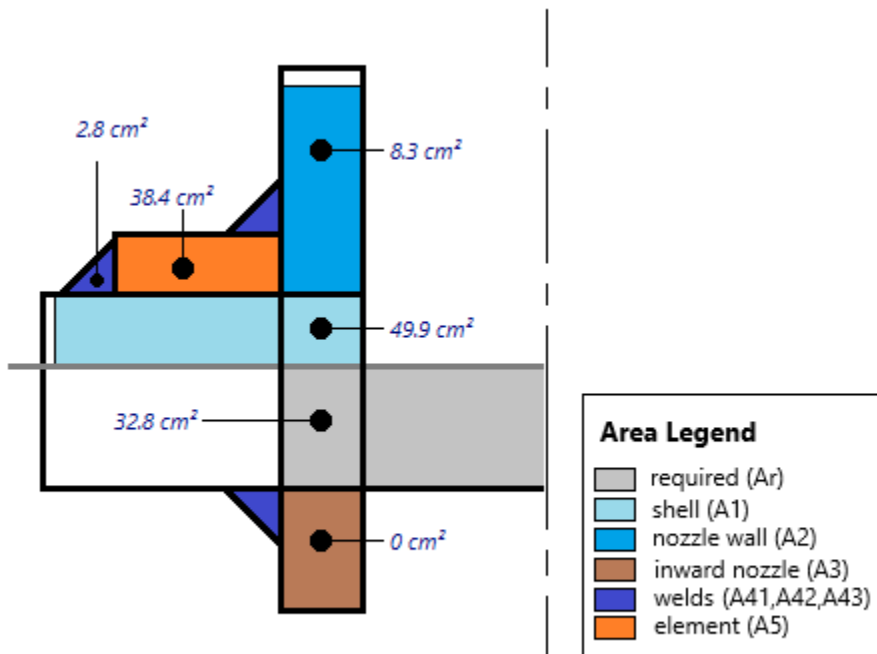


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	32.761	18.908	NA
Area in Shell	A1	49.905	44.956	NA
Area in Nozzle Wall	A2	8.272	8.717	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	2.815	2.815	NA
Area in Element	A5	38.400	38.400	NA
TOTAL AREA AVAILABLE	Atot	99.391	94.888	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (331 \cdot 9.78 \cdot 1 + 2 \cdot 12.1 \cdot 9.78 \cdot 1 \cdot (1 - 0.86))$$

$$= 32.761 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 331(1 \cdot 25 - 1 \cdot 9.78) - 2 \cdot 12.1$$

$$(1 \cdot 25 - 1 \cdot 9.78) \cdot (1 - 0.86)$$

$$= 49.905 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 46.2) \cdot (12.1 - 1.62) \cdot 0.86$$



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$$= 8.272 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 $= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 14^2 \cdot 1$
 $= 2.815 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot fr_4$
 $= (596 - 356) \cdot 16 \cdot 1$
 $= 38.400 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.6227 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 12.7815 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 12.7815 \text{ mm}$
Wall Thickness, shell/head, external pressure	$tr_{b2} = 3.9320 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[11.3, \max(12.8, 4.5)]$
 $= 11.3312 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(4.62, 11.3)$
 $= 11.3312 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 15.088 = 13.202 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 27.5,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 267.5 N./mm ²	Passed
Occasional	: 7.1,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 15.7,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 13.2$, $t_r = 1.62$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.16$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A	0°C
Min Metal Temp. at Required thickness (UCS 66.1)	-104°C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 13.2$, $t_r = 1.62$, $c = 3 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.16$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20°C
Min Metal Temp. at Required thickness (UCS 66.1)	-104°C
Min Metal Temp. w/o impact per UG-20(f)	-29°C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Govrn. thk, $t_g = 13.2$, $t_r = 1.62$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 16$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.39$, Temp. Reduction = 53 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 13.2$, $t_r = 1.62$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -48 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1080.77 / 1960.01 = 0.551$

Weld Size Calculations, Description: N2-Shell Outlet

Intermediate Calc. for nozzle/shell Welds	T_{min}	12.0876 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	16.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$8.0000 = 0.5 * T_{minPad}$	$9.8980 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - t_r)) \cdot Sv)$
 $= \max(0, (32.8 - 49.9 + 2 \cdot 12.1 \cdot 0.86 \cdot (1 \cdot 25 - 9.78)) \cdot 138)$
 $= \max(0, -19697.34) \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (8.27 + 38.4 + 2.82 - 0 \cdot 0.86) \cdot 138$
 $= 69635.54 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$



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$$= (8.27 + 0 + 0.86 + (5.17)) \cdot 138$$

$$= 20114.71 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (8.27 + 0 + 2.82 + 38.4 + (5.17)) \cdot 138$$

$$= 76907.57 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 356 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 32932. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 596 \cdot 14 \cdot 0.49 \cdot 138$$

$$= 90311. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 172) \cdot (15.1 - 3) \cdot 0.7 \cdot 118$$

$$= 54935. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 356 \cdot 16 \cdot 0.74 \cdot 138$$

$$= 93063. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 356 \cdot (16 - 0) \cdot 0.74 \cdot 138$$

$$= 93063. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (90311 + 54935) = 145246 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (32932 + 93063 + 93063 + 0) = 219058 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (90311 + 93063 + 0) = 183374 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 145245 Kgf, must exceed W = 0 Kgf	or W1 = 69635 Kgf
Path 2-2 = 219057 Kgf, must exceed W = 0 Kgf	or W2 = 20114 Kgf
Path 3-3 = 183373 Kgf, must exceed W = 0 Kgf	or W3 = 76907 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2160 KPa.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 12.8134 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 340.8134 mm

Input Echo, WRC297 Item 1, Description: N2-Shell Outle:



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Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2480.000	mm
Shell Thickness	28.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	355.600	mm
Nozzle Thickness	15.0876	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	595.600	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:
External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	1080.77	KPa.
Radial Load	P	-2676.93	Kgf
Circumferential Shear	Vc	1784.62	Kgf
Longitudinal Shear	Vl	1784.62	Kgf
Circumferential Moment	Mc	1748.26	Kg-m.
Longitudinal Moment	Ml	1748.26	Kg-m.
Torsional Moment	Mt	2497.50	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 355.6 + 2 \cdot 1.65 \cdot \sqrt{1263.5(28.0 - 3.0)}$
 $= 942.105 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2527.000	mm
Nozzle Outside Diameter	(d) =	355.600	mm
Vessel Thickness used	(T) =	41.000	mm
Nozzle Thickness used	(t) =	12.088	mm
T / t	=	3.392	
d / t	=	29.419	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.105$$

Nr/P	=	0.077
Mr/P	=	0.088



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MØ/P = 0.090
 NØ/P = 0.235
 MrD/Mc = 0.155
 NrDL/Mc = 0.114
 MØd/Mc = 0.202
 NØDL/Mc = 0.173
 MrD/Ml = 0.123
 NrDL/Ml = 0.097
 MØD/Ml = 0.147
 NØDL/Ml = 0.346

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-9	-9	9	9
Outplane Bending (ML)	-25	25	25	-25
Normal Pressure Stress	32	33	32	33
Outplane Stress Summary	9	43	79	13

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-21	21	21	-21
Inplane Pressure Stress	16	16	16	16
Inplane Stress Summary	1	27	49	-9

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	3	3	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	10	44	79	22

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	3	3	3	3



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Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-34	34	34	-34
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16
Outplane Stress Summary	-11	41	68	-18

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-26	26	26	-26
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	32	33	32	33
Inplane Stress Summary	11	49	71	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	23	50	73	22

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-9	-9	9	9
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	18	17	18
Hoop Stress Summary	10	12	30	32

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	53	-53	53	-53
Axial Membrane (Mc)	0	0	0	0



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Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-15	-15	15	15
Axial Bending (ML)	-147	147	147	-147
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-99	88	227	-174
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	12	12	7	7

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	113	90	227	207

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
hoop direction)				
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	18	17	18
Hoop Stress Summary	15	17	25	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
pipe cross-section)				
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	53	-53	53	-53
Axial Membrane (Mc)	-15	-15	15	15
Axial Bending (Mc)	-193	193	193	-193
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-144	134	272	-220

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10
Shear Stress Summary	7	7	12	12

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl



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(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	161	134	273	248

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 355.6 + 2 \cdot 1.65 \cdot \sqrt{1255.5(28.0 - 3.0)}$$

$$= 940.245 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2511.000 mm
Nozzle Outside Diameter	(d) =	595.600 mm
Vessel Thickness used	(T) =	25.000 mm
Nozzle Thickness used	(t) =	132.088 mm
T / t	=	0.189
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.377$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.065
Mr/P	=	0.128
M0/P	=	0.038
N0/P	=	0.069
MrD/Mc	=	0.281
NrDL/Mc	=	0.093
M0d/Mc	=	0.085
N0DL/Mc	=	0.081
MrD/Ml	=	0.098
NrDL/Ml	=	0.082
M0D/Ml	=	0.029
N0DL/Ml	=	0.090

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	53	54	53	54
Outplane Stress Summary	53	51	77	43

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	32	-32	32	-32
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-27	27	27	-27
Inplane Pressure Stress	26	26	26	26



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Inplane Stress Summary	30	20	92	-26
------------------------	----	----	----	-----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	1	1
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	53	51	92	69
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-23	23	23	-23
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	26	26	26	26

Outplane Stress Summary	11	39	66	0
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	32	-32	32	-32
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-77	77	77	-77
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	53	54	53	54

Inplane Stress Summary	6	98	170	-48
------------------------	---	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right



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	Outside	Inside	Outside	Inside
Two * Max Shear Stress	11	98	170	48

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	33	32	33	32	33	32	33
Circ. Pl (SUS)	-6	-6	12	12	-2	-2	4	4
Circ. Q (SUS)	-17	17	33	-33	-18	18	34	-34
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-1	-1	3	3	-1	-1	7	7
Long. Q (SUS)	-13	13	29	-29	-26	26	42	-42
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	32.0	33.0	32.0	33.0	32.0	33.0	32.0	33.0
Pm+Pl (SUS)	26.0	27.0	44.0	45.0	30.0	31.0	36.0	37.0
Pm+Pl+Q (Total)	10.1	44.5	77.3	22.8	23.8	50.0	71.4	22.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	138.00	Passed
Pm+Pl (SUS)	45.00	207.00	Passed
Pm+Pl+Q (TOTAL)	77.31	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	53	54	53	54	53	54	53	54
Circ. Pl (SUS)	-2	-2	6	6	-2	-2	6	6
Circ. Q (SUS)	1	-1	17	-17	-45	45	109	-109
Long. Pm (SUS)	26	26	26	26	26	26	26	26
Long. Pl (SUS)	-1	-1	5	5	-1	-1	5	5
Long. Q (SUS)	5	-5	59	-59	-14	14	32	-32
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	53.0	54.0	53.0	54.0	53.0	54.0	53.0	54.0
Pm+Pl (SUS)	51.0	52.0	59.0	60.0	51.0	52.0	59.0	60.0



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Pm+Pl+Q (Total)	52.0	51.0	90.1	71.0	11.2	97.0	168.0	49.0
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	54.00	138.00	Passed
Pm+Pl (SUS)	60.00	207.00	Passed
Pm+Pl+Q (TOTAL)	168.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	17	18	17	18	17	18	17	18
Circ. Pl (SUS)	-6	-6	12	12	-1	-1	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-13	-13	17	17	-13	-13	17	17
Long. Q (SUS)	-94	94	200	-200	-140	140	246	-246
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	17.0	18.0	17.0	18.0	17.0	18.0	17.0	18.0
Pm+Pl (SUS)	16.5	17.5	29.8	30.7	21.4	22.4	26.6	27.0
Pm+Pl+Q (Total)	112.6	90.8	225.3	205.6	161.8	135.5	271.6	247.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	18.00	118.00	Passed
Pm+Pl (SUS)	30.70	177.00	Passed
Pm+Pl+Q (TOTAL)	271.58	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N4-Tube Outlet

From: 110

Pressure for Reinforcement Calculations	P	1080.770	KPa.
Temperature for Internal Pressure	Temp	130	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	130	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2480.00	mm
Design Length of Section	L	1757.4667	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		9545.70	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

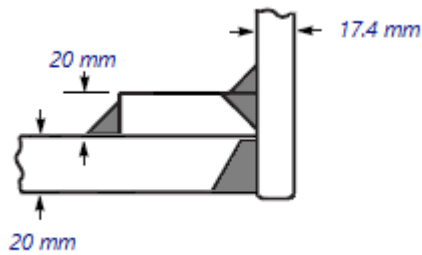
Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0157	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	17.4498	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1030.0000	mm
Thickness of Pad	te	20.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm
Reinforcing Pad Width		210.0006	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



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 Nozzle Calcs.: N4-Tube Outlet Nozl: 18 6:07pm Sep 8,2026



Reinforcement CALCULATION, Description: N4-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 24.016 in.
 Actual Thickness Used in Calculation 0.687 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1081 \cdot 1243 / (138 \cdot 1 - 0.6 \cdot 1081)$
 $= 9.7815 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1081 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 1081)$
 $= 2.3814 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.6380 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1162.1984	mm
Parallel to Vessel Wall, opening length	d	581.0992	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		42.5000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



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Nozzle Calcs.: N4-Tube Outlet

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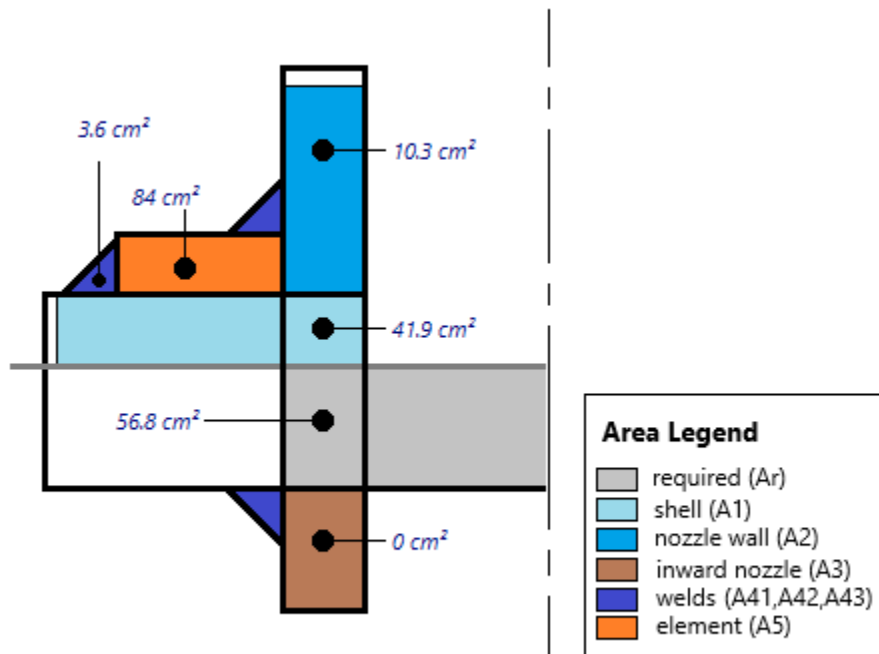


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	56.840	20.151	NA
Area in Shell	A1	41.947	58.486	NA
Area in Nozzle Wall	A2	10.258	10.890	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.560	3.560	NA
Area in Element	A5	84.000	84.000	NA
TOTAL AREA AVAILABLE	Atot	139.765	156.936	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 615.3740 20.0000 mm
Based on given Pad Diameter: 1030.0000 0.2560 mm
Based on Shell or Nozzle Thickness: 616.1595 17.4498 mm

Area Required [A]:
= (d • tr • F + 2 • tn • tr • F • (1-fr1)) UG-37(c)
= (581 • 9.78 • 1 + 2 • 14.4 • 9.78 • 1 • (1-1))
= 56.840 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d (E1 • t - F • tr) - 2 • tn (E1 • t - F • tr) • (1 - fr1)
= 581 (1 • 17 - 1 • 9.78) - 2 • 14.4
(1 • 17 - 1 • 9.78) • (1 - 1)



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$$= 41.947 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 42.5) \cdot (14.4 - 2.38) \cdot 1$$

$$= 10.258 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1 + 16^2 \cdot 1$$

$$= 3.560 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (1030 - 610) \cdot 20 \cdot 1$$

$$= 84.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.3814 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 12.7815 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 12.7815 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.9320 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(12.8, 5.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.38, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 17.4498 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 27.1,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 317.9 N./mm ²	Passed
Occasional	: 10.6,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 12.6,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 17.4, tr = 2.38, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.16, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	8 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 17.4, tr = 2.38, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.16, Temp. Reduction = 78 °C



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Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-28 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 17.4, tr = 2.38, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E*/(tg - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-28 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 20, tr = 9.78, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E*/(tg - c) = 0.58$, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 17.4, tr = 2.38, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E*/(tg - c) = 0.16$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-28 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -48 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1080.77/1960.01 = 0.551$

Weld Size Calculations, Description: N4-Tube Outlet

Intermediate Calc. for nozzle/shell Welds	Tmin	14.4498 mm
Intermediate Calc. for pad/shell Welds	TminPad	17.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$8.5000 = 0.5 * T_{minPad}$	$11.3120 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr))S_v)$
 $= \max(0, (56.8 - 41.9 + 2 \cdot 14.4 \cdot 1 \cdot (1 \cdot 17 - 9.78))138)$
 $= 23892.57 \text{ Kgf}$



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Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr}2) \cdot S_v$$

$$= (10.3 + 84 + 3.56 - 0 \cdot 1) \cdot 138$$

$$= 137645.86 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S_v$$

$$= (10.3 + 0 + 1 + (4.91)) \cdot 138$$

$$= 22755.27 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr}1)) \cdot S$$

$$= (10.3 + 0 + 3.56 + 84 + (4.91)) \cdot 138$$

$$= 144559.12 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 66067. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 1030 \cdot 16 \cdot 0.49 \cdot 138$$

$$= 178491. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 298) \cdot (17.4 - 3) \cdot 0.7 \cdot 138$$

$$= 133150. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 610 \cdot 20 \cdot 0.74 \cdot 138$$

$$= 199551. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 610 \cdot (16 - 0) \cdot 0.74 \cdot 138$$

$$= 159640. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (178491 + 133150) = 311640 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (66067 + 199551 + 159640 + 0) = 425259 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (178491 + 159640 + 0) = 338131 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 311640 Kgf, must exceed W = 23892 Kgf	or W1 = 137645 Kgf
Path 2-2 = 425259 Kgf, must exceed W = 23892 Kgf	or W2 = 22755 Kgf
Path 3-3 = 338131 Kgf, must exceed W = 23892 Kgf	or W3 = 144559 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:



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Converged Maximum Allowable Pressure in the Operating case: 1848 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 38.0952 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 458.0952 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.945 %

Input Echo, WRC297 Item 1, Description: N4-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2480.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	609.999	mm
Nozzle Thickness	17.4498	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	1030.000	mm
Thickness of Reinforcing Pad	20.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTained loads.

Design Internal Pressure	Dp	1080.77	KPa.
Radial Load	P	-4589.03	Kgf
Circumferential Shear	Vc	3059.35	Kgf
Longitudinal Shear	Vl	3059.35	Kgf
Circumferential Moment	Mc	5137.73	Kg-m.
Longitudinal Moment	Ml	5137.73	Kg-m.
Torsional Moment	Mt	7339.62	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{mean}(t - ca)} \\
 &= 610.0 + 2 \cdot 1.65 \cdot \sqrt{1261.5(20.0 - 3.0)} \\
 &= 1093.260 \text{ mm}
 \end{aligned}$$



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WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2523.000 mm
 Nozzle Outside Diameter (d) = 609.999 mm
 Vessel Thickness used (T) = 37.000 mm
 Nozzle Thickness used (t) = 14.450 mm
 T / t = 2.561
 d / t = 42.215

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.997$$

Nr/P = 0.059
 Mr/P = 0.076
 M0/P = 0.053
 N0/P = 0.172
 MrD/Mc = 0.165
 NrDL/Mc = 0.094
 M0d/Mc = 0.138
 N0DL/Mc = 0.231
 MrD/Ml = 0.093
 NrDL/Ml = 0.060
 M0D/Ml = 0.070
 N0DL/Ml = 0.330

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-19	-19	19	19
Outplane Bending (ML)	-25	25	25	-25
Normal Pressure Stress	35	36	35	36
Outplane Stress Summary	6	37	97	26

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-33	33	33	-33
Inplane Pressure Stress	17	17	17	17
Inplane Stress Summary	-2	35	72	-25

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2



Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	12	40	97	51

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-13	-13	13	13
Outplane Bending (Mc)	-49	49	49	-49
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	17	17	17	17
Outplane Stress Summary	-29	49	97	-22

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	14	-14	14	-14
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-59	59	59	-59
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	35	36	35	36
Inplane Stress Summary	-12	77	117	-30

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	30	78	118	31

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0



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Hoop Membrane (ML)	-19	-19	19	19
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	25	26	25	26
Hoop Stress Summary	11	12	50	52

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	59	-59	59	-59
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-12	-12	12	12
Axial Bending (ML)	-150	150	150	-150
Axial Pressure Stress	12	12	12	12
Axial Stress Summary	-89	92	236	-182

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8
Shear Stress Summary	10	10	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	102	93	236	234

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13	-13	13	13
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	25	26	25	26
Hoop Stress Summary	17	18	44	46

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	59	-59	59	-59
Axial Membrane (Mc)	-12	-12	12	12
Axial Bending (Mc)	-279	279	279	-279
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	12	12	12	12



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Axial Stress Summary -218 221 365 -311

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	8	8	8	8

Shear Stress Summary 6 6 10 10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	235	221	365	358

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 610.0 + 2 \cdot 1.65 \cdot \sqrt{(1251.5(20.0 - 3.0))}$
 $= 1091.341 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2503.000 mm
 Nozzle Outside Diameter (d) = 1030.000 mm
 Vessel Thickness used (T) = 17.000 mm
 Nozzle Thickness used (t) = 224.450 mm
 $T / t = 0.076$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.993$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.045
 Mr/P = 0.071
 M0/P = 0.019
 N0/P = 0.026
 MrD/Mc = 0.240
 NrDL/Mc = 0.100
 M0d/Mc = 0.072
 N0DL/Mc = 0.051
 MrD/Ml = 0.049
 NrDL/Ml = 0.048
 M0D/Ml = 0.015
 N0DL/Ml = 0.026

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-15	15	15	-15
Normal Pressure Stress	78	79	78	79



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Outplane Stress Summary	80	76	120	54
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	65	-65	65	-65
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-50	50	50	-50
Inplane Pressure Stress	39	39	39	39
Inplane Stress Summary	53	22	170	-61

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	81	76	170	116

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-72	72	72	-72
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	39	39	39	39
Outplane Stress Summary	-20	89	142	-38

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	65	-65	65	-65
Inplane Membrane (Mc)	-16	-16	16	16
Inplane Bending (Mc)	-243	243	243	-243
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	78	79	78	79
Inplane Stress Summary	-109	247	412	-206



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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			2	2	2	2
Shear Stress Summary			1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			109	247	412	206

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	35	36	35	36	35	36	35	36
Circ. Pl (SUS)	-14	-14	24	24	-4	-4	6	6
Circ. Q (SUS)	-15	15	35	-35	-45	45	73	-73
Long. Pm (SUS)	17	17	17	17	17	17	17	17
Long. Pl (SUS)	-2	-2	4	4	-8	-8	18	18
Long. Q (SUS)	-19	19	47	-47	-39	39	59	-59
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	35.0	36.0	35.0	36.0	35.0	36.0	35.0	36.0
Pm+Pl (SUS)	21.0	22.0	59.0	60.0	31.0	32.0	41.0	42.0
Pm+Pl+Q (Total)	11.7	38.9	94.3	51.4	30.5	77.3	114.4	32.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	138.00	Passed
Pm+Pl (SUS)	60.00	207.00	Passed
Pm+Pl+Q (TOTAL)	114.44	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	78	79	78	79	78	79	78	79



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Circ. P1 (SUS)	0	0	8	8	-9	-9	23	23
Circ. Q (SUS)	2	-2	32	-32	-178	178	308	-308
Long. Pm (SUS)	39	39	39	39	39	39	39	39
Long. P1 (SUS)	-1	-1	15	15	-4	-4	12	12
Long. Q (SUS)	15	-15	115	-115	-55	55	89	-89
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	78.0	79.0	78.0	79.0	78.0	79.0	78.0	79.0
Pm+P1 (SUS)	78.0	79.0	86.0	87.0	69.0	70.0	101.0	102.0
Pm+P1+Q (Total)	80.3	77.2	169.0	116.0	109.0	248.0	409.0	206.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	79.00	138.00	Passed
Pm+P1 (SUS)	102.02	207.00	Passed
Pm+P1+Q (TOTAL)	409.03	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	25	26	25	26	25	26	25	26
Circ. P1 (SUS)	-14	-14	24	24	-8	-8	18	18
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. P1 (SUS)	-11	-11	13	13	-11	-11	13	13
Long. Q (SUS)	-91	91	209	-209	-220	220	338	-338
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	25.0	26.0	25.0	26.0	25.0	26.0	25.0	26.0
Pm+P1 (SUS)	11.4	12.4	49.2	50.2	17.2	18.2	43.2	44.2
Pm+P1+Q (Total)	103.0	93.2	234.2	234.3	236.3	221.2	363.3	357.6

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	26.00	138.00	Passed
Pm+P1 (SUS)	50.16	207.00	Passed
Pm+P1+Q (TOTAL)	363.31	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



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Input Echo, Tubesheet Number 1, Description: Stationary Tube

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	520.00 KPa.
Shell Maximum Operating Pressure	Psox,max	520.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	28.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2480.000 mm
Shell Temperature for Internal Pressure	Ts	130.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	20.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2480.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	6000
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	6000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5770.00 mm
Design Temperature of the Tubes		130.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	160.40 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1130.510 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm



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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	32948.0 cm ²
Perimeter of Tube Layout	Cp	7006.7 mm

Tubesheet Data:

Tubesheet TYPE: Stationary, Gasketed both Sides, Conf. d
 Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	82.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	115.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	2562.000 mm
Dimension G for the Channel Side	Gc	2537.285 mm
Area of the Untubed Lanes	AL	0.0 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	635.910 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	2748.000 mm
Flange Inside Diameter	B	2480.000 mm
Flange Face Outside Diameter	Fod	2565.000 mm
Flange Face Inside Diameter	Fid	2480.000 mm
Gasket Outside Diameter	Go	2554.000 mm
Gasket Inside Diameter	Gi	2510.000 mm
Small end Hub thk.	g0	16.0000 mm
Large end Hub thk.	g1	35.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	5118.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	0.0000
Partition Gasket Design Seating Stress	yPart	0.00 N./mm ²
Partition Gasket Facing Sketch		Code Sketch 1a
Partition Gasket Column for Gasket Seating		Code Column II

Bolting Information:

Diameter of Bolt Circle	C	2663.000 mm
Nominal Bolt Diameter	dB	38.1000 mm
Thread Series		TEMA Thread Series
Number of Bolts	n	72
Bolt Material		SA-193 B7
Bolt Material UNS Number		G41400
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²



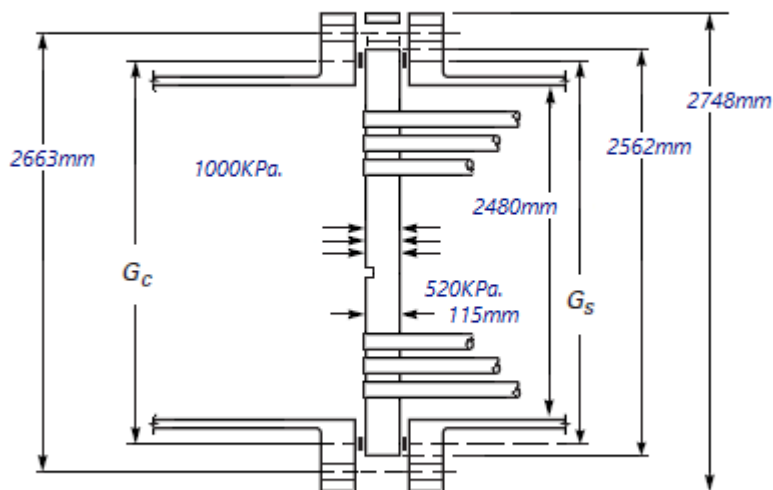
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Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm
Alternate Flange Operating Bolt Load, Wm1		570228.06 Kg _f
Alternate Flange Seating Bolt Load, Wm2		468391.50 Kg _f
Alternate Flange Design Bolt Load, W		1144638.00 Kg _f

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with	None
Tubesheet Extended as Flange	No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	130.0 °C	0.19629E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%



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Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	130.0 °C	0.19629E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	82.0 °C	0.19784E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (1000 \cdot (19/2 - 0)) / (92.4 \cdot 1 + 0.4 \cdot 1000)$
 $= 0.1026 + 0.0000 = 0.1026 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 130.00 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	5770.00	9.03	50.0000	0.0134948	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 15019 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.2905	19.05	5770.00	65.57	50.0000	0.0002558	25.58

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 25.6) / (3 \cdot 65.6) = 520 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.2905 mm
Allowable Internal Pressure at Corroded thickness	22456.99 KPa.
Required Internal Design Pressure	1000.00 KPa.
Allowable External Pressure at Corroded thickness	15019.20 KPa.
Required External Design Pressure	520.00 KPa.
Required Thickness due to Shell Side pressure	0.2905 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

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Gasket Contact Width,	$N = (Goc - Gic) / 2$	22.000 mm
Basic Gasket Width,	$b0 = N / 2$	11.000 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	8.355 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	2537.291 mm

Results for ASME Stationary Tubesheet Calculations for Configuration d:

Results for Tubesheet Calculations Original Thickness:



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ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio $[\rho]$:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/115$$

$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter $[d^*]$:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(19 - 2 \cdot 2.11 (199214080/197835184) \cdot$$

$$(92.4/138) \cdot (0); 19 - 2 \cdot 2.11)$$

$$= 19.0500 \text{ mm}$$

Equivalent outer tube limit circle diameter $[Do]$:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 1131 + 19$$

$$= 2280.070 \text{ mm}$$

Determine the basic ligament efficiency for shear $[\mu]$:

$$= (p - dt) / p$$

$$= (25.4 - 19) / 25.4$$

$$= 0.250$$

Equivalent outer tube limit radius $[ao]$:

$$= Do / 2$$

$$= 2280/2$$

$$= 1140.0350 \text{ mm}$$

Effective tube pitch $[p^*]$:

$$= p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$$

$$= 25.4 / \sqrt{ 1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 2280 \cdot 25.4) / (\pi \cdot 2280^2) }$$

$$= 25.4000 \text{ mm}$$

Effective ligament efficiency for bending $[\mu^*]$:

$$= (p^* - d^*) / p^*$$

$$= (25.4 - 19) / 25.4$$

$$= 0.2500$$

Ratio $[\rho_s]$:

$$= a_s / a_o$$

$$= 1269/1140$$

$$= 1.1128$$

Ratio $[\rho_c]$:

$$= a_c / a_o$$

$$= 1269/1140$$

$$= 1.1128$$

Compute parameter $[xt]$:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 6000 ((19 - 2 \cdot 2.11) / (2 \cdot 1140))^2$$

$$= 0.7462$$

Determine parameter $[xs]$:

$$= 1 - Nt(dt / (2 \cdot a_o))^2$$

$$= 1 - 6000 (19 / (2 \cdot 1140))^2$$

$$= 0.5812$$

Determine the value $[h'g]$:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:



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Intermediate parameters for Tubesheet Gasketed on the Shell Side:
 $\beta_s, k_s, \lambda_s, \delta_s$ are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:
 $\beta_C, k_C, \lambda_C, \delta_C$ are all = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$h/p = 4.527559$; $\mu^* = 0.250000$; $E^* = 42853184$ KPa.
 $E^*/E = 0.216611$; $\nu^* = 0.396457$

Note: As h/p (4.528) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor $[X_a]$:

$$= \left(\left(24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4}$$

$$= \left(\left(24(1 - 0.4^2)6000 \cdot 196289984 \cdot 2.11 \right. \right.$$

$$\left. \left(19 - 2.11 \right)1140^2 \right) / (42853184 \cdot 5770 \cdot 115^3) \right)^{1/4}$$

$$= 7.3654$$

Values from Table 4.18.3

$Z_d = 0.003715$; $Z_v = 0.018522$; $Z_m = 0.194582$
 $Z_a = 0.468923E+03$; $Z_w = 0.018522$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio $[K]$:

$= A / Do$
 $= 2562/2280$
 $= 1.1236$

Compute coefficient $[F]$:

$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_C + E \cdot \ln(K))$
 $= (1 - 0.4) / 42853184 \cdot (0 + 0 +$
 $197835184 \cdot \ln(1.12))$
 $= 0.3248$

Compute parameter $[\phi]$:

$= (1 + \nu^*)F$
 $= (1 + 0.4)0.32$
 $= 0.4536$

Compute parameter $[Q_1]$:

$= (ps - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$
 $= (1.11 - 1 - 0.45 \cdot 0.019) / (1 + 0.45 \cdot 0.19)$
 $= 0.095940$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor $[\gamma_b]$:

$= (G_c - G_s) / Do$ (config d)
 $= (2537 - 2537) / 2280 = 0$

Calculate parameter $[\omega_s]$:

$= ps \cdot ks \cdot \beta_s \cdot \delta_S (1 + h \cdot \beta_s)$
 $= 1.11 \cdot 0 \cdot 0 \cdot 0.000 (1 + 115 \cdot 0)$
 $= 0.0000 \text{ mm}^2$

Calculate parameter $[\omega^*s]$:

$= ao^2(ps^2 - 1) \cdot (ps - 1) / 4 - \omega_s$
 $= 1140^2(1.11^2 - 1) \cdot (1.11 - 1) / 4 - 0$
 $= 8736.4102 \text{ mm}^2$

Parameter $[\omega_c]$:

$= pc \cdot kc [\delta_C (C_m \Theta + \beta_C + C_q \Delta \cdot h \cdot \beta_C^2) -$
 $\zeta c (C_m \Theta + h \cdot C_q \Theta + h \cdot \beta_C / 2)]$



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$$= 1.11 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 115) - 0.000 (0/115 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [$\omega \cdot c$]:

$$= ao^2 [(pc^2 + 1) \cdot (pc - 1)/4 - (ps - 1)/2] - \omega c$$

$$= 1140^2 [(1.11^2 + 1) \cdot (1.11 - 1)/4 - (1.11 - 1)/2] - 0$$

$$= 8736.4131 \text{ mm}^2$$

Pressure [$P \cdot S$]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [$P \cdot C$]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$Pe = Ps - Pt$$

$$= 520 - 1000 = -480 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q_2]:

$$= [((\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt) - (\omega s \cdot P \cdot s - \omega c \cdot P \cdot c)) + W \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((8736 \cdot 520 - 8736 \cdot 1000) - (0 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + 1144638 \cdot 0 / (2 \cdot \pi)] / (1 + 0.45 \cdot 0.19)$$

$$= -392.9491 \text{ Kgf}$$

Calculate factor [Q_3]:

$$= Q_1 + 2 \cdot Q_2 / (Pe \cdot ao^2)$$

$$= 0.096 + 2 \cdot -393 / (-480 \cdot 1140^2)$$

$$= 0.10829$$

Fm Value from Table 4.18.3 = 0.075387

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.075 / 0.25) \cdot (2 \cdot 1140 / (115 - 0))^2 \cdot -480$$

$$= -85.3527 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138 = 207 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.07 / 0.25) \cdot (2 \cdot 1140 / (71 - 0))^2 \cdot -480$$

$$= -206.9883 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$

$$= 71 + 0 + 0$$

$$= 71.0260 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy)$$

$$= (2 \cdot 0.25 \cdot 115 / 1140) \cdot 110$$

$$= 5.5681 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [Pe] <= τ_{limit}

Tubesheet Shear Stress - final thickness [τ_f]:



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$$\begin{aligned}
 &= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p) \\
 &= (32948 \cdot -480) / (0.25 \cdot 8.18 \cdot 7007) \\
 &= 110.3980 \text{ N./mm}^2
 \end{aligned}$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$\begin{aligned}
 &= H + C_{ats} + C_{atc} \\
 &= 8.18 + 0 + 0 \\
 &= 8.1786 \text{ mm}
 \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:

$$\begin{aligned}
 &= \max(H_{reqB}; H_{reqS}) \\
 &= \max(71; 8.18) \\
 &= 71.0260 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:
 Ftmin = -1.18, Ftmax = 8.42

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$\begin{aligned}
 &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s) \\
 &= ((520 \cdot 0.58 - 1000 \cdot 0.75) - (-480) \cdot -1.18) / \\
 &\quad (0.75 - 0.58) \\
 &= -6.1309 \text{ N./mm}^2
 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned}
 &= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s) \\
 &= ((520 \cdot 0.58 - 1000 \cdot 0.75) - (-480) \cdot 8.42) / \\
 &\quad (0.75 - 0.58) \\
 &= 21.7959 \text{ N./mm}^2
 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 21.8 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 92.3983 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$\begin{aligned}
 &= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25) \\
 &= \max((3.25 - 0.25(0.0037 + 0.11 \cdot 0.019)7.37^4); 1.25) \\
 &= 1.2500 \text{ (Should be } \leq 2)
 \end{aligned}$$

Determine the factor [rt]:

$$\begin{aligned}
 &= (\sqrt{(dt^2 + (dt - 2 \cdot tt)^2)}) / 4 \\
 &= (\sqrt{(19^2 + (19 - 2 \cdot 2.11)^2)}) / 4 \\
 &= 6.0355 \text{ mm}
 \end{aligned}$$

Determine the factor [Ct]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot E / S_{yt})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 196289984 / 160)} \\
 &= 155.4270
 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned}
 &= k \cdot L / r \\
 &= 1 \cdot 636 / 6.04 \\
 &= 105.3620
 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned}
 &= (S_y / F_s) \cdot (1 - F_t / (2 \cdot C_t)) \\
 &= (160 / 1.25) \cdot (1 - 105 / (2 \cdot 155)) \\
 &= 84.826 \text{ N./mm}^2 \text{ (Never greater than } S_{ot})
 \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.



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Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 21.8 \cdot 1.12$$

$$= 249.57 \text{ Kgf}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19 - 2.11) 2.11$$

$$= 1.123 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (19 - 2.11) 92.4$$

$$= 1057.989 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.11 (19 + 0.67 \cdot 2.11) 92.4$$

$$= 1057.9891 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 92.4 / 92.4$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.11 (19 + 0.67 \cdot 2.11) 92.4$$

$$= 702.9343 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 19)^2 + 1.07 \cdot 2.11 \cdot (19 - 2.11) 1 \cdot 1} - 0.75 \cdot 19 \right)$$

$$= 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm

Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 1058 is $\geq$ tube strength 1058 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Computations completed for ASME tubesheet configuration: d

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	85.4 $\leq$	207.0 N./mm ²	Pass
Tubesheet Shear Stress	7.9 $\leq$	110.4 N./mm ²	Pass
Maximum Stress in the Tubes	21.8 $\leq$	92.4 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-6.1 $\leq$	-84.8 N./mm ²	Pass
Maximum Force on any one Tube	249.6 $\leq$	1058.0 Kgf	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):



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Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	71.026	115.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

Stationary Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----- Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	105.782 5.684	178 207	16 110	Ps+Pt-Th	D1 Ok
D2uc	74.158 2.956	92 207	9 110	Ps+Pt-Th	D2 Ok
D3uc	71.026 2.728	85 207	8 110	Ps+Pt-Th	D3 Ok
D1c	111.769 5.684	196 207	17 110	Ps+Pt-Th-c	D1 Ok
D2c	80.152 2.956	102 207	9 110	Ps+Pt-Th-c	D2 Ok
D3c	77.028 2.728	94 207	8 110	Ps+Pt-Th-c	D3 Ok
Max:	111.7694 5.684 mm	0.946	0.156	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Tube, Shell and Channel Stress Summary:

Case#	Ten	----- Tube Stresses Allwd Cmp Allwd	Tube Loads Ld Allwd	Shell Stress Stress Allwd	Channel Stress Stress Allwd	Pass Fail
D1uc	46	92 -12 -84	532 1057			Ok
D2uc	25	92 -25 -84	283 1057			Ok
D3uc	22	92 -6 -84	250 1057			Ok
D1c	49	92 -12 -84	566 1057			Ok
D2c	26	92 -26 -84	300 1057			Ok
D3c	23	92 -6 -84	266 1057			Ok
Max	RATIO	0.535 0.309	0.535	...	...	

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	111.769	115.000	Ok



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Tube thickness :	0.291	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	1056.66	1.000	1056.66	1.000
Tubesheet Shear Stress	6397.19	1.000	6397.19	1.000
Tube Tensile Stress	1868.32	1.000	1831.29	1.000
Tube Compressive Stress	7015.19	1.000	1681.20	1.000
Tube-Tubesheet Joint load	1868.32	1.000	1831.28	1.000
Tube Pressure Stress	22456.98	1.000	15019.19	1.000
Minimum MAWP	1056.66		1056.66	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	1166.65	1.000	1166.65	1.000
Tubesheet Shear Stress	6749.33	1.000	6749.33	1.000
Tube Tensile Stress	1976.17	1.000	1934.79	1.000
Tube Compressive Stress	7878.08	1.000	1934.79	1.000
Tube-Tubesheet Joint load	1976.17	1.000	1934.79	1.000
Tube Pressure Stress	22456.98	1.000	15019.19	1.000
Minimum MAPnc	1166.65		1166.65	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Determine the governing MDMT considering the governing condition:

Governing thickness on the shell side per figure UCS-66.3 (c):

= tubesheet thickness/4
 = 115/4
 = 28.750 mm

Thickness Ratio = 0.86, Temperature Reduction per Fig. UCS 66.1 = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 3 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -5 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:

= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
 = max(1000/1167, 520/1167)
 = 0.857

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Input Echo, Tubesheet Number 1, Description: Floatating tube

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	520.00 KPa.
Shell Maximum Operating Pressure	Psox,max	520.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	28.0000 mm
Shell Internal Corrosion Allowance	cas	3.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2480.000 mm
Shell Temperature for Internal Pressure	Ts	130.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	20.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2480.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	6000
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	6000.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5770.00 mm
Design Temperature of the Tubes		130.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	160.40 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1130.510 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm



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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	32948.0 cm ²
Perimeter of Tube Layout	Cp	7006.7 mm

Tubesheet Data:

Tubesheet TYPE: Floating, Tubesheet Gasketed, Conf. B
 Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	82.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	115.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	2465.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		No
Area of the Untubed Lanes	AL	0.0 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	635.910 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	2465.000 mm
Flange Inside Diameter	B	2297.000 mm
Flange Face Outside Diameter	Fod	2326.000 mm
Flange Face Inside Diameter	Fid	2297.000 mm
Gasket Outside Diameter	Go	2323.000 mm
Gasket Inside Diameter	Gi	2297.000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	6.3500 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	2368.000 mm
Nominal Bolt Diameter	dB	38.1000 mm
Thread Series		TEMA Thread Series
Number of Bolts	n	72
Bolt Material		SA-193 B7M
Bolt Material UNS Number		G41400
Bolt Allowable Stress At Temperature	Sb	138.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	138.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

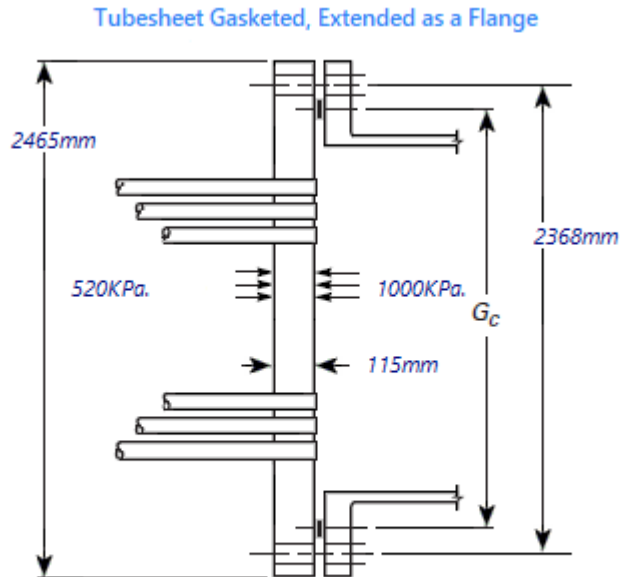
Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Extended as Flange		Yes
------------------------------	--	-----



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Thickness of Extended Portion of Tubesheet Tf 99.0000 mm
 Is Bolt Load Transferred to the Tubesheet Yes
 Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



Configuration B:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	130.0 °C	0.19629E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	130.0 °C	0.19629E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C> 0.3%



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Elastic Mod. at Design Temperature 82.0 °C 0.19784E+09 KPa.
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20132E+09 KPa.

Note:
 The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (1000 \cdot (19/2 - 0)) / (92.4 \cdot 1 + 0.4 \cdot 1000)$
 $= 0.1026 + 0.0000 = 0.1026 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 130.00 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	5770.00	9.03	50.0000	0.0134948	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 15019 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.2905	19.05	5770.00	65.57	50.0000	0.0002558	25.58

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 25.6) / (3 \cdot 65.6) = 520 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.2905 mm
Allowable Internal Pressure at Corroded thickness	22456.99 KPa.
Required Internal Design Pressure	1000.00 KPa.
Allowable External Pressure at Corroded thickness	15019.20 KPa.
Required External Design Pressure	520.00 KPa.
Required Thickness due to Shell Side pressure	0.2905 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	2310.156 mm

Results for ASME Floating Tubesheet Calculations for Configuration B:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio $[\rho]$:
 $= ltx / h$ (modified for corrosion if present)
 $= 0/115$
 $= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d^*]$:
 $= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$



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$$= \max(19 - 2 \cdot 2.11 (199214080/197835184) \cdot (92.4/138) \cdot (0); 19 - 2 \cdot 2.11)$$

$$= 19.0500 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 1131 + 19$$

$$= 2280.070 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (25.4 - 19) / 25.4$$

$$= 0.250$$

Equivalent outer tube limit radius [ao]:

$$= D_o / 2$$

$$= 2280 / 2$$

$$= 1140.0350 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{(1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2))}$$

$$= 25.4 / \sqrt{(1 - 4 \cdot \min(0 \cdot 100; 4 \cdot 2280 \cdot 25.4) / (\pi \cdot 2280^2))}$$

$$= 25.4000 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (25.4 - 19) / 25.4$$

$$= 0.2500$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 1155 / 1140$$

$$= 1.0132$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 1155 / 1140$$

$$= 1.0132$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 6000((19 - 2 \cdot 2.11) / (2 \cdot 1140))^2$$

$$= 0.7462$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot a_o))^2$$

$$= 1 - 6000(19 / (2 \cdot 1140))^2$$

$$= 0.5812$$

Determine the value [h'g]:

$$= \max(h_g - CATC; 0) \text{ (For pressure only cases)}$$

$$= \max(0 - 0; 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:
 β_s , k_s , λ_s , δ_s are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:
 β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$$h/p = 4.527559; \mu^* = 0.250000; E^* = 42853184 \text{ KPa.}$$



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$$E^*/E = 0.216611 ; v^* = 0.396457$$

Note: As h/p (4.528) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor $[X_a]$:

$$= \left(\left(24(1 - v^*)Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4}$$

$$= \left(\left(24(1 - 0.4^2)6000 \cdot 196289984 \cdot 2.11 \right. \right. \\ \left. \left. (19 - 2.11)1140^2 \right) / (42853184 \cdot 5770 \cdot 115^3) \right)^{1/4}$$

$$= 7.3654$$

Values from Table 4.18.3

$$Z_d = 0.003715 ; Z_v = 0.018522 ; Z_m = 0.194582$$

$$Z_a = 0.468923E+03 ; Z_w = 0.018522$$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio $[K]$:

$$= A / Do$$

$$= 2465/2280$$

$$= 1.0811$$

Compute coefficient $[F]$:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.4) / 42853184 \cdot (0 + 0 + 197835184 \cdot \ln(1.08))$$

$$= 0.2173$$

Compute parameter $[\phi]$:

$$= (1 + v^*)F$$

$$= (1 + 0.4)0.22$$

$$= 0.3034$$

Compute parameter $[Q_1]$:

$$= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.01 - 1 - 0.3 \cdot 0.019) / (1 + 0.3 \cdot 0.19)$$

$$= 0.007152$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor $[\gamma_b]$:

$$= (G_c - C) / Do \text{ (config b)}$$

$$= (2310 - 2368) / 2280 = -0.025$$

Calculate parameter $[\omega_s]$:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S(1 + h \cdot \beta_s)$$

$$= 1.01 \cdot 0 \cdot 0 \cdot 0.000(1 + 115 \cdot 0)$$

$$= 0.0000 \text{ mm}^2$$

Calculate parameter $[\omega^*s]$:

$$= ao^2(\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$$

$$= 1140^2(1.01^2 - 1) \cdot (1.01 - 1) / 4 - 0$$

$$= 113.8897 \text{ mm}^2$$

Parameter $[\omega_c]$:

$$= \rho_c \cdot k_c [\delta C(C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c(C_m \theta + h \cdot C_q \theta \cdot h \cdot \beta C / 2)]$$

$$= 1.01 \cdot 0 [0.000(0 + 0 \cdot 0 \cdot 115) - 0.000(0/115 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter $[\omega^*c]$:

$$= ao^2[(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c$$

$$= 1140^2[(1.01^2 + 1) \cdot (1.01 - 1) / 4 - (1.01 - 1) / 2] - 0$$

$$= 113.8891 \text{ mm}^2$$



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Pressure [P*S]:
 = 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:
 = 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$P_e = P_s - P_t$$

$$= 520 - 1000 = -480 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$= [((\omega * s * P_s - \omega * c * P_t) - (\omega_s * P * s - \omega_c * P * c)) + W * \gamma_b / (2 * \pi)] / (1 + \phi * Z_m)$$

$$= [((114 * 520 - 114 * 1000) - (0 * 0 - 0 * 0)) * 0.0001 + 918372 * -0.025 / (2 * \pi)] / (1 + 0.3 * 0.19)$$

$$= -3506.6316 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q_1 + 2 * Q_2 / (P_e * a_o^2)$$

$$= 0.0072 + 2 * -3507 / (-480 * 1140^2)$$

$$= 0.11739$$

Fm Value from Table 4.18.3 = 0.079278

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 * F_m / \mu *) * (2 * a_o / (H - h'g))^2 * P_e$$

$$= (1.5 * 0.079 / 0.25) * (2 * 1140 / (115 - 0))^2 * -480$$

$$= -89.7573 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 * S = 1.5 * 138 = 207 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 * F_m / \mu *) * (2 * a_o / (h - h'g))^2 * P_e$$

$$= (1.5 * 0.073 / 0.25) * (2 * 1140 / (72.8 - 0))^2 * -480$$

$$= -206.9849 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 72.8 + 0 + 0$$

$$= 72.7655 \text{ mm}$$

H reqd. = HReqB = 72.7655 mm

Computations completed for ASME tubesheet configuration: B

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	89.8 <=	207.0 N./mm ²	Pass

Floating Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc	86.945 55.424	130 207	...	Ps+Pt-Th	D1 Ok



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D2uc	37.934	...	43	207	...	...	Ps+Pt-Th	D2	Ok
D3uc	72.766	...	90	207	...	...	Ps+Pt-Th	D3	Ok
D1c	92.944	55.424	142	207	...	...	Ps+Pt-Th-c	D1	Ok
D2c	43.923	...	47	207	...	...	Ps+Pt-Th-c	D2	Ok
D3c	78.749	...	99	207	...	...	Ps+Pt-Th-c	D3	Ok

Max: 92.9437 55.424 mm 0.686 ... (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	92.944	115.000	Ok
Tubesheet thickness flanged extension :	55.424	99.000	Ok
Tube thickness :	0.291	2.110	Ok
Tube-tubesheet fillet weld leg :	0.000	2.110	Ok
Tube-tubesheet groove weld leg :	0.000	2.110	Ok

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	1808.00	1.000	2308.45	1.000
Tube Pressure Stress	22456.98	1.000	15019.19	1.000
Tubesheet Extension Stress	2014.18	...	No Calc	No Calc
Minimum MAWP	1808.00		2308.45	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	1963.91	1.000	2489.49	1.000
Tube Pressure Stress	22456.98	1.000	15019.19	1.000
Tubesheet Extension Stress	2014.18	...	No Calc	No Calc
Minimum MAPnc	1963.91		2489.49	



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(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Determine the governing MDMT considering the governing condition:

Governing thickness on the shell side per figure UCS-66.3 (c):
= tubesheet thickness/4
= 115/4
= 28.750 mm

Thickness Ratio = 0.51, Temperature Reduction per Fig. UCS 66.1 = 31 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-28 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:
= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
= max(1000/1964, 520/2489)
= 0.509

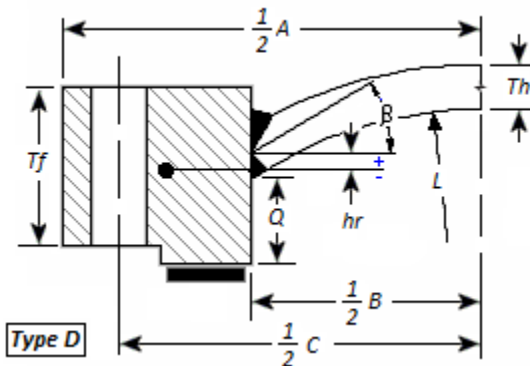
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 Floating Head: Operating Case: 1 6:07p Sep 8,2026

Input Echo, Floating Head Item 1, Description: Operating

Floating Head Type		D	
Tube Side (Internal) Design Pressure	Pts	1000.00	KPa.
Shell Side (External) Design Pressure	Pss	520.00	KPa.
Design Temperature for Spherical Head	Temp	130.00	°C
Head Material		SA-516 70	
Head Material UNS Number		K02700	
Head Allowable Stress at Temperature	Sho	138.00	N./mm ²
Head Allowable Stress at Ambient	Sha	138.00	N./mm ²
Crown Radius for Spherical Head	L	2067.000	mm
Head Thickness	Th	30.000	mm
Tube Side (Internal) Corrosion Allowance	Cats	3.0000	mm
Shell Side (External) Corrosion Allowance	Cass	3.0000	mm
Flange Material		SA-266	
Flange Allowable Stress at Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress at Ambient	Sfa	138.00	N./mm ²
Flange Outside Diameter	Fod	2465.000	mm
Flange Inside Diameter	Fid	2297.000	mm
Flange Thickness	Tf	170.0000	mm
Bolt Material		SA-193 B7M	
Bolt Material UNS Number		G41400	
Bolt Allowable Stress At Temperature	Sb	138.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	138.00	N./mm ²
Diameter of Bolt Circle	C	2368.000	mm
Nominal Bolt Diameter	dB	38.1000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		72	
Full face Gasket Flange Option	Program Selects		
Flange Face Outside Diameter	Ffod	2326.000	mm
Flange Face Inside Diameter	Ffid	2297.000	mm
Gasket Outside Diameter	Go	2323.000	mm
Gasket Inside Diameter	Gi	2297.000	mm
Gasket Factor,	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Flange Facing Sketch	Code Sketch 1a		
Column for Gasket Seating	Code Column I		
Gasket Thickness		6.3500	mm
Width of raised face		0.0000	mm
Dist. from Head Centerline to Flange Centroid hr		15.0566	mm
The Flange is not Slotted.			
This unit does not have a Backing Ring.			
Use Full Bolt Load in Flange Design		Yes	



Internal Pressure results for Spherical Heads:

ASME Code, Section VIII, Division 1, 2025, App. 1-6

Thickness Due to Internal Pressure[thr]:
 $= 5PL / 6Sho$
 $= (5 \cdot 1000 \cdot 2070) / (6 \cdot 138)$
 $= 12.5010 \text{ mm}$

Maximum Allowable Working Pressure at Given Thickness[Pa]:
 $= 6Sho \cdot (Th - Cass - Cats) / 5L$
 $= (6 \cdot 138 \cdot 24) / (5 \cdot 2070)$
 $= 1919.85 \text{ KPa.}$

Maximum Allowable Pressure, New and Cold[Pnc]:
 $= 6Sha / 5L$
 $= (6 \cdot 138 \cdot 30) / (5 \cdot 2067)$
 $= 2403.30 \text{ KPa.}$

Actual Stress at given Pressure and Thickness[Sact]:
 $= 5PL / 6(Th - Cass - Cats)$
 $= (5 \cdot 1000 \cdot 2070) / (6 \cdot 24)$
 $= 72. \text{ N./mm}^2$

Spherical head Percent Elongation $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$: 1.081 %

External Pressure results, Spherical Head ASME Code, Section VIII Division 1, 2025

External Pressure Chart CS-2 at 130.00 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Maximum Allowable External Pressure:

Corroded Thickness of Shell	TCA	24.0000	mm
Outside Crown Radius of Shell	OD/2	2093.9998	mm
Crown Radius / Thickness Ratio	$(OD/2)/T$	87.2500	
Geometry Factor, A	f(D/T, LD)	0.0014327	
Materials Factor, B	f(A, Chart)	95.5859	N./mm ²
Maximum Allowable Working Pressure		1095.48	KPa.
EMAWP = $B / ((OD/T)/2) = 95.6 / (174/2.0) = 1095$			

Results for Req'd Thickness for Ext. Pressure (Tca):

Corroded Thickness of Head	TCA	13.7756	mm
Outside Diameter of Shell	ODCA	4188.000	mm
Diameter / Thickness Ratio	(D/T)	304.0156	
Geometry Factor, A	f(DT, LD)	0.0008223	
Materials Factor, B	f(A, Chart)	79.0583	N./mm ²
Maximum Allowable Working Pressure		520.06	KPa.
EMAWP = $B / ((D/T)/2) = 79.1 / (304/2) = 520$			



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Intermediate Calculations for Flanged portion:

ASME Code, Section VIII, Div. 1, 2025 App. 2

Gasket Contact Width, $N = (G_{oc} - G_{ic}) / 2$ 13.000 mm
 Basic Gasket Width, $b_0 = N / 2$ 6.500 mm
 Effective Gasket Width, $b = \text{sqrt}(b_0) * 2.5$ 6.422 mm
 Gasket Reaction Diameter, $G = G_o - 2.0 * b$ 2310.156 mm

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:
 $= 2a + 6t / (m + 0.5)$
 $= 2 * 38.1 + 6 * 170 / (3 + 0.5)$
 $= 367.629 \text{ mm}$

Actual Circumferential Bolt Spacing [Bs]:
 $= C * \sin(\pi / n)$
 $= 2368 * \sin(\pi / 72)$
 $= 103.291 \text{ mm}$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:
 $= \max(\sqrt{Bs / (2a + t)}, 1)$
 $= \max(\sqrt{103 / (2 * 38.1 + 170)}, 1)$
 $= 1.0000$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

Total Area of Bolts 652.644 cm²

	Minimum	Actual	Maximum
Radial Distance between Bolts and Edge:	38.100	45.500	
Circ. Spacing between the Bolts:	82.550	103.291	367.629

Results for the Internal Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	427433.9	Kgf
Contact Load on Gasket Surfaces	Hp	28518.3	Kgf
Hydrostatic End Load at Flange ID	Hd	424790.1	Kgf
Pressure Force on Flange Face	Ht	2643.8	Kgf
Radial Component of Head Membrane Force	Hr	637624.7	Kgf
Operating Bolt Load:	Wm1	455952.2	Kgf
Gasket Seating Bolt Load	Wm2	327703.7	Kgf
Required Bolt Area	Am	324.024	cm ²

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calculation]:
 $= A_b * S_b / (y * \pi * (G_o + G_i))$
 $= 653 * 138 / (68.9 * 3.14 * (2323 + 2297))$
 $= 9.000 \text{ mm}$

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :
 = 38.100 mm [Note: Exceeds actual gasket width, 13.000]

Flange Design Bolt Load (Seating)	W	918372.3	Kgf
Gasket Seating Force (Operating)	Hg	28518.3	Kgf
Distance to Gasket Load Reaction	hg	28.9222	mm
Distance to Face Pressure Reaction	ht	30.7111	mm
Distance to End Pressure Reaction	hd	32.5000	mm

Summary of Moments for Internal Pressure:

Loading	Force	Distance	Bolt Corr	Moment
End Pressure,	Md	424790.	32.5000	1.0000 13806. Kg-m.



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Face Pressure,	Mt	2644.	30.7111	1.0000	81. Kg-m.
Gasket Load,	Mg	28518.	28.9222	1.0000	825. Kg-m.
Floating Hd. Load,	Mh	637625.	15.0566	1.0000	-9601. Kg-m.
Gasket Seating,	Ma	918372.	28.9222	1.0000	26562. Kg-m.

Total Moment for Operation (Internal Pressure) 5111.304 Kg-m.
 Total Moment for Gasket Seating (Int. Pressure) 26561.844 Kg-m.

Results for the External Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	222265.6	Kgf
Hydrostatic End Load at Flange ID	Hd	220890.8	Kgf
Pressure Force on Flange Face	Ht	1374.8	Kgf
Radial Component of Head Membrane Force	Hr	331564.9	Kgf
Gasket Seating Bolt Load	Wm2	327703.7	Kgf
Required Bolt Area	Am	232.884	cm ²

Flange Design Bolt Load (Seating)	W	918372.3	Kgf
Distance to Gasket Load Reaction	hg	28.9222	mm
Distance to Face Pressure Reaction	ht	30.7111	mm
Distance to End Pressure Reaction	hd	32.5000	mm

Summary of Moments for External Pressure:

Loading	Force	Distance	Bolt Corr	Moment	
End Pressure,	Md	220891.	3.5778	1.0000	790. Kg-m.
Face Pressure,	Mt	1375.	1.7889	1.0000	2. Kg-m.
Floating Hd. Load,	Mh	331565.	15.0566	1.0000	-4992. Kg-m.
Gasket Seating,	Ma	918372.	28.9222	1.0000	26562. Kg-m.

Total Moment for Operation (External Pressure) 4199.563 Kg-m.
 Total Moment for Gasket Seating (Ext. Pressure) 26561.844 Kg-m.

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 1000 \cdot 2303 \cdot \sqrt{(4 \cdot 2070 \cdot 2070 - 2303 \cdot 2303) / (8 \cdot 138(2459 - 2303))}$$

$$= 46.0079 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 5111(2459 + 2303) / (138 \cdot 2303)(2459 - 2303)$$

$$= 4814.5029 \text{ mm}^2$$

Required thickness for Main Flange, Internal Operating Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 46 + \sqrt{(46 \cdot 46 + 4815)}$$

$$= 129.2619 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 26562(2459 + 2303) / (138 \cdot 2303)(2459 - 2303)$$

$$= 25019.4629 \text{ mm}^2$$

Required thickness for Main Flange, Internal Bolt-up Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 0 + \sqrt{(0 \cdot 0 + 25019)}$$

$$= 158.1754 \text{ mm}$$

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 520 \cdot 2303 \cdot \sqrt{(4 \cdot 2070 \cdot 2070 - 2303 \cdot 2303) / (8 \cdot 138(2459 - 2303))}$$



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= 23.9241 mm

Factor [J]:

= $Mo(A + B)/(S \cdot B)(A - B)$
 = $4200(2459+2303)/(138 \cdot 2303)(2459-2303)$
 = 3955.7051 mm²

Required thickness for Main Flange, External Operating Condition [T]:

= $F + \sqrt{(F \cdot F + J)}$ per 1-6(g)
 = $23.9 + \sqrt{(23.9 \cdot 23.9 + 3956)}$
 = 91.2150 mm

Factor [J]:

= $Mo(A + B)/(S \cdot B)(A - B)$
 = $26562(2459+2303)/(138 \cdot 2303)(2459-2303)$
 = 25019.4629 mm²

Required thickness for Main Flange, External Bolt-up Condition [T]:

= $F + \sqrt{(F \cdot F + J)}$ per 1-6(g)
 = $0 + \sqrt{(0 \cdot 0 + 25019)}$
 = 158.1754 mm

Summary of Required Thicknesses(mm):

	Head	Flange
Tubeside (Internal) Pressure	12.5010	129.2619
Shellside (External) Pressure	13.7756	91.2150
Tubeside Gasket Seating Load		158.1754
Shellside Gasket Seating Load		158.1754
Maximum + Corrosion Allowance	19.7756	164.1754
Actual Thickness as Given	30.0000	170.0000

Floating Head MAWP and MAPnc (KPa.):

Stress Condition	Tubeside		Shellside	
	MAWP	MAPnc	MAWP	MAPnc
Flange Stress	1353.56	1440.87	1182.90	1412.60
Floating head Stress	1919.75	2403.20	1095.38	1447.07
Bolt Stress	2014.09	2014.09	...	...
Minimum MAWP & MAPnc	1353.56	1440.87	1095.38	1412.60

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: